

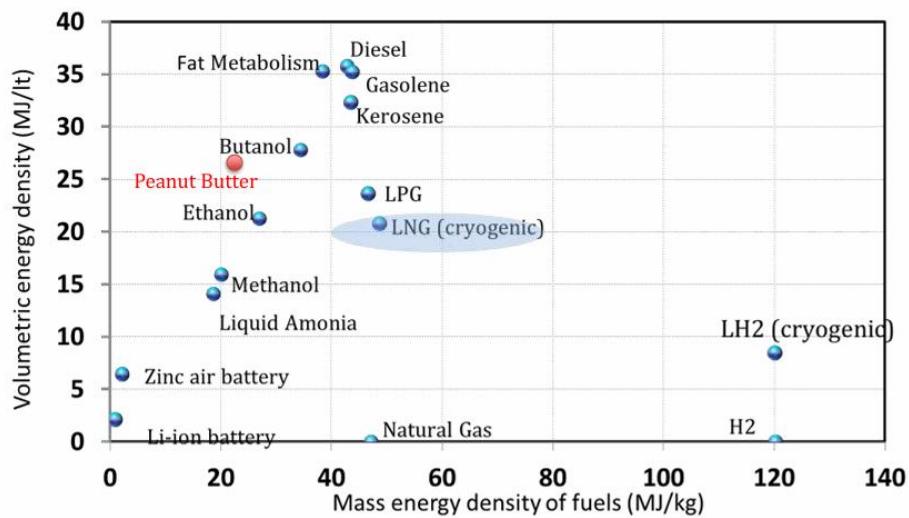
Aero engine technology

Lecture 1

A propulsion system is made up of three main components:

- The energy source, which provides energy to the engine
- Energy to work convertor (e.g. the gas turbine part of the aero engine)
- Thrust producer: takes the work generated and converts it into thrust

When it comes to energy sources, for aviation the most critical characteristic is the energy density of the source w.r.t. volume and mass.



We see that normal batteries have low energy density in both cases, while conventional fuels have much higher energy density, especially per unit volume. On the other hand hydrogen has very low energy density per unit volume, however this can be increased by condensing it (liquefying it).

Many factors need to be considered in the choice of an energy source:

- Volumetric energy density
- Specific energy density
- Emissions
- Policy
- Compatibility
- Safety
- Infrastructure
- Availability
- Cost

The only feasible alternative (to traditional fuels) energy sources for civil aviation in modern times can be classified into three classes:

- **Drop-in fuels:** this contains sustainable alternatives such as biofuels and synthetic kerosene. They are compatible with current aero engines;
- **Batteries:** electric powered engines are still only a concept but are being developed. The main disadvantage is the low specific and volumetric energy density of batteries.
- **Cryogenic:** alternatives with low volumetric energy densities can still be used in aviation in liquid form.

The energy provided by the fuel is then turned into useable power. In aviation this is done by the gas turbine.

Depending on the mission requirements there are different types of engine with varying power outputs:

Type	Power Range	Speed Range (Mach No)	Power to weight (KW/kg)	Approx. Efficiency (%)
Military Engines	< 20 MW	Upto Mach 2	~20	30
Large Turbofans	~ 50 MW	~ Mach 0.9	~10	40
Micro gas turbines	~ 50 kW	< Mach 0.5	~6	10
Turbo Props	< 10 MW	< Mach 0.7	~6	40
Electric Motors	< 250 kW	< Mach 0.5	~5	95
Wankel Engine	< 200 kW	< Mach 0.5	~2.5	25
Radial engines	< 3000 kW	< Mach 0.5	~1.8	30

For aero engines, this power needs to be then converted into thrust. This can be done in two ways:

- **Nozzle:** the exhaust gasses pass through a convergent channel and their speed is greatly increased.
- **Propeller:** the power produced is used to spin a propeller which generates thrust. Typically characterized by very high propulsive efficiency but only works for low Mach numbers.

Thermodynamics summary

To fully understand how an aero engine works it is important to recap some basic thermodynamic concepts.

There are 4 fundamental laws in thermodynamics, which state:

Zeroth Law Thermal equilibrium & the concept of temperature

First Law Conservation of Energy and its interconvertability

Second Law Entropy of any isolated system always increases

Third Law Entropy of a system is zero at absolute zero

The first law states the conservation of energy in a closed system, in other words energy is converted from form to another, however the total value is conserved. This can be expressed as:

$$\Delta E = \Delta KE + \Delta PE + \Delta U$$

Where E is the total energy, U is the internal energy (which is a function of temperature), KE is the kinetic energy in the system and PE is the potential energy. The **internal energy** is a state variable, which means that it's value depends on the first and last thermodynamic states of the system and not on the transformation. It can be expressed as:

$$du = c_v \cdot dT$$

Another important thermodynamic quantity is **enthalpy**, which can be defined as:

$$h = u + p \cdot v$$

Just like the internal energy it can be expressed as:

$$dh = c_p \cdot dT$$

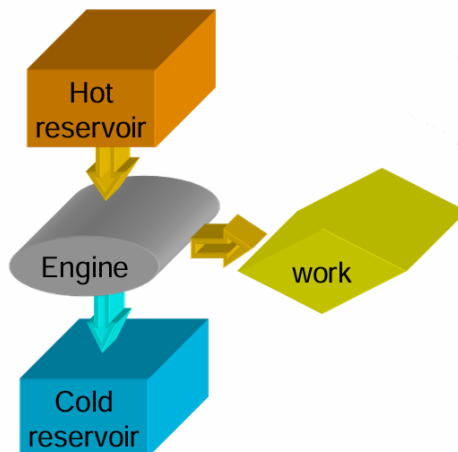
The second law introduces an additional quantity called **entropy**, which can be thought of as a measure of the disorder of a system. Just like enthalpy and internal energy, entropy is also a state variable so its value only depends on the initial and final thermodynamic states of a transformation. Specifically the second law states that the entropy of the universe always increases in a spontaneous (natural) process and remains unchanged in an equilibrium process (an ideal process in which all thermodynamic states are equilibrium states).

In other words the second law defines the concept of irreversible processes, or processes during which the entropy of the system increases. These processes cannot be reversed and are caused by dissipation and losses.

Entropy can be evaluated as:

$$dS = (\partial Q/T)_{\text{rev}}$$

Going back to the engine, let's now look at the thermodynamics of a heat engine. A **heat engine** is a machine that operates between two heat reservoirs and completes a thermodynamic cycle. A generator is a heat engine which takes heat from the hot source, produces work and releases heat into the cold source.

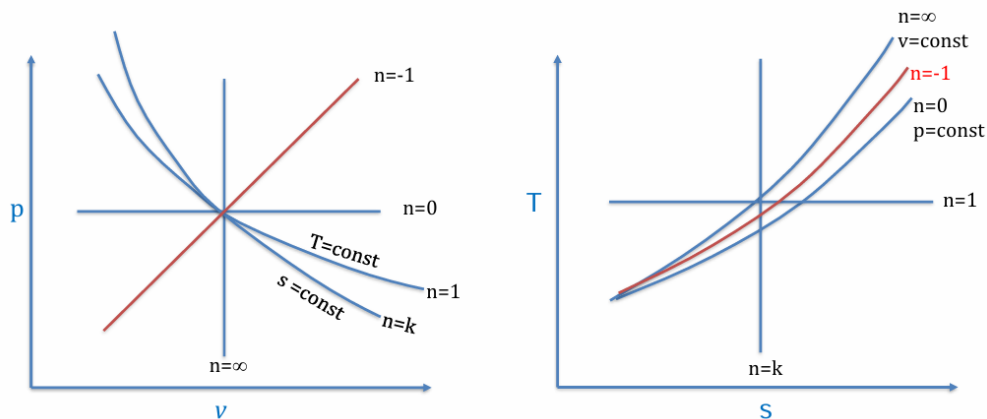


Thermodynamic cycles are made up of different thermodynamic processes. These are usually **polytropic processes**, in other words processes which can be defined by the following relation:

$$pV^n = C$$

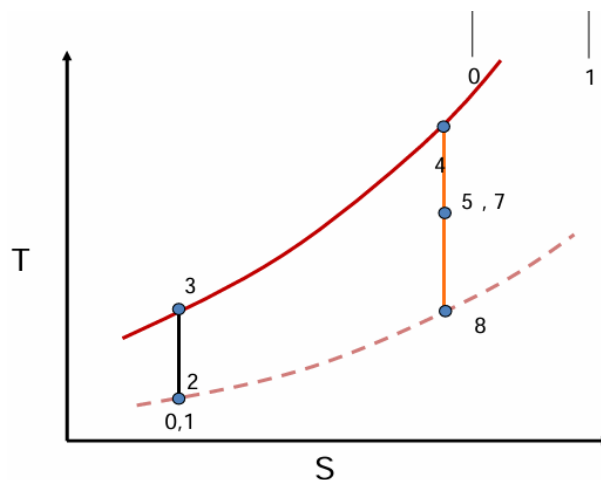
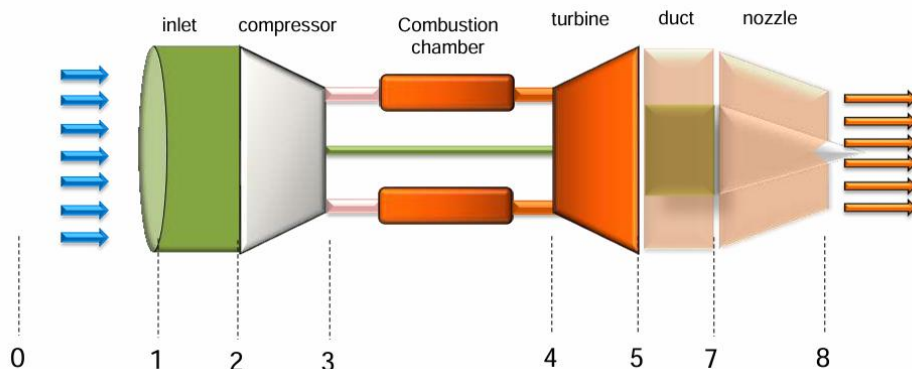
Where n depends on the process:

- $n = 0$: **isobaric process**
- $n = 1$: **isothermal process**
- $n = k$: **isentropic process** (where k is the specific heat capacity coefficient ratio)
- $n = \infty$: **isocoric**



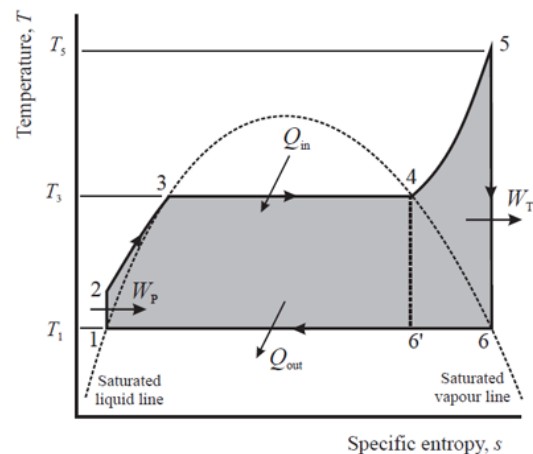
A gas turbine is nothing more than a heat engine in which a fluid (air) completes a thermodynamic cycle. This cycle is the **Joule-Brayton cycle** which is ideally made up of:

- **Isentropic compression:** this occurs in the inlet and the compressor. In reality the process is adiabatic since there is an associated entropy increase due to losses. Work is added to the cycle with no heat involved.
- **Isobaric combustion:** this occurs in the combustor. In reality it is not perfectly isobaric since there are pressure losses. Heat is added to the cycle with no work involved.
- **Isentropic expansion:** this occurs in the turbines and the nozzle. In reality the process is adiabatic since there is an associated entropy increase due to losses. Work is extracted from the cycle with no heat involved. Part of the work extracted is used to run the compressors, while the remaining part is used to generate thrust.
- **Isobaric refrigeration:** this is the last process required to close out the cycle, however it is not executed by the aero engine. It represents what occurs in the atmosphere: the hot exhaust gasses release heat into the atmosphere at constant atmospheric pressure.

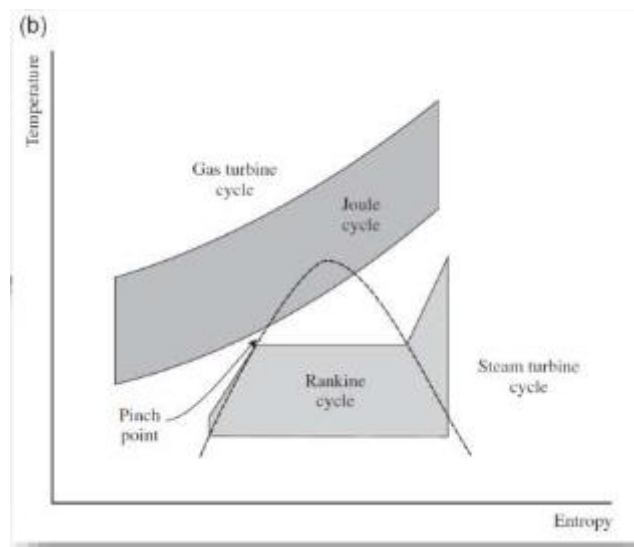


Another example of a thermodynamic cycle is the **Rankine cycle**, which is made up of:

- **Isentropic compression:** this occurs in the liquid phase
- **Isobaric heating:** heat is added into the system. The fluid transitions and becomes vapor at a constant temperature and pressure. Once all of the vapor has transitioned more heat can be added. This last phase is called superheating (4-5).
- **Isentropic expansion:** work is extracted by letting the vapor expand in a turbine
- **Condensation:** heat is extracted from the system. All of the vapor transitions back to its liquid form.



Some power plants use these two cycles combined to obtain very high thermodynamic efficiencies. The exhaust gases coming out of a Joule-Bryton cycle are quite hot, so they can be used to heat up the liquid in the Rankine cycle. This allows for very high efficiencies in the order of 60%.



Lecture 2

Brayton cycle

The assumptions of the ideal Brayton cycle are the following:

- Ideal gas with constant C_p , C_v and chemical composition
- Mass flow constant
- Isentropic expansion and compression
- No pressure losses in ducts or during heat supply

- No change in kinetic energy per each component (i.e. there is no need to use total quantities since the differences and ratios of total and static quantities coincide)

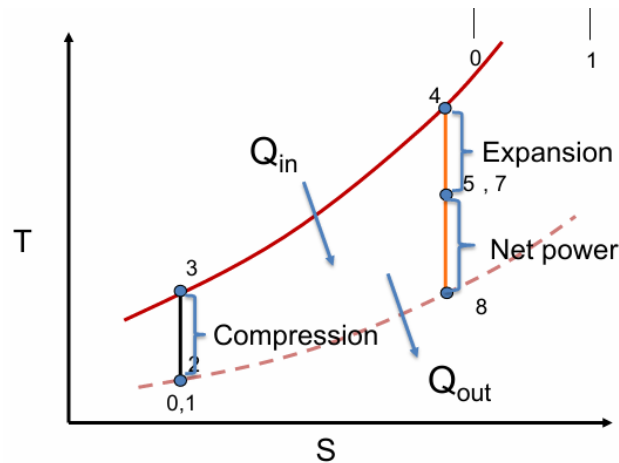
If we consider the whole cycle, from the first law:

$$\Delta U_{2-3-4-8} = (Q_{in} - Q_{out}) + (W_{in} - W_{out}) = 0$$

The change in internal energy is zero since it's a state variable and the initial and final thermodynamic state coincide.

By again using the first law, the work (compression and expansion) and the heat (heat supply and rejection) coming in/out the system can be evaluated:

Compression	$W_{in} = W_{2-3} = \dot{m} \cdot c_p \cdot (T_3 - T_2)$	
Heat supply	$\dot{Q}_{in} = \dot{m} \cdot c_p \cdot (T_4 - T_3)$	
Expansion 1	$W_{4-5} = \dot{m} \cdot c_p \cdot (T_4 - T_5)$	} W_{out}
Expansion 2	$W_{net} = W_{5-8} = \dot{m} \cdot c_p \cdot (T_5 - T_8)$	



With these relations a power balance equation can be setup:

$$W_{2-3} + \dot{Q}_{3-4} = W_{4-5} + W_{5-8} + \dot{Q}_{8-2}$$

Since the power extracted by the turbine is used to spin the compressor:

$$W_{4-5} = W_{2-3}$$

$$\dot{Q}_{3-4} = W_{5-8} + \dot{Q}_{8-2}$$

The heat we need to provide to the gas in the combustion chamber is equal to the power generated in the nozzle and the heat that is released into the environment by the exhaust gasses.

By using these expressions, the thermodynamic efficiency of the Brayton cycle can be evaluated. This is defined as the ratio between the net power output and the total energy supply.

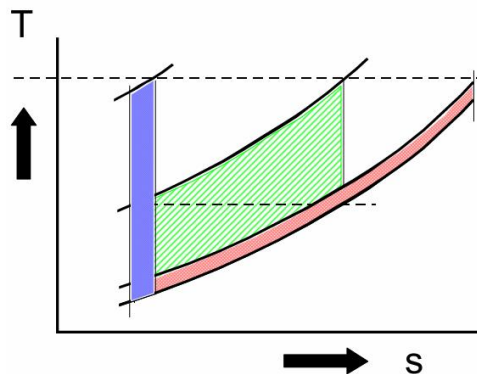
$$\begin{aligned}
\eta_{thdy} &= \frac{W_{net}}{\dot{Q}_{in}} = \left(\frac{W_{4-8} - W_{2-3}}{\dot{Q}_{3-4}} \right) \\
&= \left(\frac{c_p \cdot \dot{m} \cdot (T_4 - T_8) - c_p \cdot \dot{m} \cdot (T_3 - T_2)}{c_p \cdot \dot{m} \cdot (T_4 - T_3)} \right) \\
\eta_{thdy} &= \left(\frac{(T_4 - T_8) - (T_3 - T_2)}{(T_4 - T_3)} \right)
\end{aligned}$$

Since compression and expansion are assumed to be isentropic, we can use the isentropic relations for temperature and pressure:

$$\begin{aligned}
\left(\frac{T_4}{T_8} \right) &= \Pi^{\frac{k-1}{k}} & \left(\frac{T_3}{T_2} \right) &= \Pi^{\frac{k-1}{k}} \\
\eta_{thdy} &= \left(\frac{T_4 \left(1 - \frac{T_8}{T_4} \right) - T_3 \left(1 - \frac{T_2}{T_3} \right)}{(T_4 - T_3)} \right) = \left(\frac{T_4 \left(1 - \frac{1}{\Pi^{\frac{\gamma-1}{\gamma}}} \right) - T_3 \left(1 - \frac{1}{\Pi^{\frac{\gamma-1}{\gamma}}} \right)}{(T_4 - T_3)} \right) \\
\eta_{thdy} &= \left(\frac{(T_4 - T_3) \cdot \left(1 - \frac{1}{\Pi^{\frac{\gamma-1}{\gamma}}} \right)}{(T_4 - T_3)} \right) & \boxed{\eta_{thdy} = \left[1 - \frac{1}{\Pi^{\frac{\gamma-1}{\gamma}}} \right]}
\end{aligned}$$

We can see that the thermodynamic efficiency of the Joule-Brayton cycle is only dependent on the pressure ratio, the pressure difference between the max pressure (reached after the isentropic compression) and the min pressure (reached after the isentropic expansion).

If we only cared about the maximum thermodynamic efficiency then we would need to maximize the pressure ratio and the ideal cycle would be the purple one.



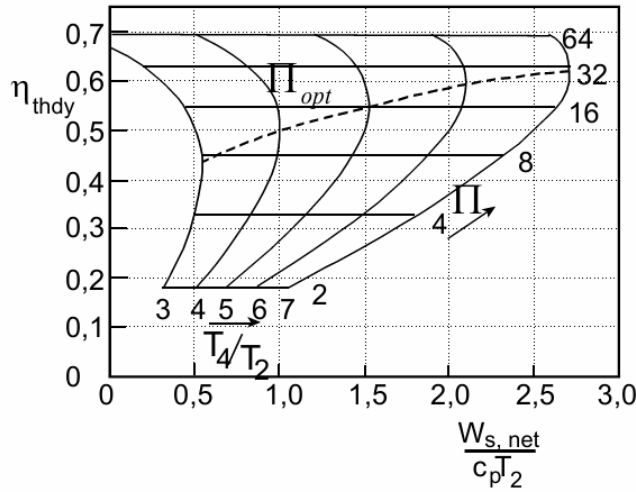
However, additional factors need to be considered. The net power is defined as the total power generated in the expansion process, minus the power required to fuel the compressor:

$$\begin{aligned}
 W_{net} &= W_{4-8} - W_{2-3} \\
 &= \dot{m} \cdot c_p \cdot (T_4 - T_8) - \dot{m} \cdot c_p \cdot (T_3 - T_2)
 \end{aligned}$$

With a similar procedure as the one used to obtain the thermodynamic efficiency:

$$\frac{W_{s,net}}{c_p T_2} = \frac{T_4}{T_2} \left[1 - \frac{1}{\Pi^{\frac{\gamma-1}{\gamma}}} \right] - \left[\Pi^{\frac{\gamma-1}{\gamma}} - 1 \right]$$

Where $W_{s,net}$ is the specific net power, or the power generated per unit of mass flow, which is then normalized w.r.t. the ambient enthalpy. We can see that this parameter, which is as important as the thermodynamic efficiency is not only dependent on the pressure ratio but also the temperature ratio. For an ideal cycle we have that the pressure ratio, for a given temperature ratio, has to be chosen at the optimum:



At a given pressure ratio, the higher the temperature ratio, the higher the specific power will, while the thermodynamic efficiency remains unchanged.

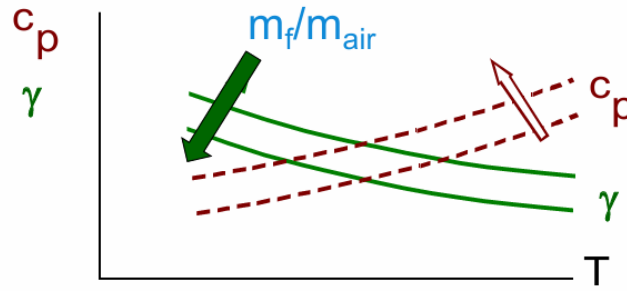
At a given temperature ratio, there exists an optimum pressure ratio for which the specific net power is highest. When it comes to the efficiency it only depends on the pressure ratio and it is highest when the pressure ratio is highest.

Real Brayton cycle

All of the previous results are based on the many assumptions listed at the beginning, which are:

- Ideal gas with constant C_p , C_v and composition
- Mass flow constant
- Isentropic expansion and compression
- No pressure losses in ducts or during heat supply
- No change in kinetic energy per each component (no energy losses)

Let's look at the validity of all these assumptions. The assumption of constant C_p and C_v is not valid for a real cycle since they both depend on temperature and also composition. The addition of fuel in the combustion chamber changes the composition of the fluid, typically reducing the heat capacity ratio and increasing the heat capacity at constant pressure.



$c_{p,a} = 1000 \text{ J/kg.K}$	$\gamma_a = 1.4$
$c_{p,g} = 1150 \text{ J/kg.K}$	$\gamma_g = 1.33$

This assumption can lead to over/underestimations of up to 5% for the efficiency of an engine. The addition of the fuel flow also breaks the assumption of constant mass flow rate. However, typically this addition is so small that it can be neglected.

Another important assumption for the ideal Brayton cycle is that there are no kinetic energy changes in each component, however this is not true for the real cycle. To take into account these kinetic energy variations, since we do not know the velocity of the fluid in all the components, total quantities are used.

$$T_t = T_s + \frac{v^2}{2} \quad p_t = p_s \left(\frac{T_t}{T_s} \right)^{\frac{\gamma}{\gamma-1}}$$

For a real Brayton cycle we also need to introduce efficiencies which take into account losses in both the compressor and the turbine. These efficiencies are defined as isentropic efficiencies since the ideal power is defined along an isentropic expansion/compression.

$$\eta_c = \frac{W_{c,ideal}}{W_{c,real}} = \frac{\dot{m} \cdot (c_{p3'} \cdot T_{t3'} - c_{p2} \cdot T_{t2})}{\dot{m} \cdot (c_{p3} \cdot T_{t3} - c_{p2} \cdot T_{t2})}$$

$$\eta_t = \frac{W_{t,real}}{W_{t,ideal}} = \frac{\dot{m} \cdot (c_{p4} \cdot T_{t4} - c_{p5} \cdot T_{t5})}{\dot{m} \cdot (c_{p4} \cdot T_{t4} - c_{p5'} \cdot T_{t5'})}$$

For both compression and expansion, in the real case an increase in entropy is associated. For the real compressor more work is required to achieve the same pressure ratio, while for the turbine less work is extractable for the same pressure ratio.

When solving a real Brayton cycle these efficiencies can be used to evaluate the final temperature/pressure reached. For the compressor:

$$T_{t3} = \frac{1}{\eta_c} \cdot T_{t2} \cdot \left(\frac{T_{t3'}}{T_{t2}} - 1 \right) + T_{t2} = T_{t2} \cdot \left(\frac{\pi_c^{\frac{\gamma}{\gamma-1}} - 1}{\eta_c} + 1 \right)$$

For the turbine:

$$P_{t5} = P_{t4} \cdot \left(1 - \frac{1}{\eta_t} \cdot \left(1 - \frac{T_{t5}}{T_{t4}} \right) \right)^{\frac{\gamma}{\gamma-1}}$$

The total temperature at the turbine outlet is measured through a power balance with the compressor:

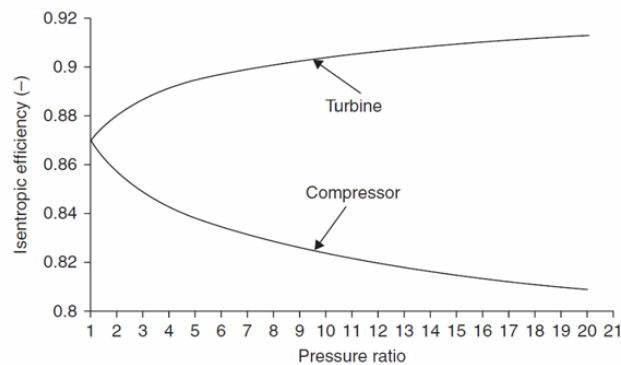
$$W_{2-3} = W_{4-5}$$

$$\dot{m} \cdot (c_{p3} \cdot T_{i3} - c_{p2} \cdot T_{i2}) = \dot{m} \cdot (c_{p4} \cdot T_{i4} - c_{p5} \cdot T_{i5})$$

$$\dot{m} \cdot c_{pa} \cdot (T_{i3} - T_{i2}) = \dot{m} \cdot c_{pg} \cdot (T_{i4} - T_{i5})$$

$$T_{i5} = T_{i4} - \frac{\dot{m} \cdot c_{pa} \cdot (T_{i3} - T_{i2})}{\dot{m} \cdot c_{pg}}$$

In general as the pressure ratio increases the isentropic efficiency of the turbine increases as well, while the isentropic efficiency of the compressor decreases, so the benefits of increasing the pressure ratio are reduced.



Mechanical losses occur between the turbine and the compressor, so a mechanical efficiency can be defined (although it is typically very high ~99.5%).

$$\eta_m = \frac{W_t - \text{mechanical losses}}{W_t}$$

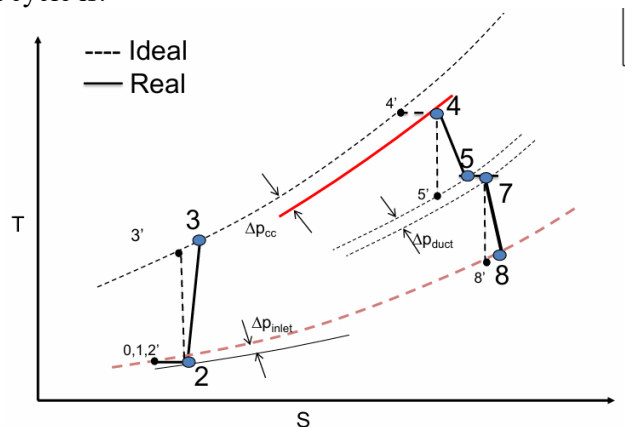
Pressure losses occur in every component and have to be accounted for by introducing pressure ratios.

Inlet: $\Delta p_{inlet} = p_{t1} - p_{t2}$

Combustion chamber: $\Delta p_{cc} = p_{t3} - p_{t4}$

Duct: $\Delta p_{duct} = p_{t5} - p_{t7}$

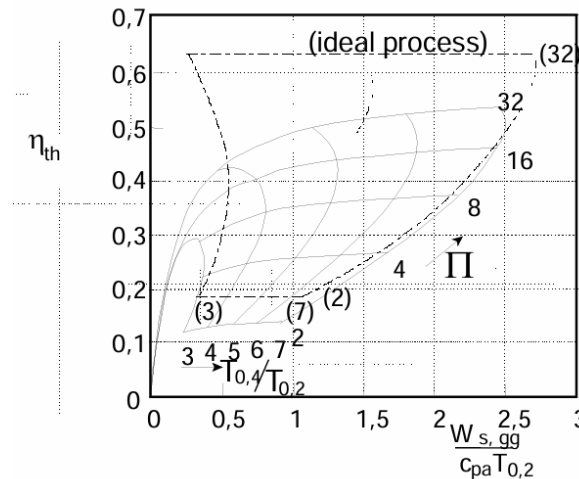
Overall the real Brayton cycle is:



Pressure losses in the inlet bring state 2 to a lower pressure. Losses in the compressor tilt the compression curve so that entropy increases and the power required increases. In the combustion

chamber, due to pressure losses, we move to a curve with lower pressure. In the turbine losses dictate an increase in entropy and a decrease in the total work that is extracted. Then pressure losses in the duct reduce the pressure from 5 to 7. Finally the expansion in the nozzle is also tilted to a higher entropy.

By introducing losses the relations obtained for an ideal Brayton cycle break down. We see that for high pressure/temperature ratios they still roughly hold, but for lower temperature ratios this is not the case:

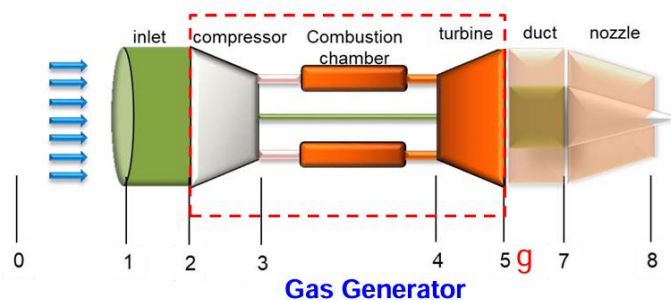


For lower temperature ratios, very little heat is being added into the system, while at the same time high pressure ratios are trying to be obtained. This is not possible since the cycle cannot maintain itself (work extracted is not enough to fuel the compression) and therefore both the net power and the efficiency go to zero. We also see that at constant pressure ratio now the thermodynamic efficiency increases slightly with temperature ratio.

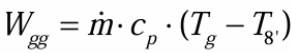
Lecture 3

Gas Generator

The "gas generator" part of the engine is the part which generates the **gas power**, or the power that is then transformed into kinetic energy (and then thrust). This transformation process into kinetic energy occurs after station g (in the nozzle). In particular, the gas power is the maximum amount of power that can be used, which is therefore defined along an isentropic expansion curve.

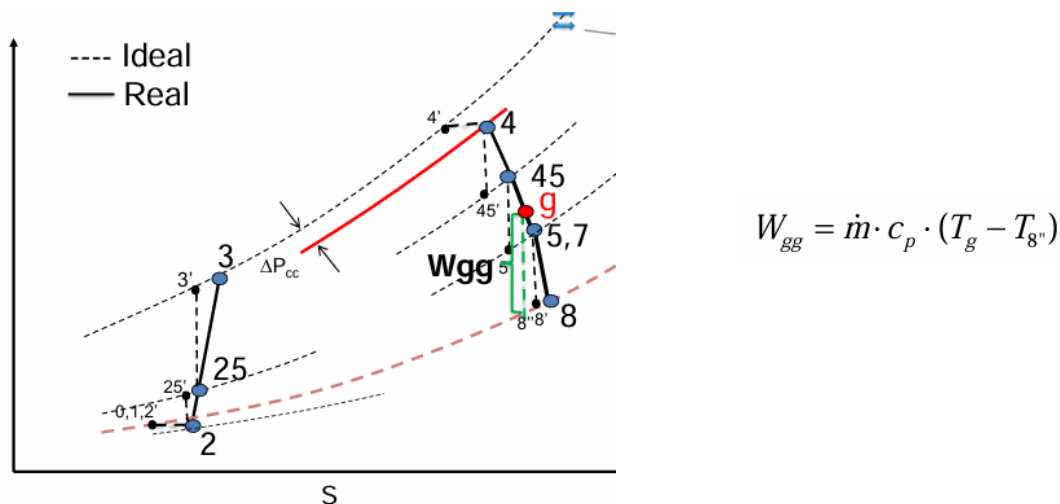


We can evaluate the power generated by this section as the gas power, or the power generated by the equivalent isentropic expansion from station g to 8'.



The diagram illustrates the flow path of a gas turbine engine. The flow starts at the inlet (0), passes through the FAN (1), then the compressor (2 to 25), the combustion chamber (CC), the high-pressure turbine (HPT), the low-pressure turbine (LPT), the duct (45), and finally the nozzle (5 to 8). The diagram also shows the cold air flow path (purple arrows) and the hot air flow path (orange arrows). The gas generator section is highlighted in red.

We also note that the part of the fan which compresses the core flow is part of the gas turbine.



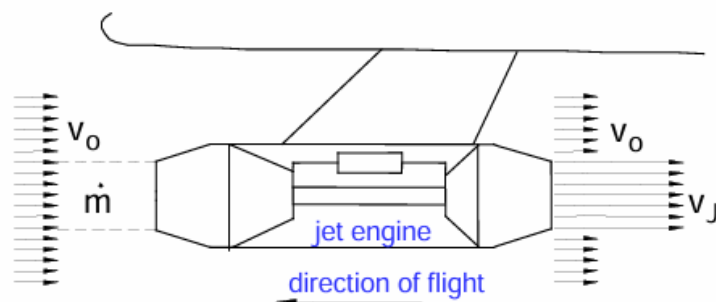
Some useful definitions for aero engines (assuming ideal expansion conditions $p_e = p_{out}$):

Gross thrust $F_G = \dot{m}v_j$; The total force produced at the exhaust nozzle

Net thrust $F_N = \dot{m} * (v_j - v_0)$;

This is given as the gross thrust minus the **ram drag**, or the force needed to slow down the incoming air to speeds the engine can handle

Propulsive jet power	$\dot{W}_{prop, jet} = \frac{1}{2} \dot{m} (v_j^2 - v_0^2)$	Kinetic energy that can be transformed into thrust
Thrust power	$\dot{W}_{thrust} = \dot{m} (v_j - v_0) v_0$	Given by the product of net thrust and flight velocity
Power loss	$P_{loss} = \frac{1}{2} \dot{m} (v_j - v_0)^2$	Residue kinetic energy at the nozzle exit that is not transformed into thrust



From these equations we can see that we can increase the net thrust by increasing either the mass flow rate or the exhaust velocity. However, by increasing the exhaust velocity the power loss also increases with the square of the exhaust velocity, so it is much more convenient to increase the mass flow rate. This is the basic idea of the turbofan engine.

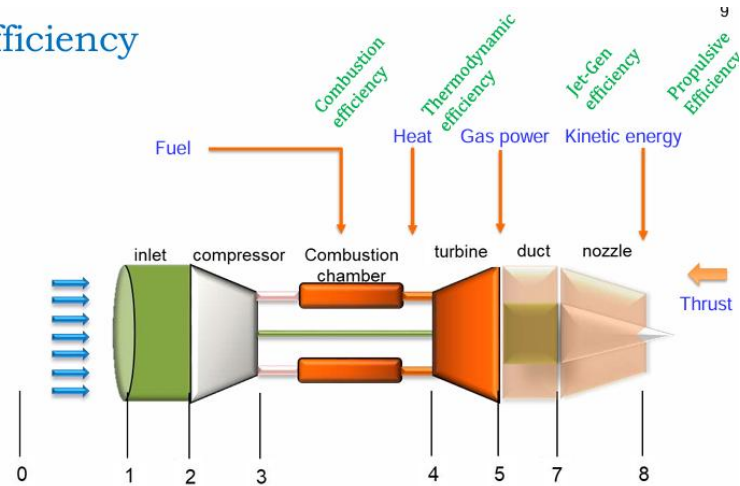
If the nozzle is not in the perfect expansion condition the pressure at the nozzle exit is different from the environment pressure and this difference generates some thrust, which is proportional to the nozzle exit area and the pressure difference. This pressure difference can also be accounted for by defining an **effective jet velocity** $v_{j,eff}$.

Gross Thrust	$F_G = \dot{m} \cdot v_j + A_{exit} \cdot (p_j - p_0) = \dot{m} \cdot v_{j,eff}$
Net Thrust	$F_N = \dot{m} \cdot (v_j - v_0) + A_{exit} \cdot (p_j - p_0) = \dot{m} \cdot (v_{j,eff} - v_0)$
Thrust power	$\dot{W}_{thrust} = F_N \cdot v_0$
Power loss	$P_{loss} = \frac{1}{2} \dot{m} (v_{j,eff} - v_0)^2$

Aero engine efficiencies

Every station of the turbojet engine introduces losses that reduce the overall efficiency.

Efficiency



- **Combustion efficiency:** the process that occurs is the transformation of chemical energy into heat. This takes into account the losses that occur in the combustion chamber due to incomplete combustion. The efficiency is given by extra amount of heat produced divided by the chemical energy contained in the fuel.

$$\eta_{comb} = \frac{(\dot{m}_{gas} \cdot c_{p,g} \cdot T_4 - \dot{m}_{air} \cdot c_{p,air} \cdot T_3)}{\dot{m}_f \cdot LHV_f} \approx \frac{\dot{m}_a \cdot c_{p,a} \cdot \Delta T_{comb}}{\dot{m}_f \cdot LHV_f}$$

Fuel Type	Lower Heating Value (LHV) [MJ/kg]
Kerosene	43
Hydrogen	120

- **Thermodynamic efficiency:** this determines the part of heat that is transformed into gas power and it's given as the ratio between gas power and total heat produced in the combustion process. Takes into account the power used by the compressor.

$$\eta_{thdy} = \frac{W_{gg}}{\dot{m}_{air} \cdot c_{p,a} \cdot \Delta T_{comb}}$$

Where the gas power can be evaluated as follows for a turbojet:

$$W_{gg} = \dot{m}_{gas} \cdot c_{p,g} \cdot (T_g - T_{8'})$$

$$W_{2-3} = W_{4-5} \Rightarrow T_{t5}$$

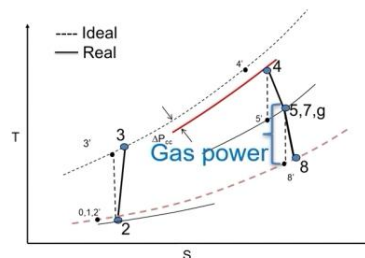
$$T_g = T_{t5}$$

$$p_g = p_{t5}$$

$$p_{8'} = p_{ambient}$$

Isentropic process

$$\frac{T_g}{T_{8'}} = \left(\frac{p_g}{p_{8'}} \right)^{1-1/\gamma} = \left(\frac{p_g}{p_{ambient}} \right)^{1-1/\gamma}$$



- **Jet generation efficiency:** the process that occurs is the transformation of gas power into kinetic energy. This takes into account the losses that occur in the duct and nozzle during the expansion.

$$\eta_{jet, gener.} = \frac{\frac{1}{2} \dot{m} (v_j^2 - v_o^2)}{W_{gg}}$$

- **Propulsive efficiency:** the process that occurs is the transformation of kinetic energy into thrust power. This takes into account the kinetic energy that cannot be transformed into thrust.

$$\eta_{prop} = \frac{W_{thrust}}{W_{kinetic}} = \frac{\dot{m} (v_j - v_o) v_o}{\frac{1}{2} \dot{m} (v_j^2 - v_o^2)}$$

$$\eta_{prop} = \frac{2}{1 + \frac{v_j}{v_o}}$$

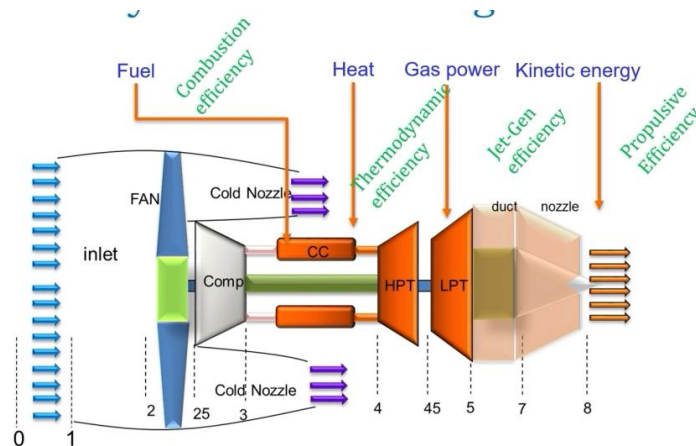
- **Thermal efficiency:** given by the combination of the combustion, thermodynamic and jet generation efficiencies. This takes into account all the losses that occur in the engine.

$$\eta_{thermal} = \frac{\frac{1}{2} \dot{m} (v_j^2 - v_o^2)}{\dot{m}_f LHV_f}$$

- **Total efficiency:** takes into account all of the losses from the combustion chamber to the generation of thrust. It quantifies how efficiently the gas turbine transforms the chemical energy of the fuel into thrust power.

$$\eta_{total} = \frac{\dot{m} (v_j - v_o) v_o}{\dot{m}_f LHV_f}$$

For a turbofan the same efficiencies can be defined and evaluated with some caveats:



- **Combustion efficiency:** defined and evaluated in the same way as we did for the turbojet.
- **Thermodynamic efficiency:** defined and evaluated in the same way as we did for the turbojet.

However, the gas power is different in this case, since station g changes as well as the final temperature $T_{8''}$. Again the temperature at the station g is evaluated by setting up the power balance between the expansion from 4-g and the power needed to drive the core compressor (including the part of the fan that compresses the core flow).

To define the pressure at station g we need to evaluate it from the turbine efficiency and the temperature of the gasses at station g.

Again the temperature at the exhaust after an isentropic expansion used to evaluate the gas power is obtained through the isentropic relations since $p_{8''}=p_8$.

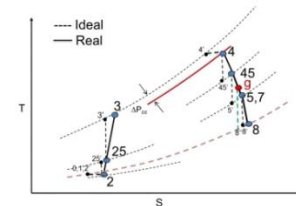
$$W_{gg} = \dot{m}_{gas} \cdot c_{p,g} \cdot (T_g - T_{8''})$$

$$W_{2-3} = W_{4-g} \Rightarrow T_g$$

$$P_g = f(\eta_t, T_g)$$

$$P_{8''} = P_{ambient}$$

Isentropic process

$$\frac{T_g}{T_{8''}} = \left(\frac{p_g}{p_{8''}} \right)^{1-1/\gamma} = \left(\frac{p_g}{P_{ambient}} \right)^{1-1/\gamma}$$


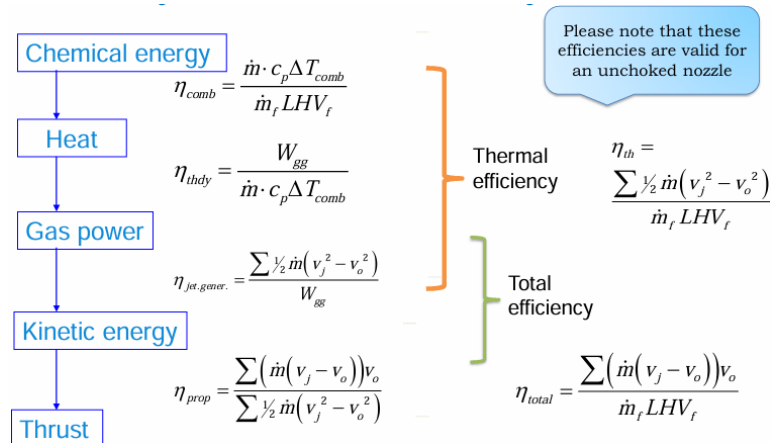
- **Jet generation efficiency:** defined in the same way as we did for the turbojet. In this case the kinetic energy not only takes into account the core flow but also the bypass flow.

$$\eta_{jet, gener.} = \frac{\sum \frac{1}{2} \dot{m} (v_j^2 - v_o^2)}{W_{gg}}$$

- **Propulsive efficiency:** defined in the same way as we did for the turbojet. In this case the both flows need to be taken into account.

$$\eta_{prop} = \frac{W_{thrust}}{W_{kinetic}} = \frac{\sum (\dot{m} (v_j - v_o)) v_o}{\sum \frac{1}{2} \dot{m} (v_j^2 - v_o^2)}$$

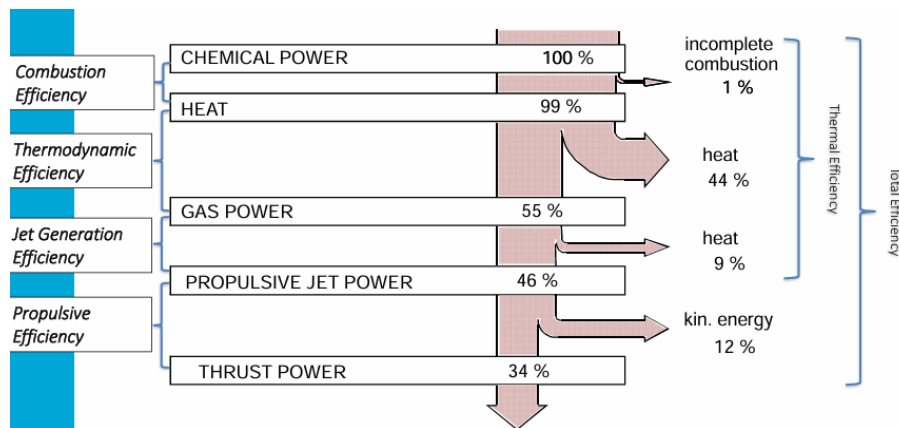
Overall for an unchoked nozzle we have:



Where the thermal efficiency can be obtained from the product of the combustion, thermodynamic and jet generation efficiencies. The global efficiency is still given by considering all of the losses encountered from the combustion chamber to the production of thrust.

For an choked nozzle the jet velocity needs to be substituted by the effective jet velocity.

For a generic jet engine we can visualize all of the losses through a Sankey diagram:



Although the actual values might change from engine to engine, the largest amount of energy will always be lost through heat/to power the compressor and will be taken into account by the thermodynamic efficiency. Another common trend is the low losses due to incomplete combustion.

Let's now introduce some useful parameters to compare engines:

- Net Thrust $F_N = \sum [\dot{m}(v_j - v_o) + A_{exit}(p_j - p_o)]$

- Specific Thrust $F_s = \frac{F_N}{\dot{m}_a}$

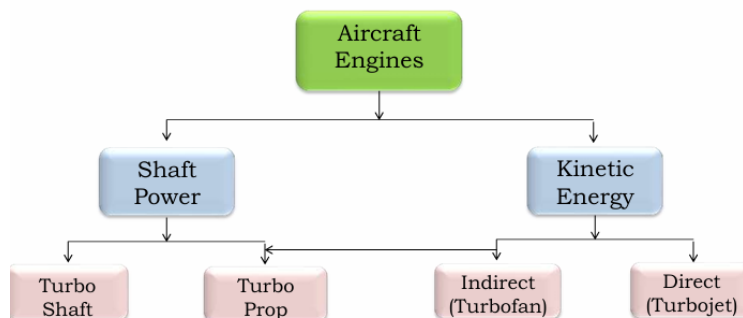
- Thrust specific fuel consumption (TSFC)

$$TSFC = \frac{\dot{m}_f}{F_N} \propto \frac{v_0}{\eta_{tot}}$$

Specific thrust allows to quantify how efficiently an engine can generate thrust w.r.t. its size by dividing the net thrust by the mass flow rate. The thrust specific fuel consumption is instead given by the fuel flow ratio divided by the net thrust and it is inversely proportional to the global efficiency. These two values allow the comparison between engines of different sizes.

Aero engine classification

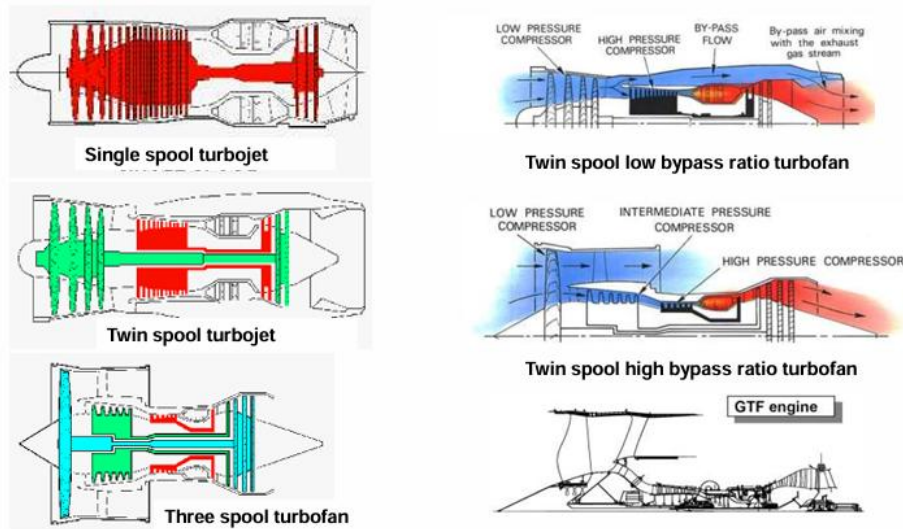
Aero engines can be classified in many ways. One is to distinguish them based on how the gas power is used.



The gas power can either be converted into shaft power or kinetic energy. Both the turbofan and the turboprop actually lie between the two categories since they use the gas power to both generate shaft power and thrust.

The other classification is based on the architecture of the engine. The simplest architecture is the single spool turbojet, which has been phased out by twin spool turbojets/turbofans (which have a high/low pressure compressor and a high/low pressure turbine) or even three spool turbofans.

This is because in the single spool turbojet, as the pressure increases the compressor stage loading is unbalanced.

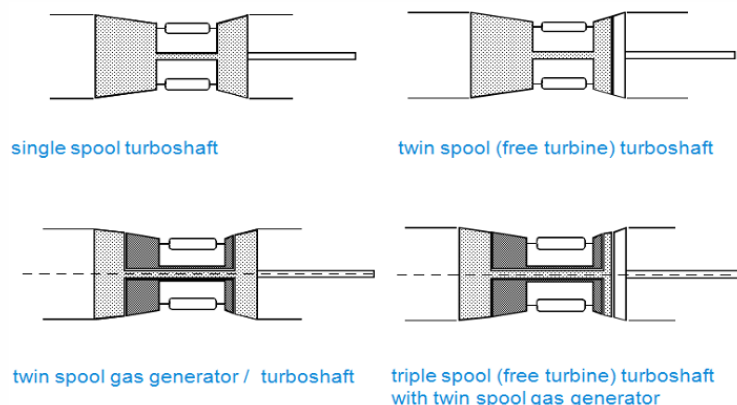


The different spools allow each part of the compressor to run at their optimal speed.

Turbofan engines can also be classified based on how large their bypass ratio (ratio between the mass flow rate in the fan and in the core) is.

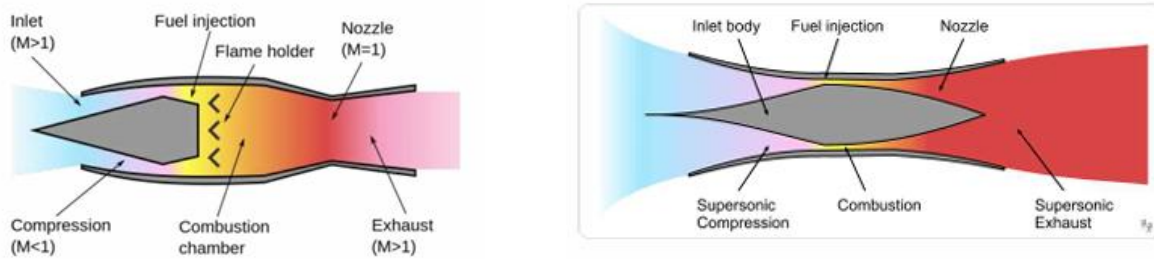
An additional engine that is outside of this classification is the Gear TurboFan engine GTF, a two spool turbofan with a gearbox which decouples the fan and the low pressure compressor. This engine has the advantage of a three spool turbofan while effectively only having two spools. The advantages of this decoupling will be explained in depth later.

A similar classification can also be applied to turboshaft/prop engines. When the load is directly connected to both the compressor and the turbine it might drastically reduce the engine's performance since their operational point is determined by the load and not by the turbomachinery optimal operating conditions. By inserting a free turbine which is only connected to the load this can be avoided and the compressor and HPT can both work at their optimal operating condition.

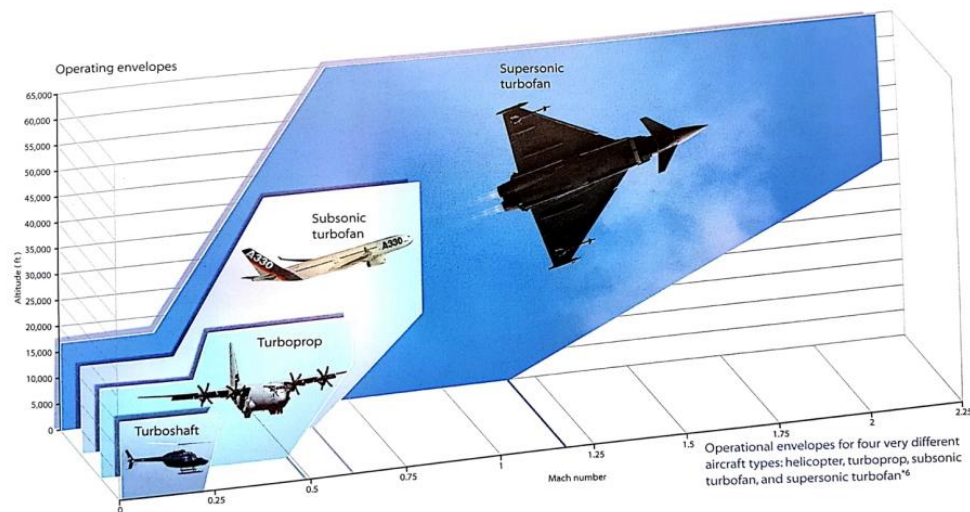


There are two other types of engines which follow the Brayton cycle that do not fit into these classifications : the ramjet and scramjet. These two types of engines do not have any mechanical components since they compress the air though dynamic compression while flying at supersonic speeds, yet they still follow the Joule-Brayton cycle. Therefore the compressor stage is not needed and neither the fan stage which drives it is.

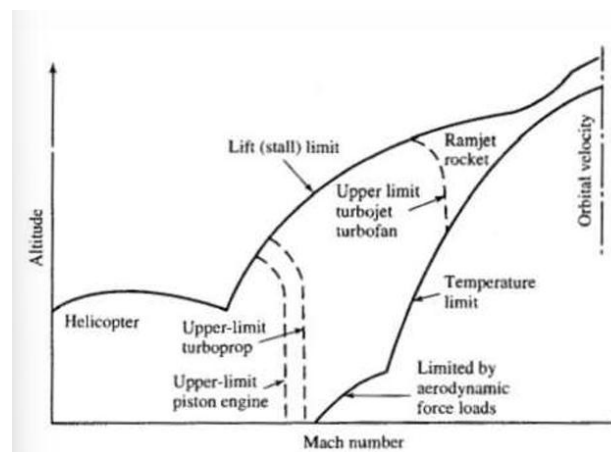
These engines only work at high supersonic speeds at which air compresses dynamically in virtue of the high speeds.



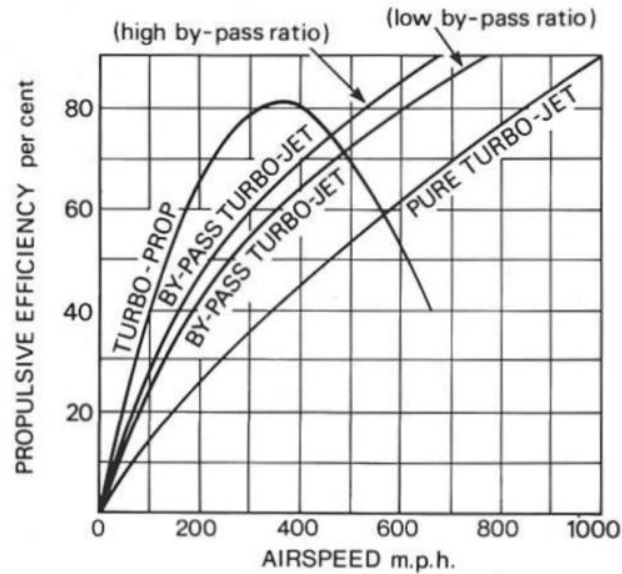
Different engine types have different operating envelopes as shown in the figure below.



The limits of this flight envelope are defined by different phenomena. At high flying altitudes and low speeds the limiting factor is the stall of the wings, while at low altitudes and high speeds it is the airframe/engine loads limits.



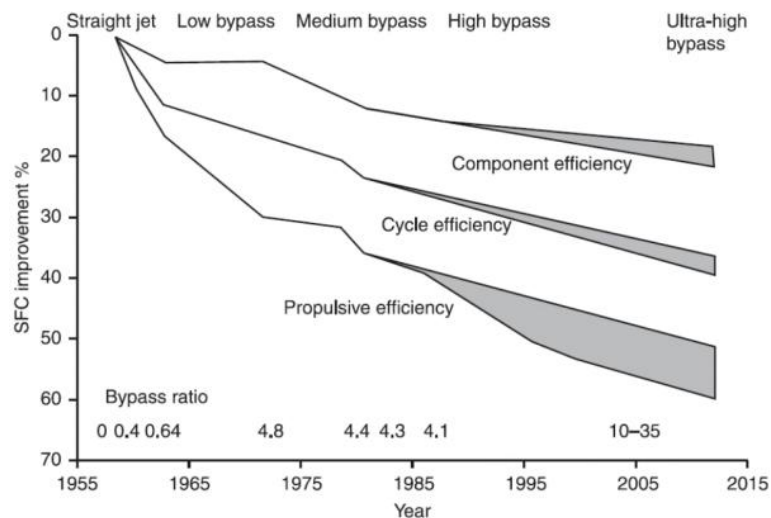
Another thing that distinguishes different engine architectures is the behaviour of propulsive efficiency to airspeed. In absolute terms, the turboprop has the highest propulsive efficiency at low speeds, however this drops drastically at transonic flight speeds due to tip shock on the propeller blades. The turbofan has the highest propulsive efficiency in the transonic and supersonic regimes, beating out the turbojet. We also observe that as the bypass ratio increases the propulsive efficiency also increases.



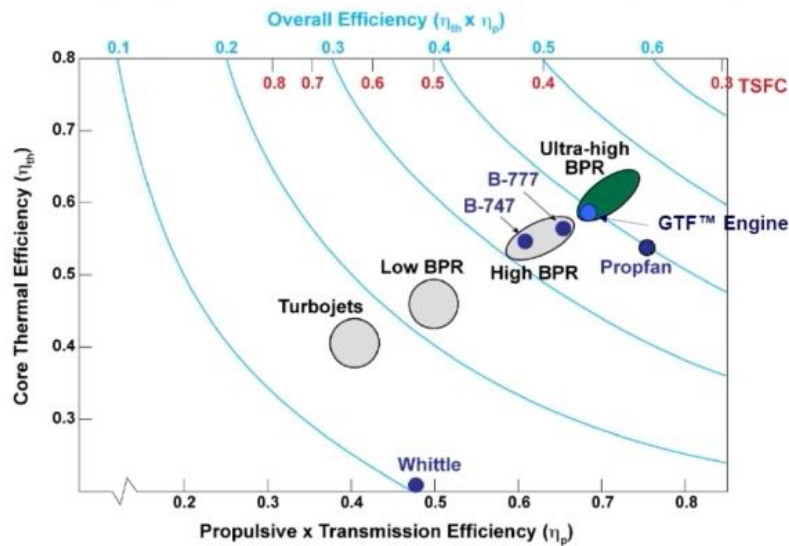
Engine efficiency trends

In general we see an increase of all the efficiencies with the development of new technologies. The efficiency of aero engines increased by 70% in the past decades, while the thrust specific fuel consumption of aero engines decreased by 50%.

The thrust specific fuel consumption reduction is due to the improvements of engine component efficiency, cycle efficiency and a large part being the propulsive efficiency. The main contributors to this increase are the increased bypass ratio, TIT, compression ratio and the overall reduction of losses.

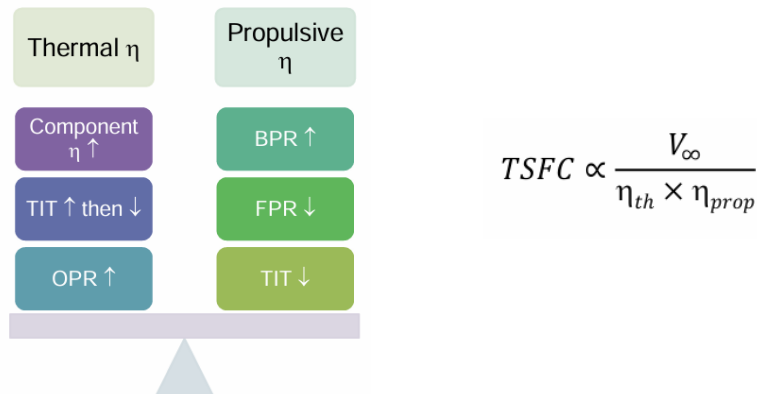


Nowadays typical values for the thermal efficiency are around 50-60%, 65-75% propulsive efficiency (for high bypass ratio turbofan engines, for which the bypass ratio has also doubled in the past decades) and total efficiencies of 40-50%.



Design considerations

To minimize the specific fuel consumption of new engines we want to maximize both the thermal efficiency and the propulsive efficiency, which are dependent on the following parameters:



To increase the thermal efficiency we need to increase the efficiency of all engine components as well as the Overall Pressure Ratio OPR. For the Turbine Inlet Temperature TIT there is a specific value for which the thermal efficiency is maximized.

Meanwhile for the propulsive efficiency we want to increase the ByPass Ratio BPR with a combined reduction of the Fan Pressure Ratio FPR. Also we want the TIT to decrease.

We can see that we have opposing design goals, such as the increase in TIT to increase the thermal efficiency which leads to a decrease of propulsive efficiency.

Therefore design choices need to be made.

We can see how the design choices have lead to improvements over the years by comparing an engine from the '50s and a modern engine.

We observe an increase in thrust generated, combined with a decrease in weight and in specific fuel consumption. This is due to the higher TIT and OPR while having fewer compressor stages. We also can observe an increase in diameter together with the BPR, while the overall length decreased.

Performance F104 Starfighter ↔ F16

Engine Parameter	Unit	J79	F100
Thrust (wet)	kN	81	108
Thrust (dry)	kN	53.5	68
Gas generator airflow	kg/s	77	61.5
Sp. Thrust	kN/kg/s	0.69	1.1
Sp. Fuel Cons, SFC	g/Nh	84	68
Bypass Ratio	-	0	0.7
Turbine Inlet Temp	°C	990	1395
Comp pr. Ratio	-	13.5	25
No. of stages Comp	-	17	3+10
No. of stages of Turbine	-	3	2 + 2
Rotor speed	Rpm	7690	9600/11900
Weight	kg	1750	1360
Diameter	cm	77.8	100
Diameter	cm	530	488

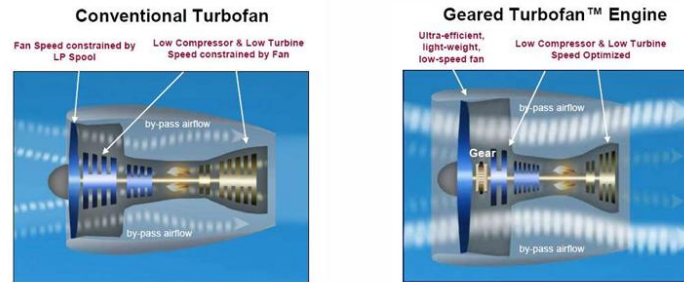
This analysis is valid for military engines, however similar trends can be observed for civil engines.

Parameters	JT 8D	CFM56	Leap-1A
Bypass Ratio	-	1.62	5.2
Core Mass Flow Rate kg/s	81.2	52.2	41.5
Fan Pressure Ratio	1.9	1.6	1.4
HPC Pressure Ratio	3.5	8	13
Turbine Inlet Temperature °C	1030	1270	1450
Overall Pressure Ratio	16.9	32.5	33
Fan, LPC & HPC Isentropic Efficiency	0.83	0.88	0.92
LPT and HPT Isentropic Efficiency	0.88	0.91	0.92
Combustor Efficiency	0.985	0.995	0.995
NR. OF STAGES IN COMPRESSOR / TURBINE	2+6+7 / 1+3	1+3+9 / 1+4	1+3+10 / 2+5
DIAMETER m	1.25	1.55	1.75

One small thing is that some newer engines are actually heavier than the old ones. This is due to the fact that fuel consumption (and emissions) has become one of the most important driving design parameters, while the thrust requirements have pretty much stayed the same.

Property:	CFM56-5B	LEAP-1A26
Entry in service:	1996	2016
Take-off thrust [kN]:	≈120	≈120
Weight [kg]:	2381	3008

A new introduction is the GTF engine which has a few advantages over conventional turbofans.

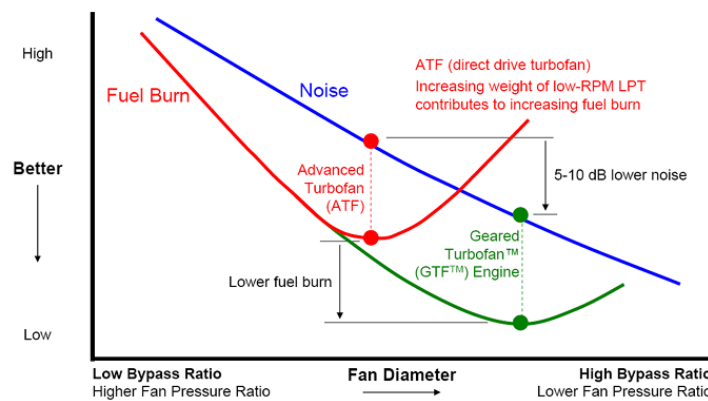


- De-couple the fan from low pressure spool by reduction gear system.
- High bypass ratio turbofan engine
- Low fan speed
- Higher low pressure spool speed

An ultra-high bypass ratio can be achieved this way, since the fan speed can be drastically reduced thanks the gearbox, thus reducing the fan noise and increasing the propulsive efficiency.

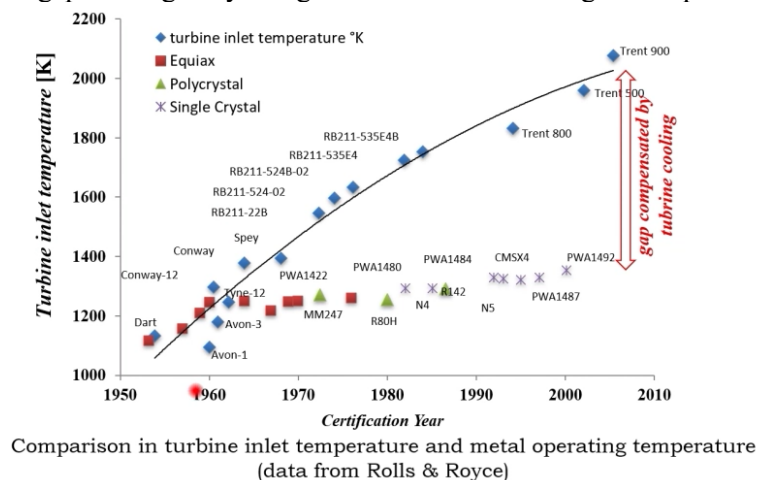
At the same time the LPT can rotate at much higher speeds, greatly increasing its efficiency while decreasing the number of stages needed and thus the weight.

This reduction of fuel burn can only be achieved with this GTF architecture, since for a normal turbofan an increase in BPR eventually leads to higher fuel burn due to the increased weight of the LPT which has to spin at low speeds (and this requires more stages) since it's coupled with the fan.

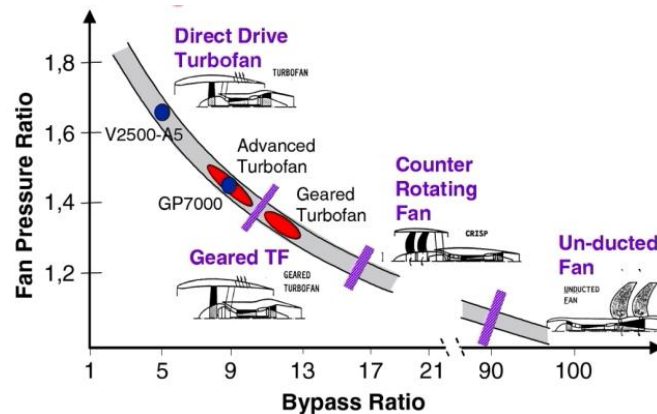


To increase the thermal efficiency we can increase the OPR. If we look at the trend over the years for the OPR this is exactly what we see, with modern engines having an OPR of around 50, almost double the OPR of engine from the '70s.

The TIT also shows a general trend to increase over the years with modern TIT reaching values of 2000 K. This is actually much higher than the melting point of the materials used for the turbine blades, however this gap is bridged by using advanced blade cooling techniques.

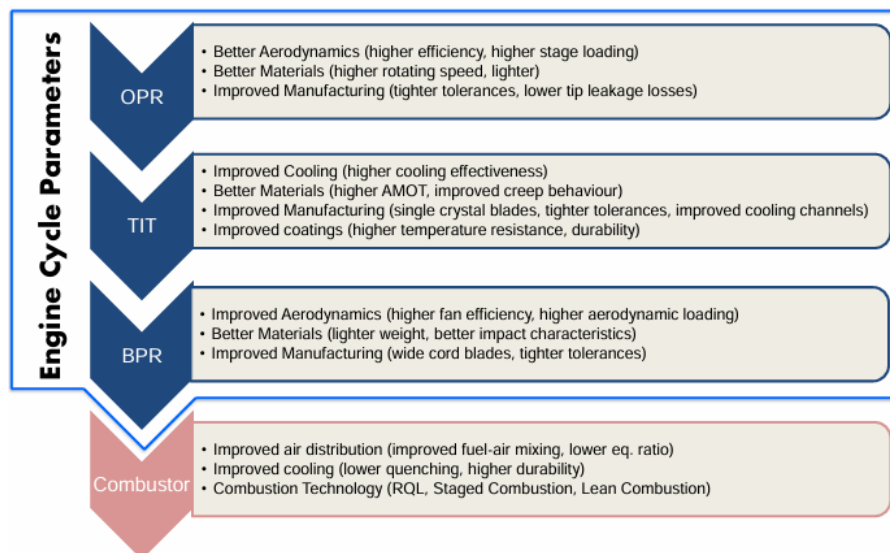


The BPR has had a much steeper increase over the last decades reaching values of 11-12, compared to BPRs of around 2-5 in the '70s and '80s. At the same time the FPR has been decreasing. Further increases in the BPR however require new architectures, since the limit for the conventional turbofan has been reached.



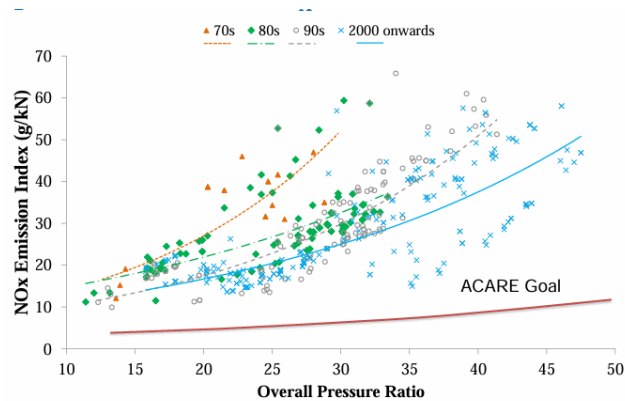
The single components efficiencies have also been increasing, with HPC efficiencies reaching values of almost 90% in the last couple of years, while also having higher OPR and lower number of stages. The design trends for the compressor have been the following:

- Higher speeds
- Higher spool pressure ratio
- Higher aerodynamic loading
- Higher pressure rise per stage
- Lower aspect ratio
- Higher solidity
- Improved configurations and blade shapes
- Lower weight



Future design trends

While older designs were mainly driven by increased thrust and lower fuel consumption, modern goals also include the reduction of NOx emissions, which tend to increase with OPR and TIT. We see that over the years the correlation between NOx emissions and OPR is decreasing thanks to better combustion technologies.



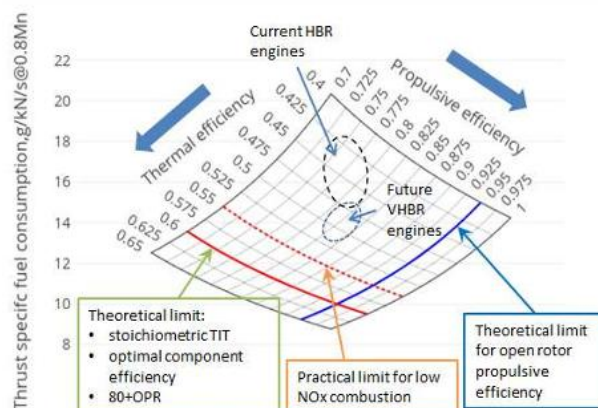
The ACARE goal is the goal to cut in half NOx emissions by 2050.

The main products which have environmental effects are the NOx and CO2 emissions, as well as **contrails** left behind by the engine.

As we increase the design parameters of the engine (TIT, BPR and OPR) the fuel consumption and therefore the CO2 emissions decrease, while contrails formation and NOx emissions increase.

Other limitations exist for the future engines. The fuel consumption is driven by thermal and propulsive efficiency. The former has a maximum theoretical limit due to limitations on the TIT (stoichiometric TIT, above which combustion becomes less effective), on the efficiency of single components and on the OPR due to structural reasons. However, this limit is never going to be reached if we want to keep low NOx emission, as that will become the limiting factor.

The latter has a theoretical limit that is imposed for open rotor engines.



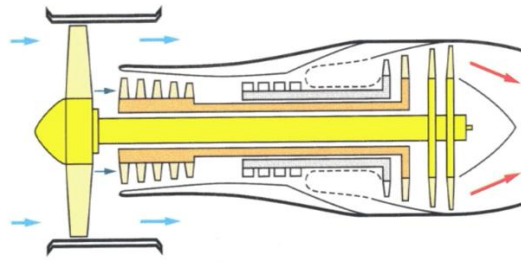
Overall however there is still room for improvement.

The main design drivers for future development are going to be emissions, efficiency, energy sources and noise.

Lecture 4

3 spool engine

As we can see from the image below the three spool turbofan engine has three different shafts. The fan is connected to the low pressure turbine since it wants to rotate more slowly, the booster (LPC) is connected the intermediate pressure turbine and the HPT is connected to the HPC.

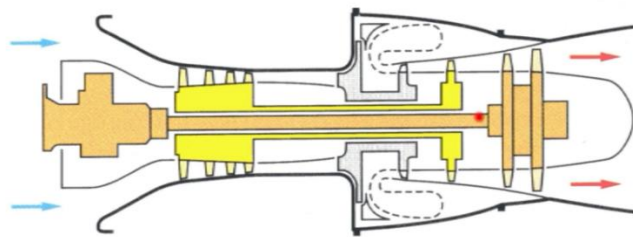


Three shaft by-pass turbo-jet

This architecture is not very common and is mostly used by Rolls-Royce engines. The main advantage of this architecture is that the LPC (and therefore also the intermediate pressure turbine) is decoupled from the fan. In the design phase the fan is the most constricted component since the tip speed can easily reach critical speeds, increasing the noise produced and introducing shock losses in the fan exhaust (decreasing the overall efficiency of the engine). Therefore its rotational speed is strictly limited, which also limits the LPC in a two spool turbofan (due to the low pressure increase it can provide it's typically called a "booster" stage).

With three spools, the LPC is free to rotate faster and allow for higher OPR.

Three spool architecture is also common for turboprops.



Twin spool free turbine turbo-engine, with front power drive

In this architecture the gearbox that drives the propeller is connected to a power turbine (free turbine), which is decoupled from the other turbines. The other two shafts connect the HPC to the HPT and the LPC to the LPT.

The main advantage of having three spools for a turboprop is that it's really easy to start the engine: the core is easier to start than the propeller, so by starting it, it can provide power to the power turbine, which drives the propeller.

The other general advantage is the same that makes two spool engine more popular than one spool engines nowadays. The LPC and HPC have different optimal rotating speeds, so by decoupling them we can obtain a higher overall efficiency.

The main disadvantage is the higher system complexity (thinner shafts, vibrations,).

In summary the advantages and disadvantages of a three-spool turbofan engines are the following:

Advantages

- The Fan can rotate slower as compared to 2-spool turbofan
- The LPC can rotate at a higher rpm
- The pressure ratio of LPC increases
- The distribution of pressure ratio between LPC and HPC is improved
- The Overall Pressure Ratio of the engine can be slightly increased (when compared to 2 spool turbofan)

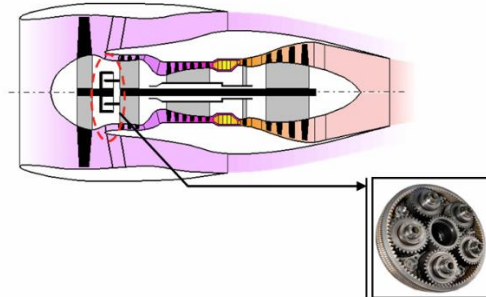
Disadvantages

- Increase in the mechanical complexity of the engine
- There is an additional shaft and bearings, which might increase the engine weight
- The rotodynamics of a 3 spool engine is more complicated
- The lubrication system becomes complicated as compared to 2spool turbofan engine

Gear turbofan engine

As we have seen before the GTF is a turbofan which has two main spools that connect the LPC and the LPT, as well as the HPT and the HPC. However, the GTF is different than conventional twin spool turbofans, because the fan and the LPC are decoupled by a gearbox. This solves the same issue that is solved by the three spool turbofan with a lower system complexity (although mechanical losses are introduced in the gearbox). This is why the GTF is called a 2.5 spool engine.

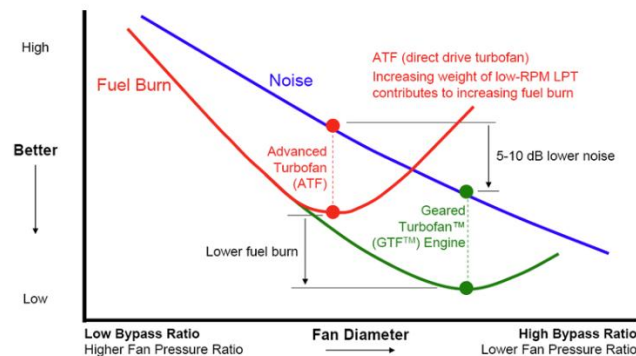
The reduction ratio of the gearbox is around 2-2.5, which means that the fan is rotating at around half the speed of the LPC. This allows for higher OPR while avoiding tip shock at the fan.



GTF also has less LPT stages (around 3) compared to the 5-6 stages that normal twin spool turbofans have. This is possible thanks to the fact that the LPT can now rotate much faster, extracting the same amount of power from fewer turbine stages. The number of LPC is also generally lower since the increased rotational speed allows for higher pressure ratios for each stage.

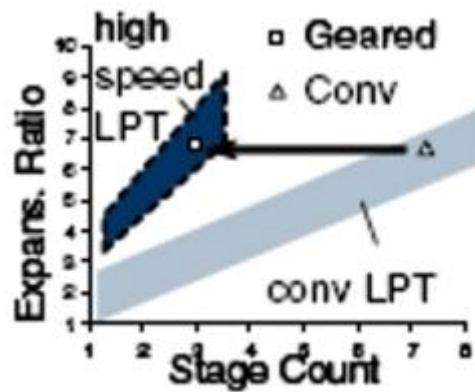
Revisiting the graph from the previous lecture, for a twin spool turbofan, after a certain BPR (around 10-11) the fuel consumption actually starts to increase. The positive effect of the increased BPR is negated by the decrease in efficiency of the fan and the LPC.

This limit can be overcome by introducing a gearbox between the fan and the LPC. This allows to increase the BPR even more, while maintaining the gains on the fuel consumption.



Let's now summarize the advantages of decoupling the fan and the LPC:

- Low speed of the fan tips: leads to great noise reduction (as a rule of thumb the noise produced in the inlet is about proportional to the 4th power of the tip speed, while the noise produced by the nozzle is proportional to the 8th power of the nozzle speed. By increasing the BPR we can choose which contribution to reduce)
- Optimum speed for the fan and the LPC: this increases the efficiency and lower the fuel consumption
- High speed LPC and LPT: this allows the reduction in LPT stages (to about half) while increasing the pressure ratio of the LPC at the same time. From the figure we can see that for the same expansion ratio the number of stages required is drastically less for a geared architecture compared to a conventional architecture.



These benefits come at the price of increased weight and system complexity. Also the larger BPR and engine diameter might lead to an increase in installation penalty (aerodynamic losses caused by the interaction between the nacelle-pylon-wing could increase. This is caused by the larger engine, which can lead to a shorter pylon to satisfy ground clearance requirements).

Due to the higher LPT rotation the turbine blades have higher requirements, since the centripetal forces acting on them are increased.

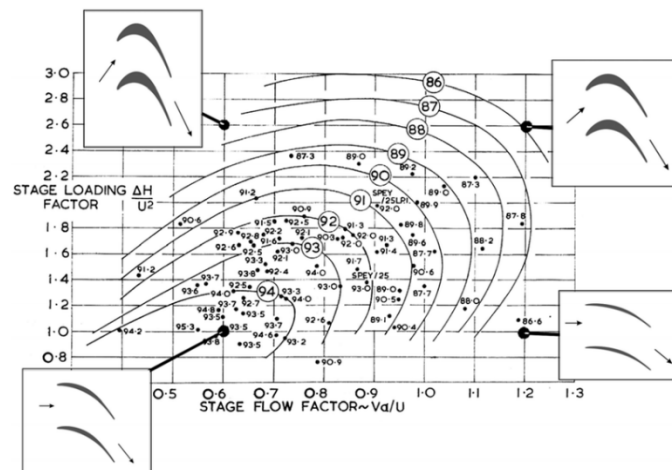
Finally the heat rejection of the engine becomes complicated (increased cooling system complexity and requirements).

A big challenge in designing the gearbox is maximizing the mechanical efficiency. This is not only too reduce power losses but also to reduce the cooling requirements.

Note: GTF engine have the lowest specific fuel consumption between all the different architectures.

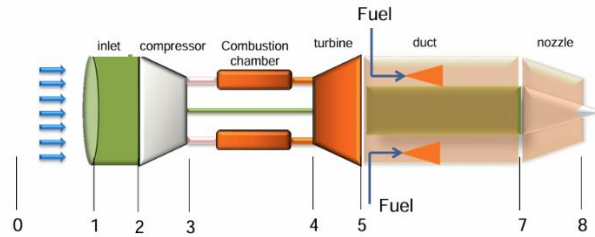
A three spool engine is slightly less efficiency since the turbine coupled with the fan still has to rotate at a slower velocity, which is not the case for a GTF.

From a smith chart we can see that a reduction in rotational velocity increases the stage loading (total enthalpy jump divided by the square of the velocity) and flow coefficient (proportional to the flow velocity), thus reducing the efficiency. Typically we want high velocity to reduce the stage loading and flow coefficient. See later lectures on turbomachinery.

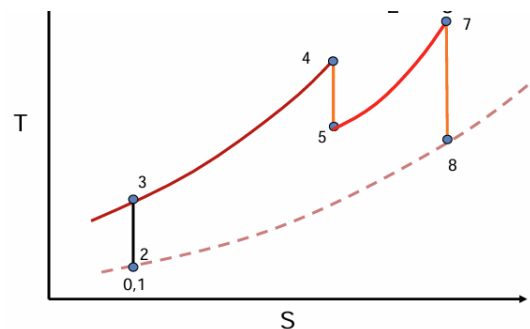


Afterburner

The afterburner is a long secondary combustion chamber, which is added to military turbojets/turbofan engines to reheat the exhaust gasses that come out of the turbine to generate more thrust in the subsequent nozzle (typically a convergent-divergent nozzle with a variable geometry).



Since we are reheating the gasses, the pressure slightly drops due to pressure losses, while the temperature increases rapidly to a temperature that is higher than the one reached in the primary combustion chamber. This is possible because the TIT is limited by the heat resistance of the turbine blades, however there are no turbomachinery components behind the afterburner.



From the T-S diagram of the ideal cycle with afterburner we can quickly see that the power generated (and thus thrust, which is typically increased by 40-50%) is much higher since the expansion process from 7 to 8 is much longer than from 4 to 5. This comes at the cost of an enormous reduction to the efficiency of the engine and increased length.

If we consider afterburning turbofans, we see that they typically have a lower BPR than conventional turbofans. The bypass flow is mixed with the exhaust gasses coming from the turbine before the reheating process occurs.

To summarize the features of an afterburning engine are:

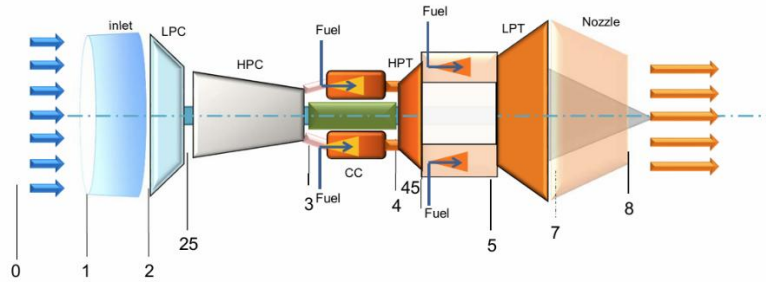
- The specific thrust can be increased instantaneously
- The thrust to weight ratio is very high
- The off design performance can be significantly increased without penalizing the design point performance (e.g. short take off, evading manoeuvres)
- The diameter of the engine is not increased

These come at some disadvantages:

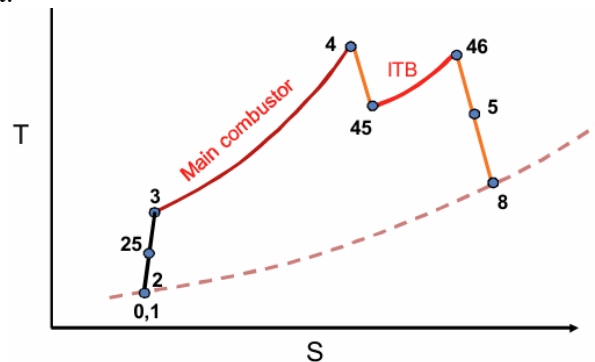
- The specific fuel consumption is increased significantly (up to 300%) when using after burner.
- The length of the engine is more than a conventional engine
- The weight of the engine is more than a conventional engine
- The infrared signature is increased immensely when using afterburner
- There is a need for variable area nozzle with using afterburner

Inter-turbine burner engine

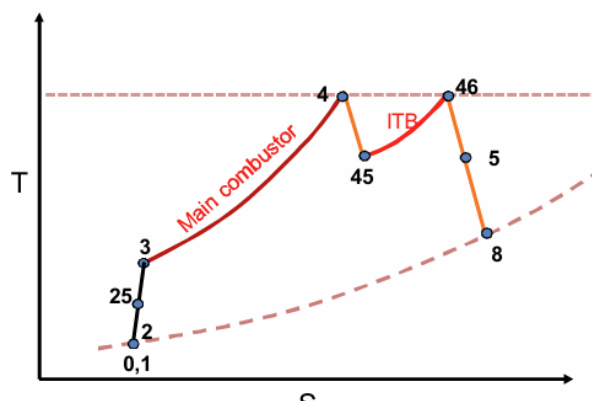
The idea at the basis of the inter-turbine burner engine is that we can insert the secondary combustion chamber ahead of the LPT instead of after it.



Just like the afterburner the inter-turbine burner is burning fuel at an higher initial temperature (when compared to the main combustor), however in virtue of the fact that it is doing it at a higher pressure w.r.t. the afterburner, the thermodynamic efficiency of the process is much higher (although it is lower than that of the main combustor). This occurs because increasing the temperature of a gas at a higher pressure is more thermodynamically efficient than doing it at a lower pressure (at higher pressure a bigger part of the added energy through heat can be transformed into useful kinetic energy and the entropy generation is less). This can also be seen directly from the T-S diagram. In conclusion this engine can produce the higher thrust than a conventional turbojet at a lower fuel “cost” than a conventional afterburner, however the process is still less efficient than a normal turbojet which produces the same thrust.

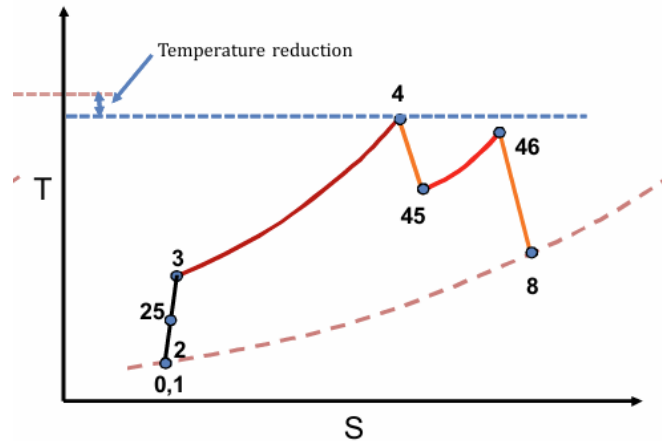


Thanks to the presence of an additional burner, the cycle can be optimized in many ways. If the goal is to maximize the specific thrust produced, then the thermodynamic cycle would probably look like the following:



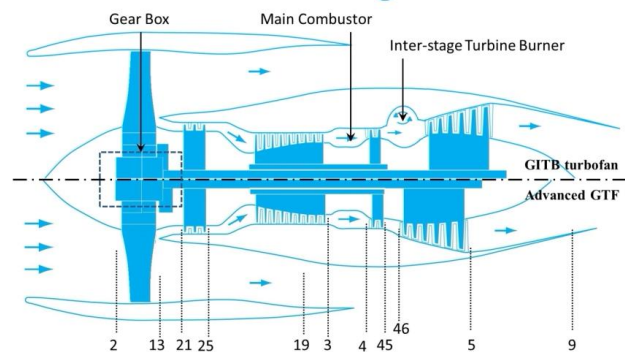
The amount of fuel added to the ITB has to so that the end temperature is the same as the TIT. By doing so the specific power of the system is maximized. This is because the temperature at the exit of the ITB is still limited by the LPT blades thermal resistance. By going as close as possible to this limit we maximize the power produced in the nozzle.

However, if we want to minimize the NO_x emissions, then the cycle would look like:



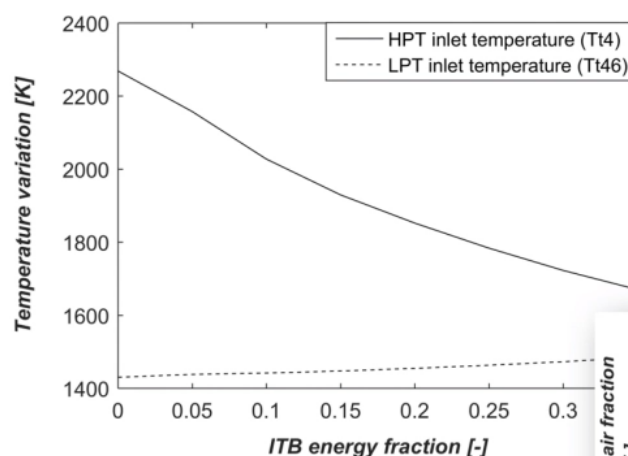
To reduce the NOx emissions we want to lower the TIT (NOx production increases almost exponentially with the combustion temperature after a certain threshold). To compensate for the lower TIT, we rise the temperature in the ITB, allowing for similar thrust generation (although at a higher TSFC).

Another effect which leads to NOx emissions reduction is the dissociation of NOx produced by the first combustor in the second combustor.

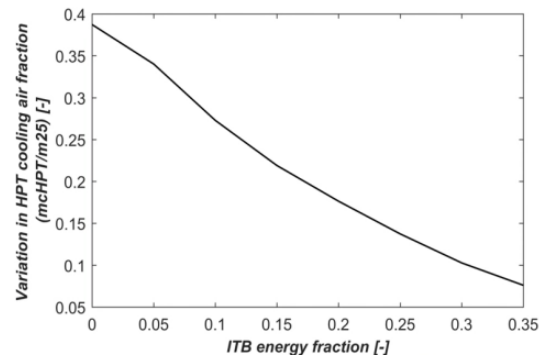


The volume of the ITB has to be higher than the volume of the first combustion chamber because the gasses are at a lower pressure, so to obtain the similar residence time (amount of time the gas stays in the combustion chamber to absorb heat) we need a larger volume.

Another advantage of the ITB is the possibility of lowering the TIT for the same thrust generation.



This allows for a lower amount of cooling air, which is typically needed since the HPT blades are operating at higher temperature than their melting point. This air is taken from the compressor, effectively reducing the overall efficiency of the engine. By reducing the amount of air taken from the compressor, the overall efficiency is further increased.



Note: the ITB energy fraction refers to how much of the fuel is allocated to the ITB instead of the main combustion chamber.

Obviously the disadvantages of this architecture is the increased size, weight and system complexity (it is very hard to design the ITB so that complete combustion is achieved).

To recap:

Features:

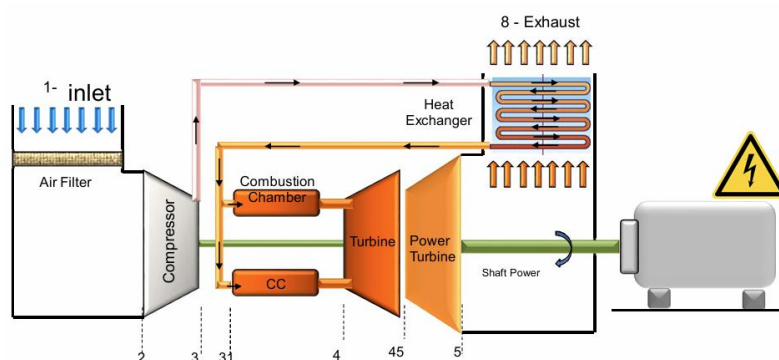
- The specific thrust generated by engine with ITB is higher
- The specific fuel consumption is lower for an engine with ITB than engine with afterburner
- The off design performance is better than a conventional turbofan engine
- The NOx emission can be significantly reduced

Challenges:

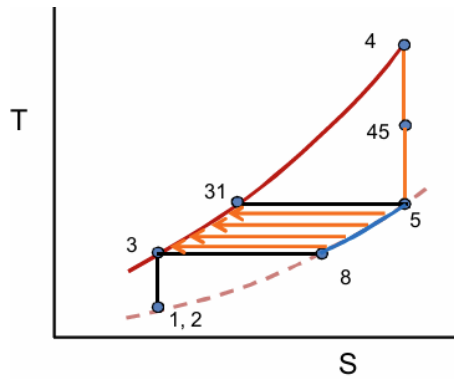
- The ITB design is quite challenging. Achieving complete combustion is difficult.
- The length of the engine is more than a conventional engine
- The weight of the engine is more than a conventional engine

Recuperated gas turbine engine

Let's first look at an industrial example of a gas turbine.

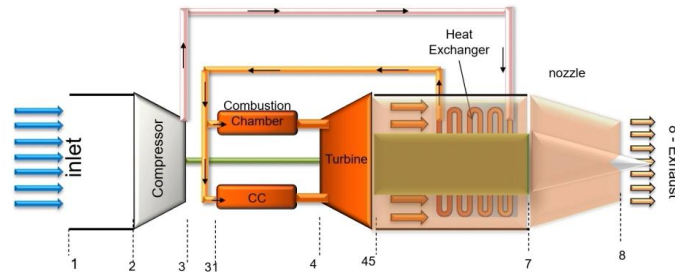


To increase the thermodynamic efficiency we can take the exhaust gasses, which are typically at a higher temperature than the gasses passing from the compressor to the combustion chamber, to heat the gas before it enters the combustor. This reduces the required heat that the combustion chamber has to provide, thus increasing the efficiency. This process is called **recuperation**.

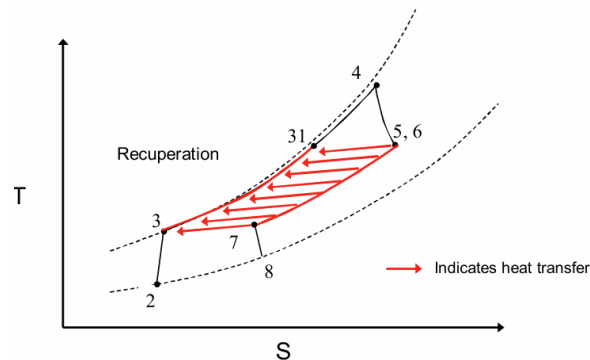


The recuperation is most effective when the pressure ratio is low, since more of the heat can be extracted from the exhaust gasses.

In an industrial power plant recuperation occurs after the expansion, however in an aero engine the recuperation can only occur before the expansion in the nozzle. This is not as efficient as the recuperation in the industrial gas turbine since the process of recuperation introduces temperature and pressure losses which affect the flow in the nozzle.



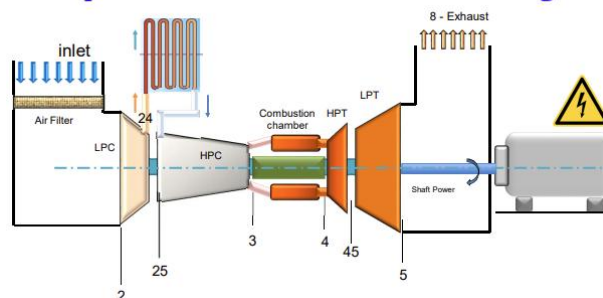
Recuperation has never been implemented in an aero engine also because it disturbs the air before it comes into the nozzle, reducing thrust generated. Also the weight and size required to implement such a concept is not compatible with civil aviation standards.



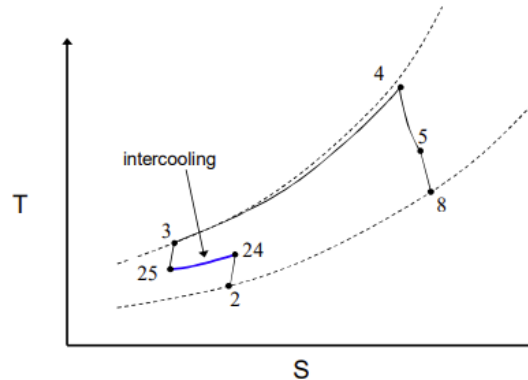
Lecture 5

Intercooling and recuperation

2 Spool Gas Turbine with Intercooling



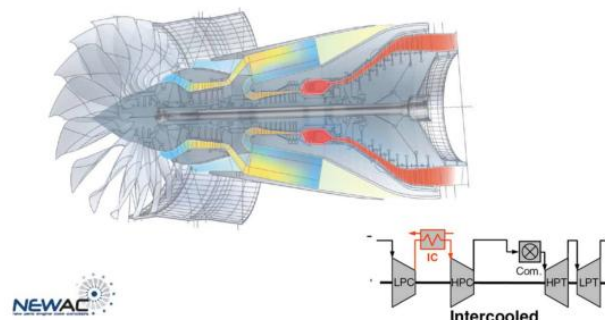
In order to reduce the amount of work required by the HPC, **intercooling** can be implemented. The air after the LPC is taken to an intercooler, where it is cooled through a heat exchanger filled with either water or surrounding air. Once the temperature of the air is reduced, it is then fed to the HPC. This significantly reduces the work required for the same pressure jump, since it is “cheaper” (it cost less power) to compress a fluid a lower temperature. This greatly increases the efficiency of the cycle.



A downside to intercooling can be directly seen from the T-S diagram. Since the temperature of the fluid is now colder when it enters the combustion chamber more heat is going to have to be added into the system. This does decrease the specific net power.

In the following example the core air after the LPC is intercooled by using the bypass air.

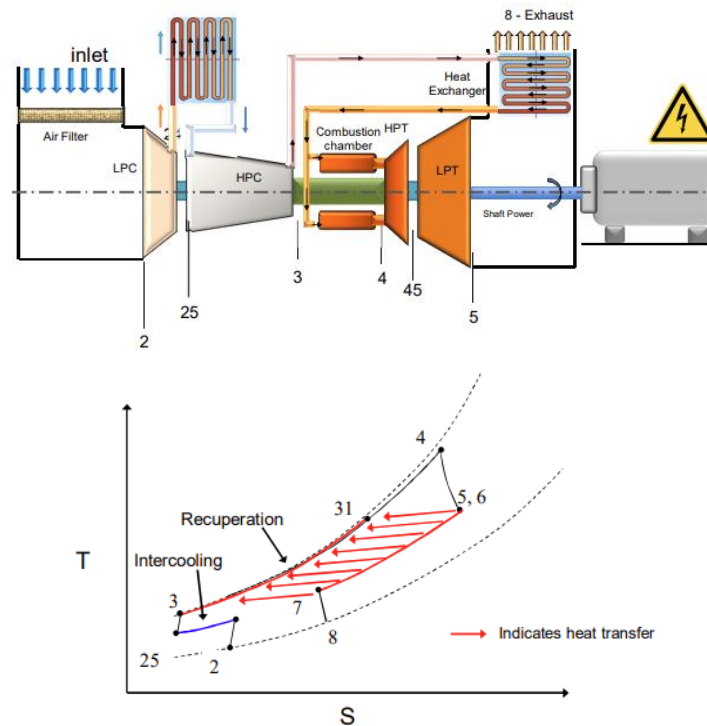
Intercooled Turbofan



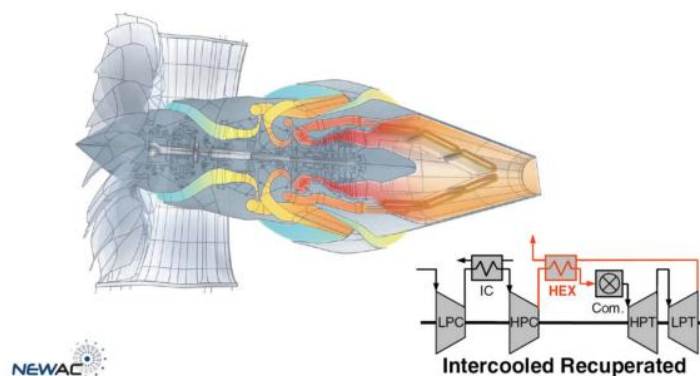
Since as we have previously seen the amount of NO_x produced in the combustion chamber is proportional to the temperature at which the combustion process occurs, by lowering the temperature the emissions of NO_x can also be reduced.

Intercooling however it's not used in aero engine due to the increased engine weight and volume. Moreover, the pressure losses that occur in the heat exchangers are typically very high and destroy almost all gains that come from the lower power required by the compressor. Also the decrease of the combustor inlet temperature decreases the specific thrust generated by the engine (although it does also decrease the TSFC).

Intercooling can actually be combined with recuperation, since the second process is actually limited by the temperature difference between the exit air at the HPC and the exhaust coming out of the LPT. The intercooling reduces the first temperature, increasing the temperature difference and making recuperation possible and more effective.



In a way recuperation also negates the disadvantage of intercooling (lower combustor inlet temperature) since it reduces the amount of heat that needs to be transferred to the gas. If we applied these concepts to a turbofan:



From conceptual designs we know that the expected temperature rise of air after the HPC in the recuperator would be in the order of 200 K, while the temperature drop in the intercooler would be around 70 K. Also the heat exchanged with the bypass air adds energy to it and gets accelerated. This third flow of cooling air could also be discharged by a third nozzle increasing overall thrust.

In the heat exchangers there are significant pressure losses, which can actually be offset by the temperature rise in the recuperator. However, pressure losses are a major limiting factor and in some cases can negate all the benefits. Also system weight is increased, while reliability is reduced.

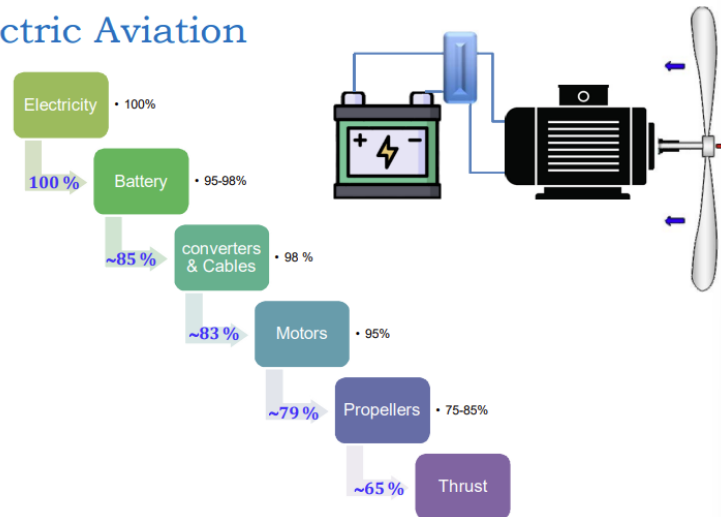
Since recuperation is typically more efficient than intercooling (the temperature jumps are different) the temperature at which combustion occurs can actually become higher and therefore the NO_x emissions could also increase.

Hybrid electric propulsion

As we have seen previously the main components of a propulsion system are the energy source, the energy to work converter and the thrust producer.

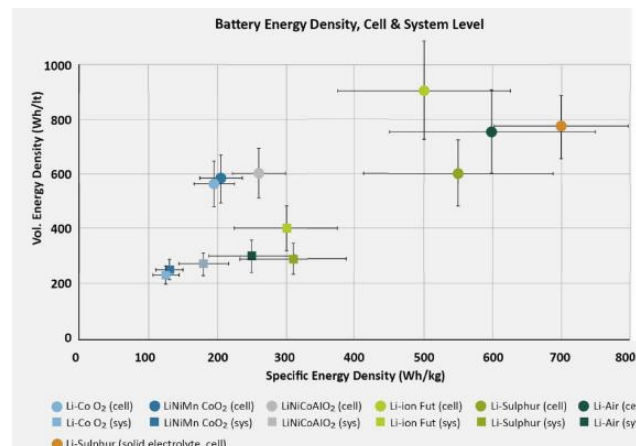
Electric aviation is one of the many options for alternative propulsion systems and use batteries as the energy source, electric motors as energy to work convertors and typically propellers as thrust generators. Let's look at how efficient a system of this kind is:

Electric Aviation



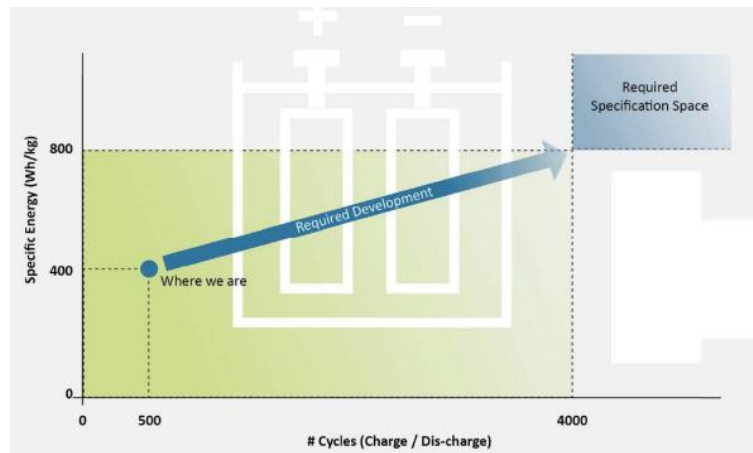
Every step of the process introduces small losses (which lead to a global efficiency of around 60%), which overall are better than the losses that occur inside a conventional aero engine.

One of the main constraints when it comes to electric energy propulsion is the energy density of batteries.

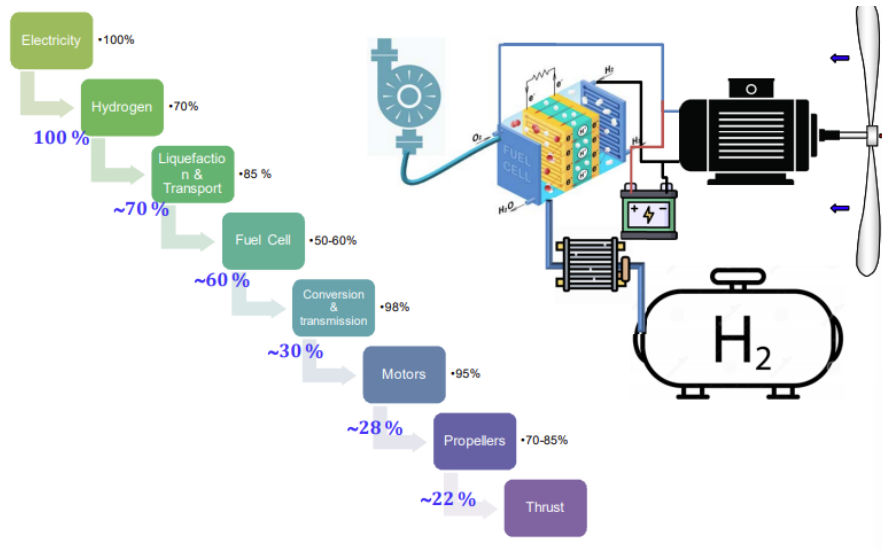


At the cell level the specific energy density and the volumetric energy density are not too bad actually, however, at the system level (along with cooling systems and other systems needed) the energy density drops. Nevertheless, even at the cell level, the volumetric energy density of kerosene is still around 40 times that of the best batteries.

Another major limitation is the battery life of current batteries. If we were to use completely a battery each recharge cycle (meaning we use their full capacity) we see that the number of cycles for which they can be used is greatly reduced and not satisfactory.



While for a car such battery life can be acceptable, since they do not need to be charged that often, for a plane, which needs to be charged on average 5-6 times a day this is unacceptable. To offset these drawbacks, the implementation of electric energy can also be done together with another form of energy. Fuel cell electric systems use hydrogen, which is passed through a condenser and is fed into a fuel cell, together with air coming from a compressor. Inside the fuel cell chemical energy is transformed into electrical energy which then powers an electric motor. The efficiency line looks like:



By adding this one step to the thrust generation process the total efficiency drops to around ~20%. However, this does partially solve the two issues of pure electric power systems.

In summary, electric propulsion systems have the following advantages:

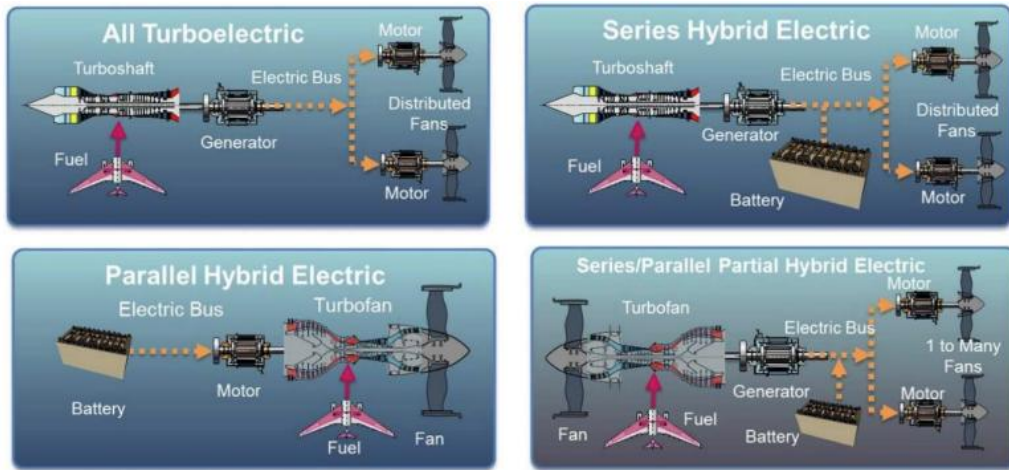
- Higher efficiency, especially for systems which use batteries
- Low emissions
- Lower noise (only regarding the component that generates the useful work, the propeller still creates a large amount of noise)
- Architectural flexibility (can be used for many different aircraft architectures)
- Reduction in the control surface area
- One engine out condition is not a constraint anymore
- Possibility of thrust vectoring
- Suited for urban mobility

However, there are a few disadvantages:

- The energy density of electrical components is not great
- The propulsion system weight is high
- The integration of propulsion systems with the aircraft can become complex

- The reliability of such systems is not known yet
- Electromagnetic interference is an open issue
- Lightning strikes and other issues need to be investigated

Yet another option is to integrate electrical power systems in conventional aero engines. Hybrid electric architecture can be classified as follows:



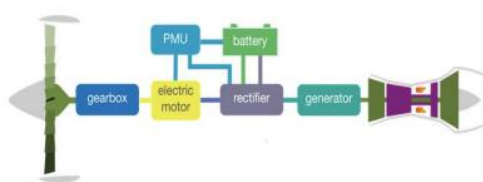
In the **all turboelectric engine**, the gas turbine is not producing any thrust, but only electric energy power which is then used to drive propellers. This is basically a turboprop.

A battery can be added in series to the turboshaft to aid in power generation to drive the propellers. Depending on the flight phase the power generated by the turboshaft can be used to recharge the batteries. The batteries in this case act as buffers to smoothen out transition phases and can help to reduce the total emissions. This architecture is called the **series hybrid electric**.

If the batteries are added in parallel then typically we have a turbofan instead of a turboshaft. In this case the battery is driving an electrical motor, which is assisting a gas turbine (it can boost the efficiency in many ways by providing energy to all components). This architecture is called the **parallel hybrid electric**.

Finally, a combination of the two previous system can be obtained. Batteries are assisting a turbofan, which is producing both thrust and electrical energy which is used to drive propellers or recharging the batteries, depending on the flight phase. This architecture is called the **series/parallel hybrid electric**.

Series HEPS



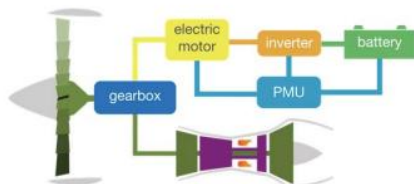
Advantages:

- Engine operates at optimal RPM
- Higher effective BPR
- Design freedom in positioning of engine and fan

Disadvantages:

- Heavy
- Less efficient

Parallel HEPS



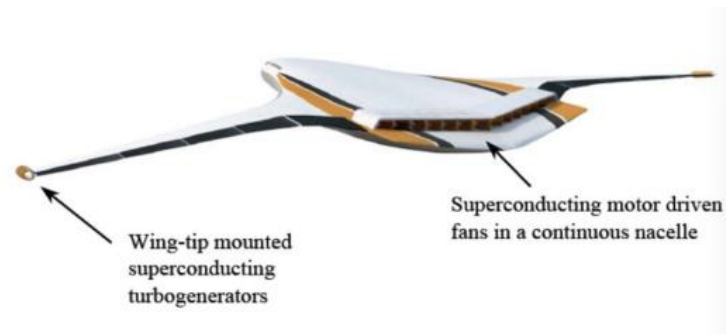
Advantages:

- Independent operation between engine and electrical system
- Independent design of power share between both subsystems
- Less radical than a series system
- Higher power to weight ratio

Note: for the HEPS the fan is decoupled from the compressors, therefore the overall efficiency of the engine is greatly increased.

In the series architecture the fan and the core are completely separated.

An example of HEPS is the NASA N3-X, a concept aircraft with a blended wing-body, distributed propulsion and hybrid electric power systems.



The gas turbine engines are at the tips of the wings and they only produce power, which is then used to run smaller electric engine that are distributed over the body.

This distributed propulsion systems have quite a few advantages, such as:

- Higher effective bypass ratio
- Lower noise
- Better suited for boundary layer ingestion
- Architectural flexibility
- Reduction in the control surface area
- One engine out condition is not a constraint anymore
- Possibility of thrust vectoring

As we can see the advantages match up with the advantages of HEPS. The disadvantages are:

- More complex system architecture
- The integration with the aircraft becomes more complex
- The reliability of such systems is not known
- The energy density of electrical components is not good
- Electromagnetic interference is an open issue
- Lightning strikes and other issues need to be investigated

An example of parallel HEPS engines are electrically assisted turbofans, which feature an electric motor that drives the fan. This motor brings great advantages during take-off, climb, descent and taxiing. Thus for short range aircraft the efficiency is increased by quite a bit, meanwhile for longer ranges no particular benefit are obtained due to the increase weight of the system and low energy density of batteries.

Open rotor propulsion

An open rotor engine, also known as a **propfan** is an aero engine which combines features of turbofans and turboprops to obtain the thrust capabilities of a turbofan in lower transonic regimes ($M \sim 0.7$), while maintaining turboprop efficiency.

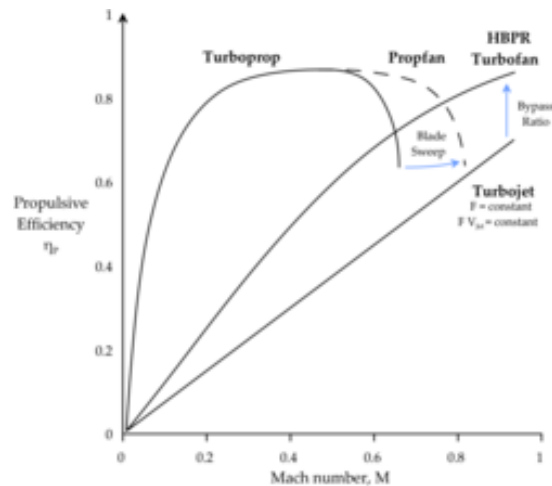


Figure 3. Propulsive efficiency for different engine types.

Although an interesting concept propfan engines never caught on mainly due to the great cabin noise they produce when compared to normal turbofans.

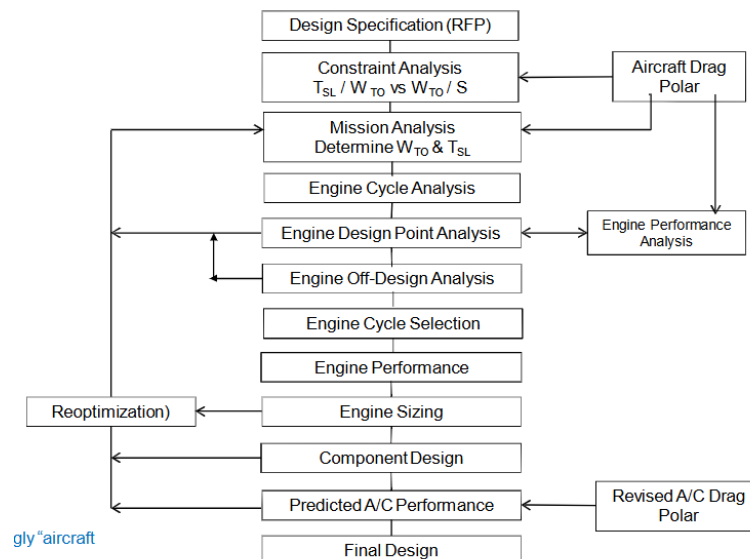
The main advantages of such an aero engine architecture are:

- Large savings in fuel burn due to the large BPR (80-90)
- Very low FPR, no duct and low downstream swirl when compared to conventional turboprops
- Reduced nacelle drag

These come at a cost:

- Noise emission is greatly increased when compared to a normal turbofan
- New engine installation needed with unprecedented issues
- Balde containment issues
- Power transmission to rotors (introduced mechanical losses)

Note: engine design cycle

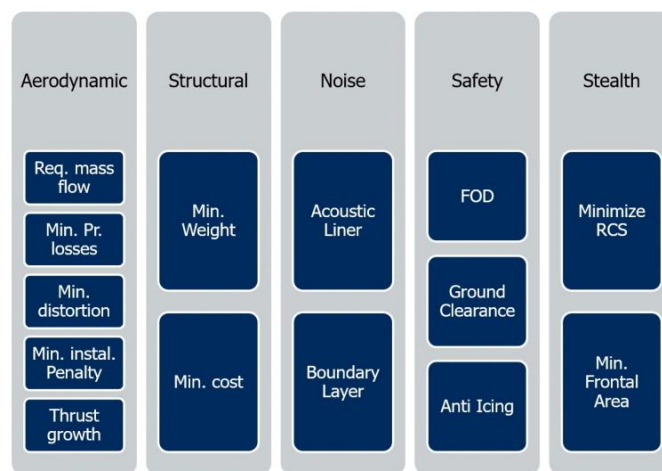


Lecture 6

Engine intake

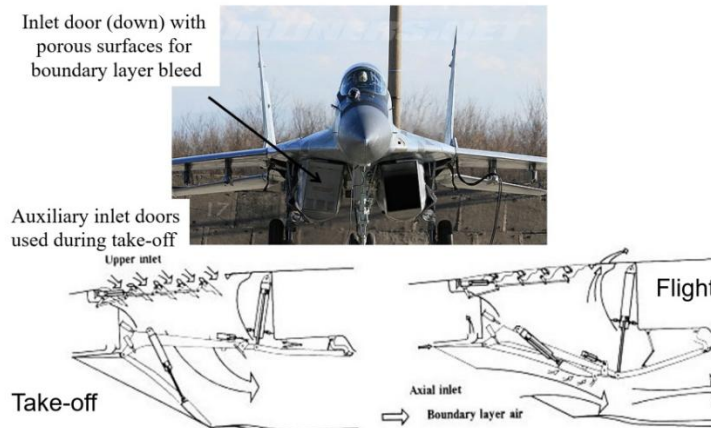
The design criteria of the inlets can be classified into 5 main types:

- **Aerodynamics:** the main requirement of an inlet is that it has to provide the adequate mass flow rate to the engine with minimal pressure loss and minimum distortion. The **installation penalty** (interaction between nacelle, pylon and wing) also has to be minimized. Finally, the thrust growth is also an important factor. When designing a family of aircrafts (e.g. A320) the thrust growth has to be taken into account, since future aircraft will require more powerful engines and ideally this wouldn't require a whole new design for the intake.
- **Structural:** the objective is to minimize the weight and the cost.
- **Noise:** the objective is to reduce fan noise. The important elements are the acoustic liner inside the intake and the management of the boundary layer.
- **Safety:** foreign object damage is particularly important for military aircrafts. Ground clearance and anti icing systems drive the design of the intake. An example of the importance of ground clearance is the intake of the Boeing 737 NG, which has an oval shape to maximize the mass flow rate coming into the engine, while maintaining a certain clearance.
- **Stealth** (only important for military aircrafts): we want to minimize the radar cross section, the infrared emissions and the frontal area.



Note: **foreign object damage** FOD refers to object present on the runway that get accidentally ingested by the engine and can greatly damage the engine. The intake has to be designed to minimize the risk of foreign object ingestion. An extreme example of an inlet designed to minimize the risk of FOD, are the top inlets of the Mig 29, which can provide enough mass flow rate for the aircraft to take off with the main inlets closed. This feature can also be used during high speed manoeuvring to increase the stall margin of the compressor.





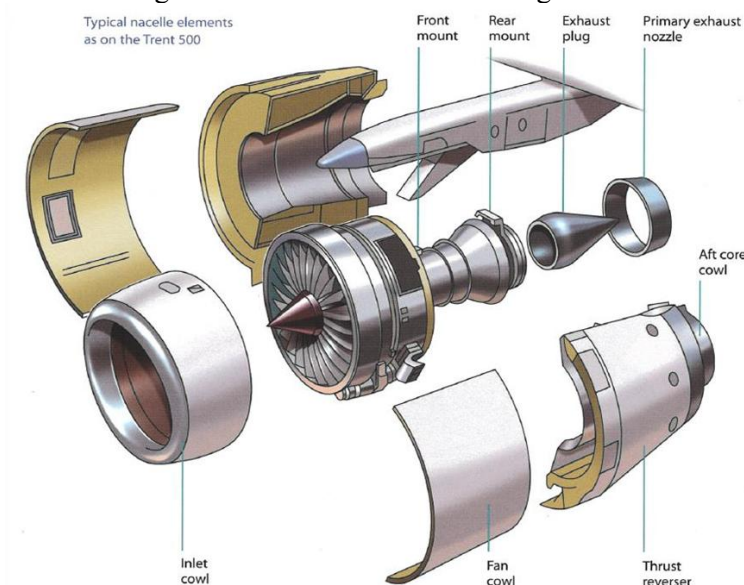
Some important considerations for inlets:

- The intake has to always provide the required mass flow rate in every condition. Therefore the main flight conditions which drive the inlet design are the take-off condition and the cruise condition.
- During cruise one important factor to take into consideration is the **ram effect**, or dynamic compression of the fluid. Since the inlet slows down the entering air to around $M=0.6$ at the fan face, some kinetic energy is transformed into potential energy. This translates into an increase in static pressure (while total pressure decreases slightly due to losses).
- Intake can be distinguished into subsonic and supersonic intakes. For the former, boundary layer behaviour is of the utmost importance, while for the latter the shock structure through which the flow is slowed down to subsonic conditions is much more important. Another important difference is that for a subsonic flow a divergent duct decelerates it, while for a supersonic flow a convergent duct decelerates it.

$$\frac{dA}{A} = (M^2 - 1) \frac{dU}{U}$$

Another example of inlet design mainly driven by one discipline is the inlets of the F117 nighthawk, which are mounted over the wing and are protected by a mesh. All of these measures are taken to improve the stealth of the aircraft.

The classic schematic of an engine nacelle can be seen in the figure below.



We see that the **thrust reverser** system is integrated with the engine nacelle. These systems (which can be of bucket, cascade or plume type) reverse part of (or all of) the flow to create reverse thrust and aid the braking on icy/wet runways. These systems cannot however be used to certify landing distance on dry runways due to their low reliability.

Nevertheless, they are used to reduce break wear. Most of the modern thrust reversers on civil a/c are of the cascade type and reverse only the fan flow.

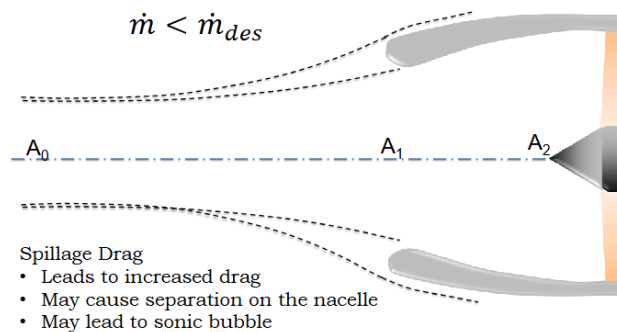
Another important part of the nacelle is the **inlet cowl**. This component is lined on the inside with a material that absorbs acoustic waves to reduce the noise produced called **acoustic liner**. Also inside the inlet cowl is a heating system, which is part of the pneumatic system, and prevents the formation of ice in the intake, which could get ingested by the engine.

The geometry of the cowl can be defined by total diameter of the nacelle, the diameter of the highlight (which is the “leading edge” of the airfoil that makes up the cowl), the ramp angle and the diameter of the throat. The last two parameters define the **mass flow rate ratio** of the intake MFR:

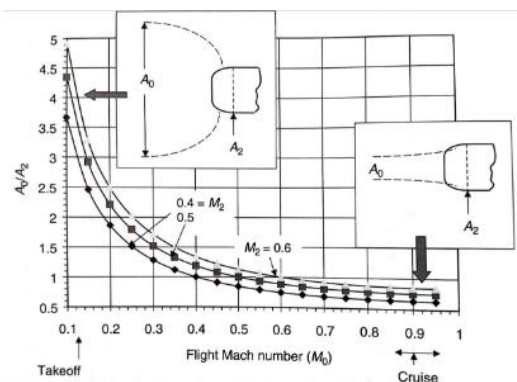
$$MFR = \frac{A_\infty}{A_h}$$

Where A_∞ is the cross section of the stream tube that gets ingested by the intake. In the two conditions that drive the design of the inlet we have:

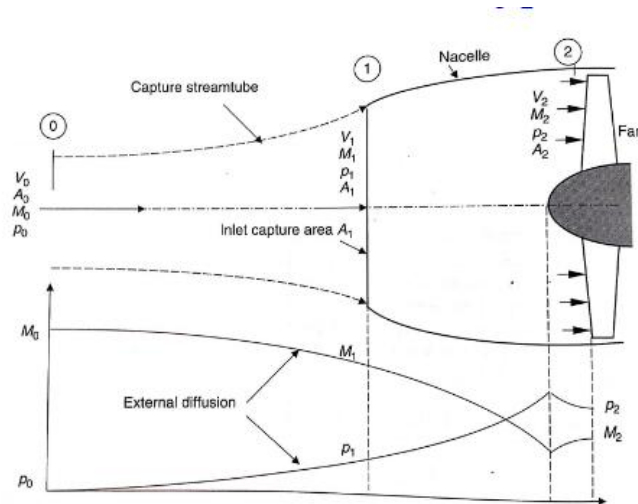
- **Cruise:** low MFR, which means that the stream tube expands over the cowl, leading to an increased risk of shock outside the cowl. The design for this phase focuses mainly on the external geometry. For mass flow rates different than the design value (lower values for example), the stream tube ingested might be slightly larger, leading to spillage drag and possible separation on the nacelle.



- **Take-off:** high MFR, which means that the stream tube converges into the intake and separation can occur on the inside of the cowl. The design of this phase focuses mainly on the internal geometry.



The pressure and Mach number variations in the inlet are as follows:



The inlet geometry controls the balance between:

- **External diffusion** (ahead of the nacelle): at cruise, part of the diffusion occurs outside the nacelle and is nearly adiabatic, reducing internal losses (there is no wall friction).
- **Internal diffusion** (inside the inlet): occurs in virtue of the converging geometry of the inlet (in the subsonic case). This is where the pressure losses are introduced.

This diffusion can be seen in the Mach number and static pressure trends depicted in the figure above. The flow is slowed down and the pressure increases.

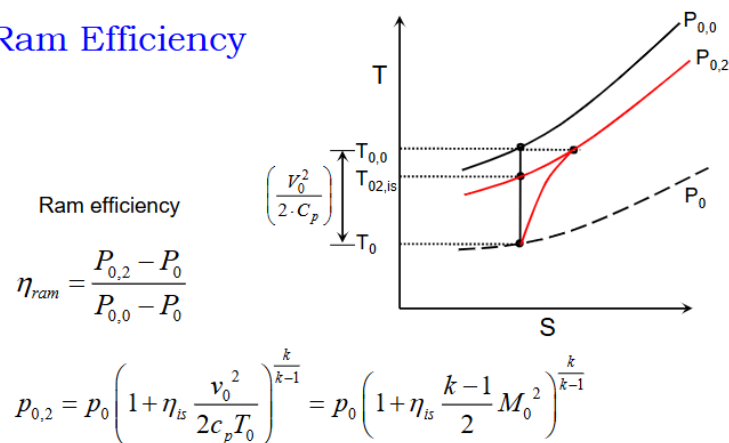
To take into account losses that occur during ram compression a **ram efficiency** (or **ram recovery factor**) can be defined, while an **isentropic efficiency** can be defined to take into account losses in mechanical energy w.r.t. an ideal compression (i.e. isentropic).

The first is defined as the ratio between the difference between the total and static pressure at the fan face and the difference between the total and static pressure at the intake entrance. The second one is defined as the ratio between the real enthalpy jump that occurs across the compression and the ideal enthalpy jump that would occur across an isentropic compression.

Up until $M = 0.8$ the two are pretty much equivalent, however the introduction of shocks at higher Mach numbers changes things and the ram recovery factor is preferred (ratio between the total pressure after the inlet and before). Empirical relations can be used to evaluate the ram recovery factor (typical values of 0.6-0.7) in supersonic conditions:

$$P_{02}/P_{00} = 1.0 - 0.075 (M-1)^{1.35}$$

Ram Efficiency



Ram efficiency

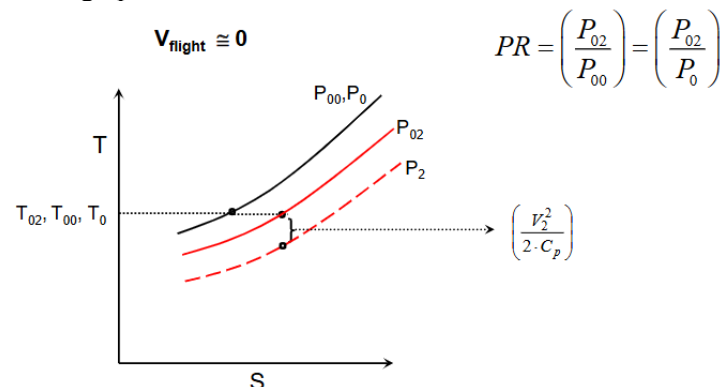
$$\eta_{ram} = \frac{P_{0,2} - P_0}{P_{0,0} - P_0}$$

$$p_{0,2} = p_0 \left(1 + \eta_{is} \frac{v_0^2}{2c_p T_0} \right)^{\frac{k}{k-1}} = p_0 \left(1 + \eta_{is} \frac{k-1}{2} M_0^2 \right)^{\frac{k}{k-1}}$$

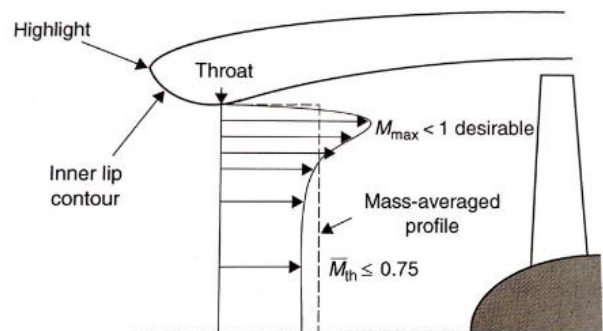
At take-off since the flight speed can be approximated to zero, the total pressure is equal to the ambient static pressure and there is almost no ram effect. In this case the inlet pressure ratio is basically 1 and the intake acts like a **Bellmouth inlet**.



A Bellmouth inlet is a rounded inlet with a trumpet-shaped entrance that has a gradual increase in cross-section from free stream to duct. This special geometry guides streamlines smoothly into the duct and prevents flow separation at the inlet lip, thus reducing inlet losses. In ideal conditions it has almost no pressure losses and that is why it is used in wind tunnel tests to achieve uniform flow at the fan face, maximizing mass flow rate. Nevertheless, they are not actually used since they have very high drag and poor off-design performance.



The curvature of the inside of the inlet throat causes a non uniform velocity profile. This is why we typically use an average value for the Mach number in the inlet and at the fan face. The higher the curvature the stronger the velocity gradients and the more the flow will be non uniform at the fan face. The shape of the lip has to be a compromise between resistance to separation at low speeds and wave drag control at high speeds.



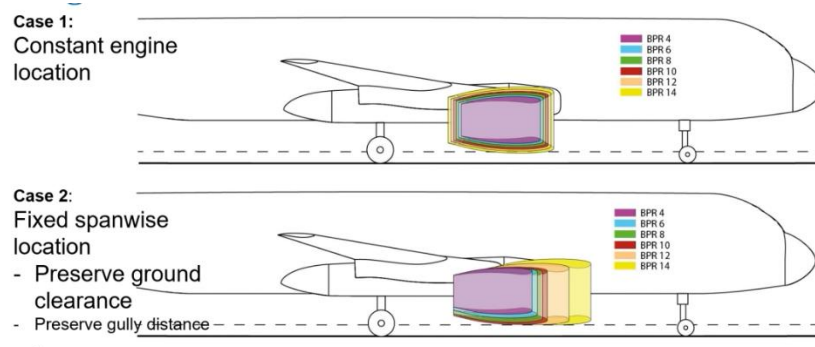
Flow curvature at the throat causes a non uniform velocity profile

All of this is particularly important for off-design operations such as low speed operations with high crosswind. These conditions can cause flow separation at the lip and drastically reduce the mass flow rate provided to the engine and cause the compressor to surge. Another issue that needs to be considered is flow distortion which can be quantified through distortion parameters. These are used to evaluate the operability margins.

Installation effects

As aviation has evolved we can clearly see a tendency for the engine sizes to increase. This is to accommodate for higher and higher BPR.

Engine can be mounted on the wing in two different ways:

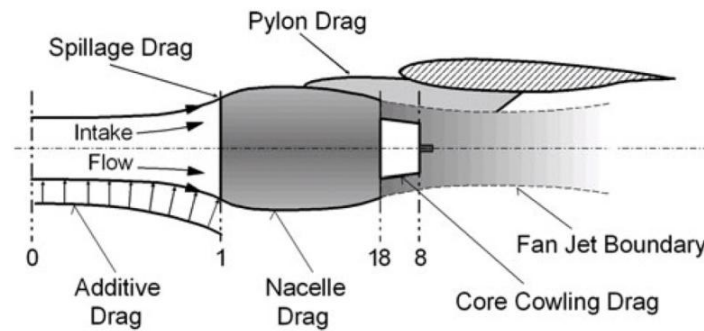


In the second case, to maintain the required ground clearance the position of the engine is shifted forward and up so that interference drag doesn't increase drastically. This is the most common solution.

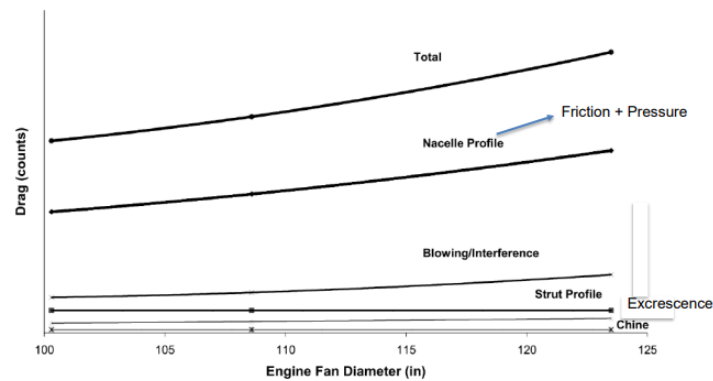
When we talk about installation penalties we are referring to the interactions between all of the components that are in close proximity to the installation point of the engine. Interference effects between the nacelle-pylon-wing/fuselage can lead to great super-velocities and therefore increased drag and risk of shock-induced separation.

However, this is not the only drag component the inlet design influences. The drag generated by a nacelle can be divided into:

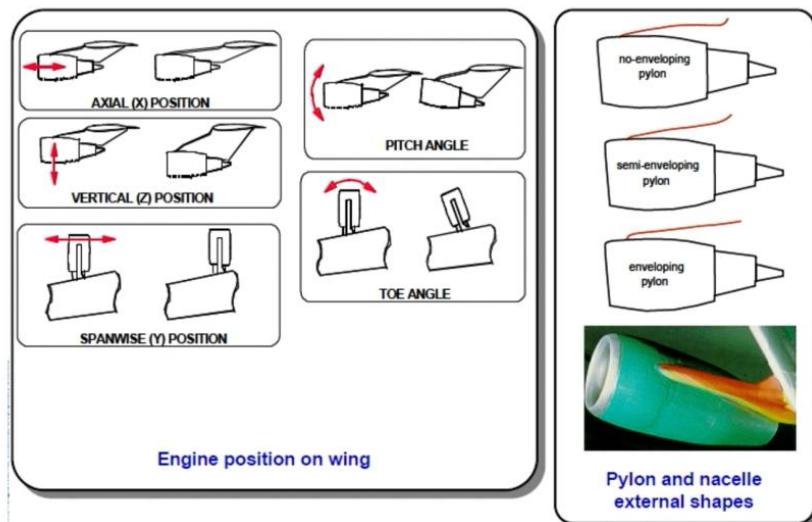
- **Profile drag:** the air flow has to move around the nacelle. This creates pressure and friction drag known as profile drag. This is the biggest contributor to total nacelle drag.
- **Spillage drag:** when the mass flow rate is different than the design mass flow rate, the stream tube coming into the engine has to converge/diverge more. This can lead to mass flow rate spillage over the nacelle which increases drag.
- **Wave drag:** super-velocities over the nacelle can lead to shocks and shock-induced separation.
- **Windmilling drag:** this drag component is only relevant when the engine is inoperative and the mass flow rate through it is drastically lower than in normal operating conditions. The windmilling drag component takes into account the increase in drag from the OEI condition.
- **Additive drag:** is the pressure drag generated from the pressure difference between the air inside the stream tube and the air around it.
- **Interference drag:** the interaction of the flow around the components that are in close proximity to each other leads to an increase in super-velocities and therefore drag.



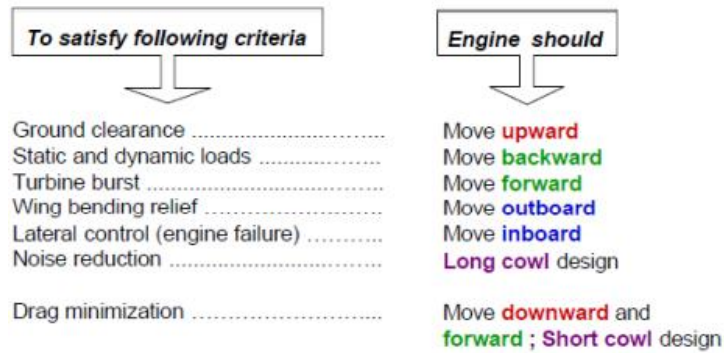
The main design parameter that effect drag generation of the nacelle is obviously the nacelle diameter. As BPR increases, all of the drag components tend to increase, so the gains brought on the TSFC are slightly offset by this increase in drag. This balance is what determines the optimal BPR of an engine.



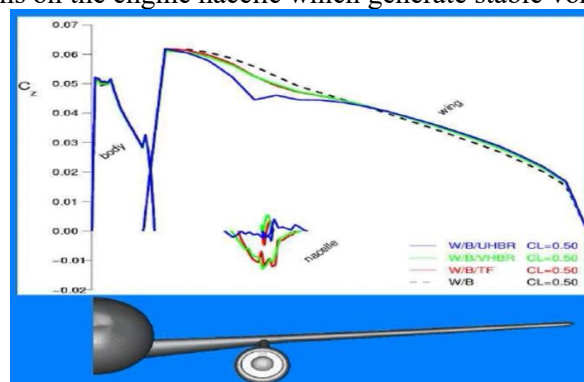
In civil aircrafts the installation of an engine on the wing can vary greatly:



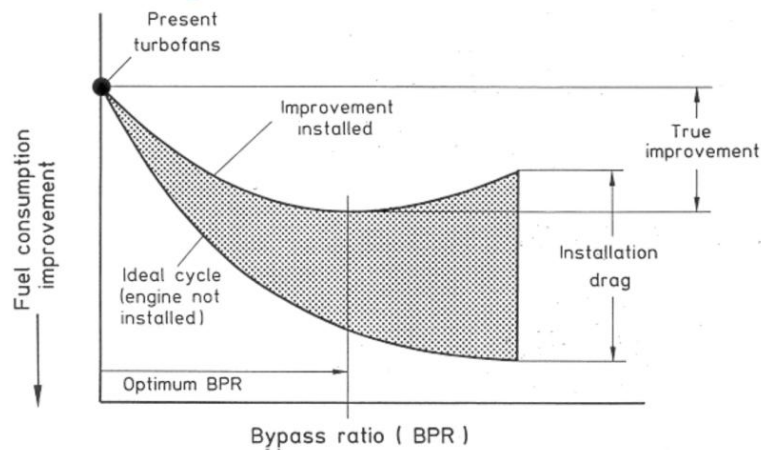
These movements are obviously constrained. For example, the axial position of the engine is determined by both the requirements on interference drag and blade off containment, the vertical position effects the ground clearance, the spanwise position is determined by the tail size for the OEI condition and so on. The pitch and toe angle mainly affect the total thrust and the stability. The general design criteria for determining the engine installation position are:



Another important aspect that needs to be taken into account when choosing the engine installation position is the fact that the presence of the nacelle “spoils” the flow behind it that goes over the wing. This reduces the local lift produced and can be observed from the spanwise distribution of lift. This can be offset by adding fins on the engine nacelle which generate stable vortices.

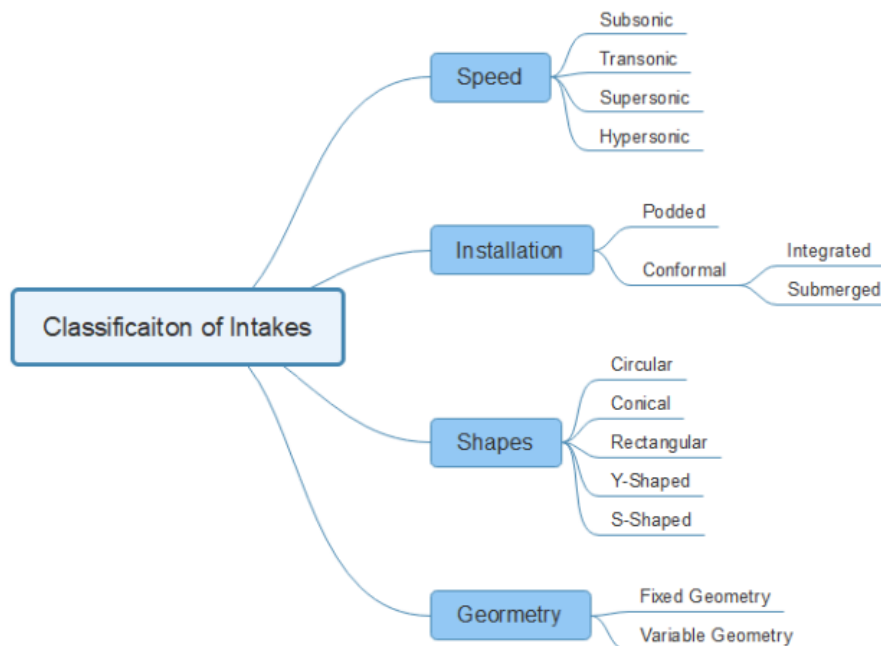


In conclusion, while for an isolated turbofan an increase in BPR to very high values can lead to great fuel consumption improvements, in reality the installation penalty determines a much lower optimal BPR to minimize fuel consumption.



Intake classification

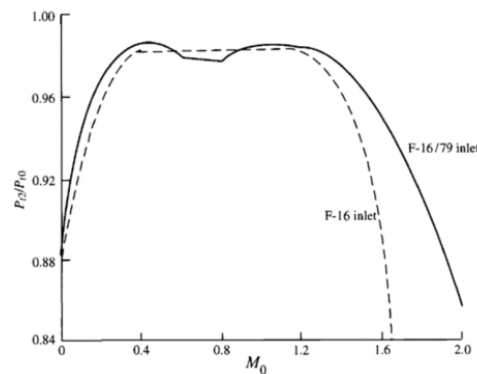
Inlets can be classified in many different ways.



Military aircrafts typically use supersonic, conformal (meaning that the inlet is flush with the aircraft geometry), rectangular/ Y-shaped and variable geometry inlets.

For civil aircrafts the inlets are instead podded, circular and have a fixed geometry.

The **inlet efficiency** (the losses of pressure that occur in the inlet) varies over the mission profile. The inlet can be designed to optimize one of the phases of a typical mission; for example supersonic inlets tend to have very thin lips to avoid shocks over the cowling during cruise. This however greatly reduces the inlet performance at lower Mach numbers since separation can occur on the inside lip. If we look at two different inlets developed for the F-16 (one with variable geometry and one not), we see that the variable geometry inlet is able to maximize the pressure ratio over a larger operational envelope.



Despite the better performance the F-16 was eventually fitted with the fixed geometry inlet due to the lower system complexity-cost-maintenance.

Inlet efficiency is extremely important, since for every percentage loss we have an overall decrease in the performance of the engine of about .5%.

Inlet distortion

Let's consider a Y-shaped inlet, in other words an inlet which has two intakes , which join together to provide the engine with only one mass flow rate.

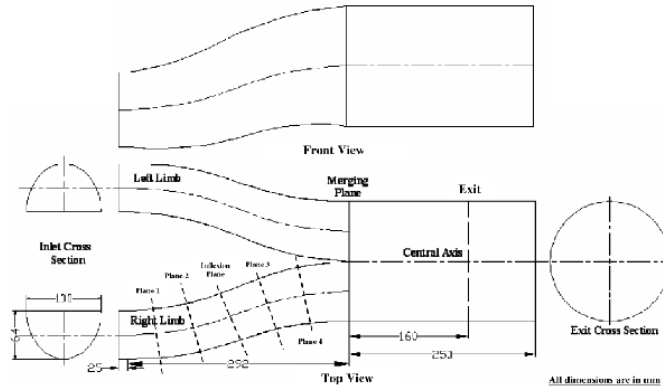
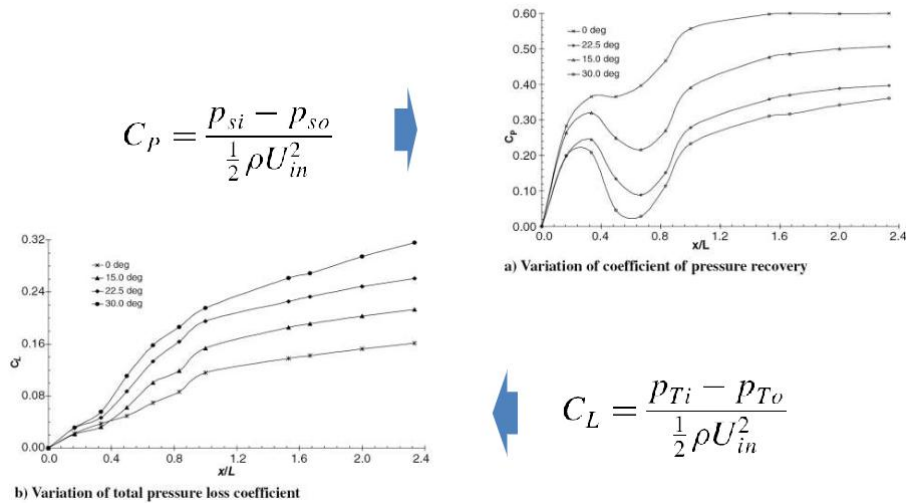


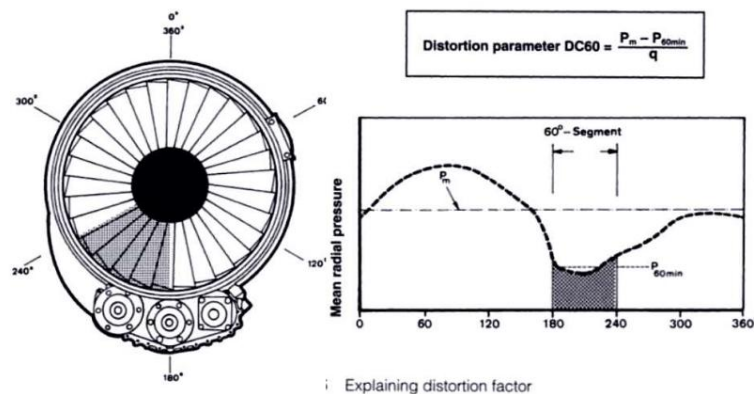
Fig. 3 Schematic layout of the Y duct showing double offsets.

In this kind of duct a prominent boundary layer forms in both the branches and does not disappear when they merge. This, together with the peculiar geometry, creates a non uniform pressure and velocity distribution.

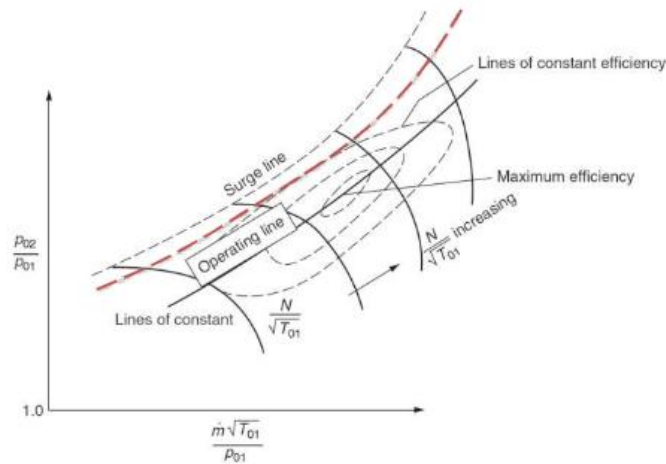
This leads to an erratic behaviour in the pressure recovery coefficient with respect to sideslip angle



This means that the engine is facing non uniform flow at the inlet. If we look at the pressure profile along the inlet walls we see high and low pressure regions. A **distortion parameter** can be defined to quantify the difference between the max pressure and min pressure in any 60 degree section w.r.t. the dynamic pressure.



When the engine is facing non uniform flow at the inlet, the compressor characteristics change. In particular the surge line of the compressor map comes down, which reduces the operational margin of the engine (increasing the risk of surge in the engine and catastrophic accidents).



The way distortion travels around the inlet can be tested with a fan test rig by putting a distortion screen over a certain 60 degree region in front of the fan. This screen creates located distortion, that allows us to evaluate the effects of the distortion on the duct, fan and compressor.

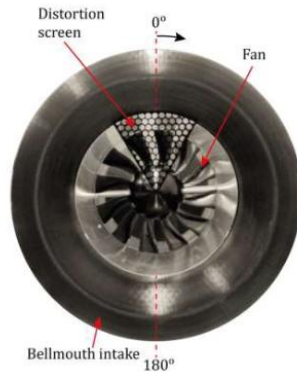
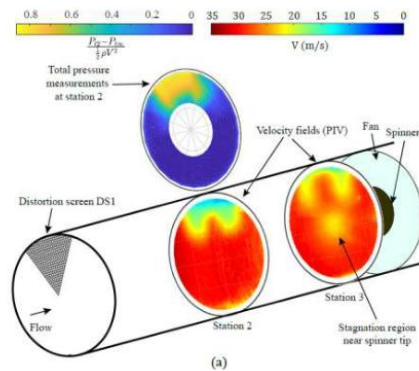


FIGURE 2. Front view of the test rig



Experimental flow field measurements using Particle Image Velocimetry

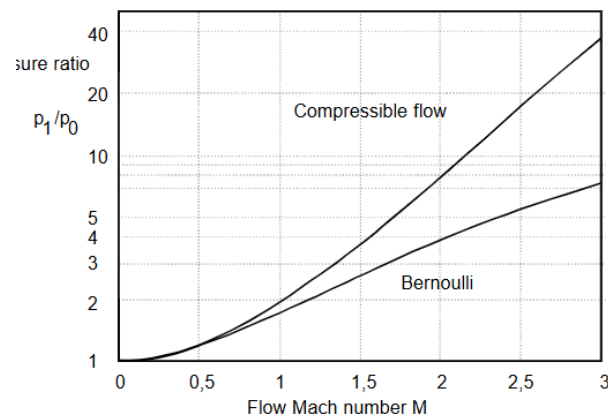
From the image on the right we can clearly see that distortion in the inlet makes the flow in the duct non-uniform and this introduces losses in all the downstream components.

Distortion can come from many sources. On some novel aircraft architectures distortion at the inlet can come from boundary layer ingestion (e.g. Boundary layer ingestion inlets prototypes).

One big issue created by distortion is the fact that once the flow is not uniform anymore, asymmetrical loads on the fan appear. This can lead to cyclical loads that promote fatigue. A way to reduce the distortion effects is to increase the inlet length. This allows the flow to become more stable by the time it reaches the fan.

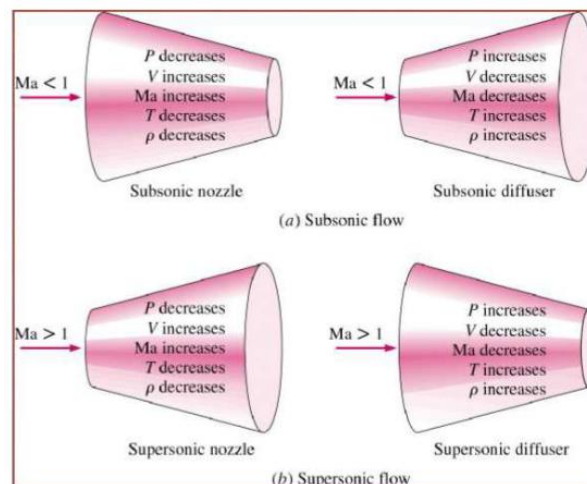
Supersonic inlets

For supersonic inlets the difference between the static pressure and the total pressure increases as the Mach number increases. As Mach increases also this difference cannot be evaluated with Bernoulli's equation since compressibility and shocks invalidate the assumptions. We can see the Bernoulli prediction deviate more and more as Mach number increases above $M = 1$.



For very high Mach numbers, the difference between static and total pressure difference becomes so big, that when the flow is slowed down to subsonic velocities in the inlet, the static pressure increases drastically and the compressor has to do little to no work (ram compression).

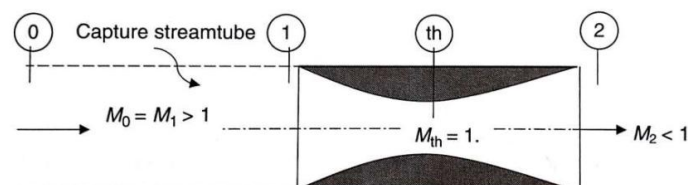
The geometry of the duct is the factor which has the most influence on the flow behaviour in an inlet. We see that the flow behaves differently depending on the regime:



$$dA/A = (M^2 - 1) * (dV/V)$$

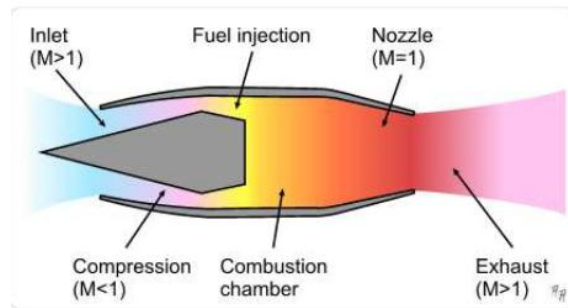
We see that supersonic flow behaves in an opposite manner to subsonic flow. Thermodynamic properties always change in the opposite way to the velocity.

As an example let's consider a supersonic convergent-divergent inlet. The supersonic flow slows down in the convergent part of the inlet and reaches the throat in sonic conditions. Now since it's subsonic, in the divergent part the flow will continue to diffuse, exiting the duct in subsonic conditions. This geometry is what is typically used for supersonic inlets.

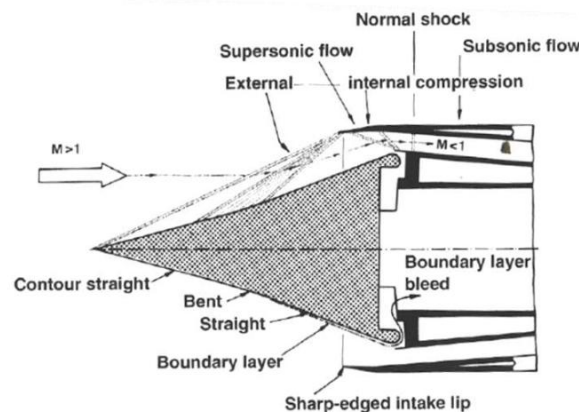


One of the applications of this kind of inlet is the ramjet engine, in which all of the compression is done by the ram compression. The compressor is therefore taken out, together with the turbine which drives it. In high supersonic regime these kinds of engines are the most efficient, however they cannot work in sonic/subsonic conditions. Both the inlet and the nozzle have a convergent-divergent

geometry, one to slow down supersonic flow to subsonic conditions and the other to speed back up to supersonic conditions.



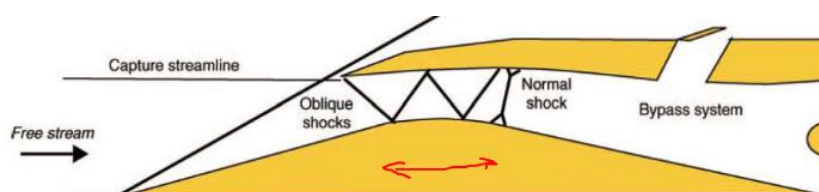
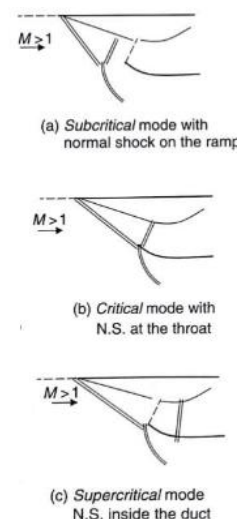
One of the possible geometries of such an inlet is the **spike geometry**. In supersonic conditions, the shock waves attach to the tip of the spike. Then the position of the spike is determined so that these shockwaves all attach to the lip of the cowling ensuring maximum mass flow rate with very little spillage. A system of oblique shocks gradually slows down the flow until a last normal shock inside the inlet brings it to subsonic conditions.



For this kind of inlet we can define three operational regimes:

- **Subcritical:** the Mach number is lower than the design value. The normal shock occurs ahead of the cowl lip increasing the spillage drag.
- **Critical:** the Mach number is equal to the design value. The normal shock is located at the throat.
- **Supercritical:** the Mach number is higher than the operational Mach number. The shock moves aft but the total mass flow rate into the engine stays the same.

To avoid the increase in spillage drag that occurs in the subcritical condition due to the normal shock being outside of the inlet we can have a variable geometry inlet, for which the spike position can be moved forward or aft. This type of systems typically also have a bypass system that allows for spillage of the mass flow rate in the inlet to control the position of the normal shock.

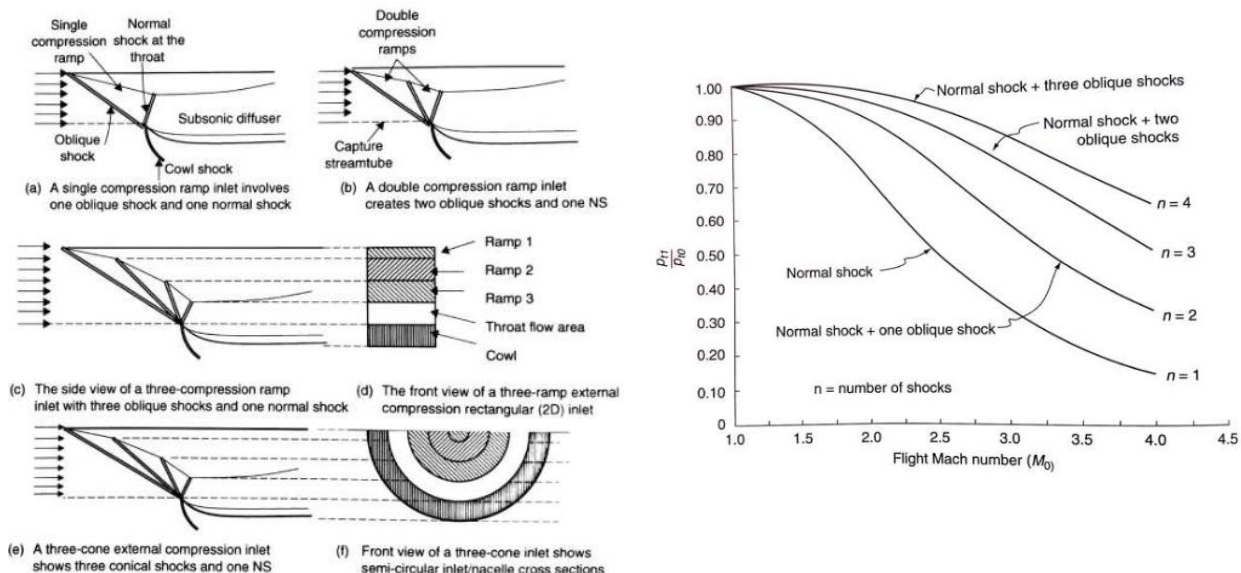


Another way to control the normal shock position is through active flow control (providing suction through the skin of the inlet).

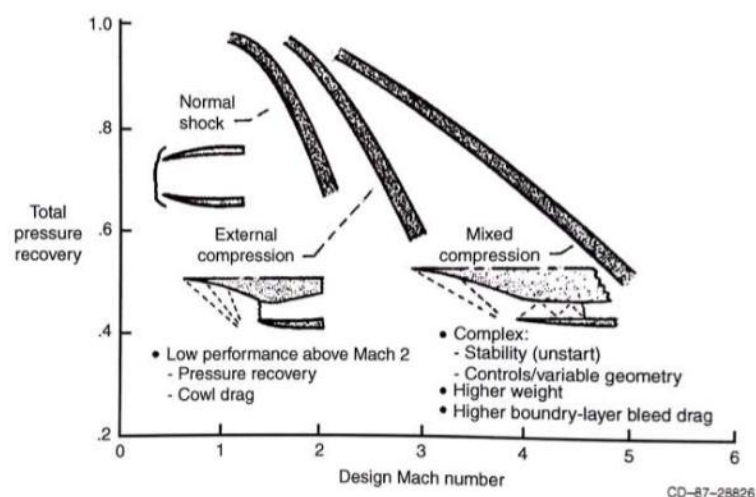
One of the main issues of fixed convergent-divergent inlets is **unstart**. For a fixed geometry inlet, the C-D duct chokes at a specific Mach number and the position of the last normal shock changes as the Mach number changes. In these conditions small perturbations can actually make it so that the normal shock comes out of the inlet forming a bow shock in front of it. This increases the spillage drag greatly and reduces the incoming mass flow rate to the engine decreasing overall thrust.

Having a strong normal shock in the inlet creates great pressure losses since the compression that occurs across it is not isentropic but is instead associated with a great increase in entropy. Therefore to reduce pressure losses we prefer to have a system of oblique shocks followed by a weak normal shock. To have this we can have an inlet which has different ramps with different ramp angles, each creating an oblique shock.

When the oblique shocks occur only outside the inlet (bow shock followed by a series of oblique shock outside the inlet) the drag is increased quite substantially and the pressure recovery drops. This can be improved by adapting the geometry so that the oblique shocks attached to the spike tip are captured by the cowl lip and the system of oblique shocks continues in the inlet and terminates with a weak shock. This is called a **mixed compression inlet**.



To summarize the total pressure recovery for different inlets:



Lecture 7

Engine exhaust

The main purpose of the nozzle is to convert the potential energy that is available after the turbine into kinetic energy (and then thrust) at a minimum pressure loss. Ideally the exit pressure at the nozzle is equal to the ambient pressure to reduce losses in the generation of thrust (if this is true all of the potential energy that could be converted into kinetic energy and then thrust has been converted). When this occurs we talk about **complete expansion**, however this is typically not the case. Another design objective of the nozzle for civil aviation is to suppress jet noise.

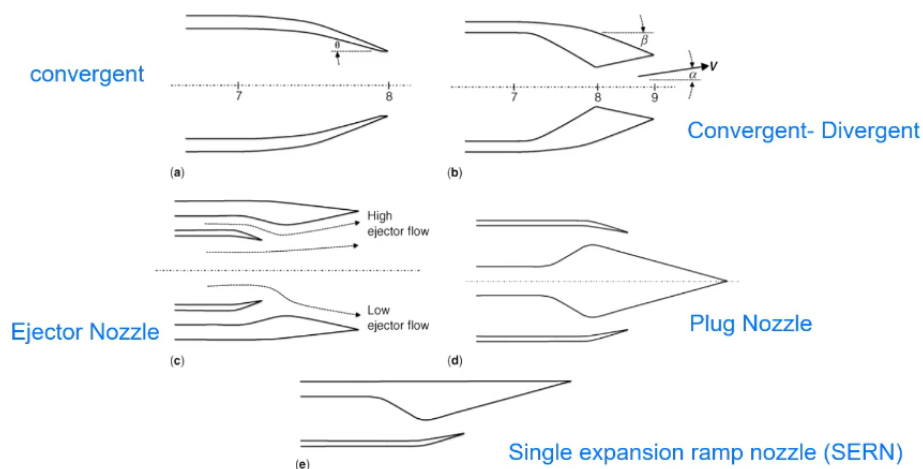
For most engines the thrust reverser system needs to be considered. This is a part of the exhaust system, which has to be designed with this in mind.

Note: for military aircrafts the exhaust system has some additional requirements such as: permit afterburner operations, allow for thrust vectoring, minimize radar cross section and permit the control of the engine operating characteristics (by changing the nozzle area the mass flow rate can be changed. Only for military a/c). For some low bypass ratio engines the core and bypass flow need to be mixed in the nozzle before expansion, therefore this is in some cases a requirement for the nozzle.

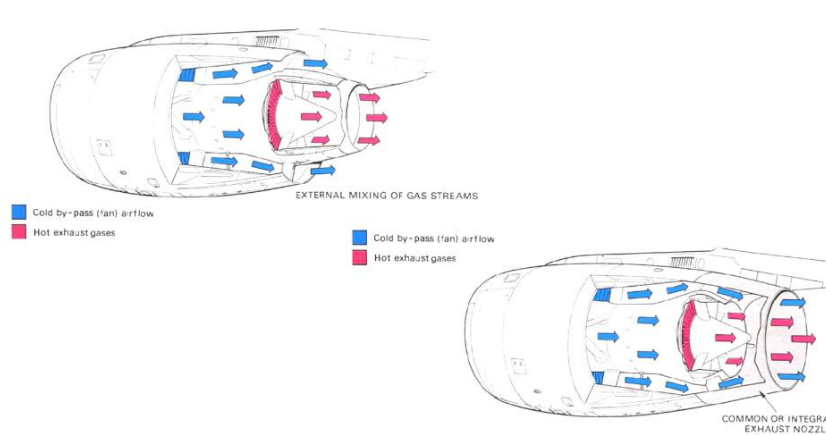
All of these requirements have to be met against minimum cost, weight and drag while meeting the reliability goals.

Nozzles can be classified as follows:

- **Convergent:** this is the most common type of exhaust system used for most civil engines. Typically at design condition in the converging section, also known as the throat, the flow is sonic. This geometry accelerates the gasses converting potential energy into kinetic.
- **Convergent-Divergent:** the convergent section is followed by a divergent nozzle. This type of nozzle is used for military aircrafts and rocket propulsion. The flow reaches sonic conditions at the throat and keeps expanding in the divergent section reaching even higher supersonic velocities.
- **Ejector:** used for supersonic aircrafts. This kind of nozzle entrains ambient air in order to reduce the jet velocity and therefore the jet noise.
- **Plug:** not used anymore. This is a kind of convergent-divergent nozzle that allows for the variation of the geometry.
- **Single Expansion Ramp Nozzle SERN:** used only for hypersonic aircrafts. The upper surface of the nozzle is actually the body of the aircraft. The geometry is designed so that the expansion continues outside the nozzle and onto the body.

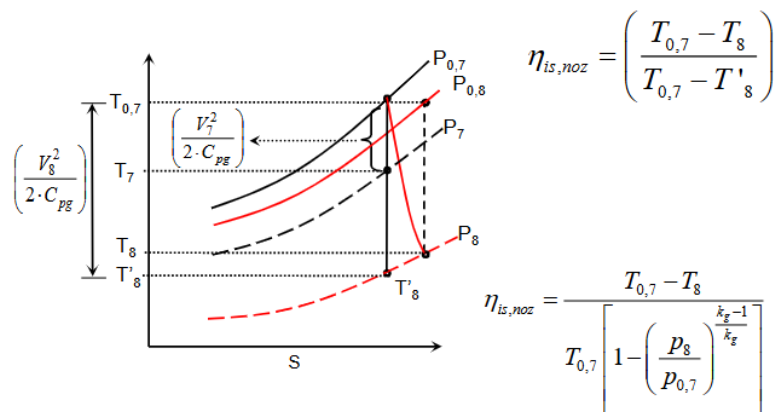


For commercial engines an unmixed convergent nozzle is typically used. The bypass and core have separate nozzles that allow for separate expansion. As we have already said, on some older jets or business jets, the two flows are actually mixed before the expansion.



From a thermodynamic point of view mixing lowers the overall exhaust velocity for the same thrust, improving propulsive efficiency (kinetic energy losses are reduced) and reducing noise. However, mixing incurs pressure losses, reducing total pressure available at the nozzle and slightly lowering ideal thrust. Also the mixer itself is heavy and requires space.

To take into account all the losses that occur in a real nozzle we can introduce the **isentropic nozzle efficiency**, which is given as the ratio between the enthalpy jump of the real expansion (between the total pressure and the ambient static pressure) and the ideal isentropic expansion:



Since no work or heat is being added to the flow, the total temperature stays constant, while the total pressure slightly decreases due to friction with the nozzle walls.

The sources of nozzle losses are:

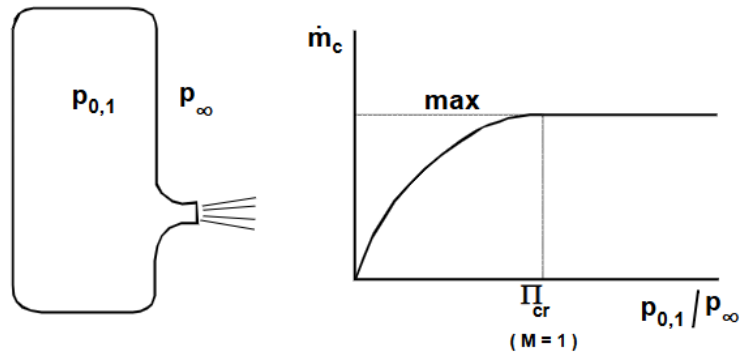
- **Exhaust velocity angularity:** due to the swirl introduced in the turbine or the nozzle itself the flow is not perfectly axial and this component cannot contribute to thrust generation. This is considered a loss.
- **Flow leakage:** relevant only for variable geometry nozzles. These nozzles are made up of petals that overlap with each other and airflow can leak through the gaps leading to losses.
- **Friction** inside the boundary layer.

In engineering practice the isentropic efficiency is hardly ever used, instead the following coefficients are defined to account for losses:

- **Discharge coefficient:** given as the ratio of the real mass flow rate and the ideal one. Can be expressed as the ratio between the effective area of the nozzle and the geometric area
- **Velocity coefficient:** ratio between the real jet velocity and the ideal one
- **Thrust coefficient:** given as the product of the two previous ones. This is equal to the ratio between the actual thrust generated by the nozzle and the ideal one.

Typical values for these coefficients are between 0.95-0.98.

To determine whether or not a nozzle is choked we have to look at the pressure ratio across the whole nozzle. If the pressure ratio P_t/P_0 is over the **critical pressure ratio** then the nozzle is choked. The critical pressure ratio can be intuitively defined by looking at a tank with a small nozzle. The flow through the nozzle is determined by the ratio between the pressure outside and inside the tank. As this ratio increases (either the pressure inside increases or the pressure outside decreases) the mass flow through the nozzle increases up until the flow reaches sonic conditions at the nozzle throat. At this point information about pressure differences cannot travel upstream since it travels at the speed of sound, meaning that any increase in the pressure ratio does not correspond to a change in mass flow rate. The only thing that changes is that the pressure of the flow at the nozzle exit is different from the outside pressure and the expansion continues outside. The expansion is **incomplete**.



$$TR_{cr} = \frac{T_{0,7}}{T_{throat}} = \frac{k_g + 1}{2} \quad \Pi_{cr} = \frac{P_{0,7}}{P_{throat}} = \left(\frac{k_g + 1}{2} \right)^{\frac{k_g}{k_g - 1}}$$

The equations above are used to evaluate the critical temperature and pressure ratios in case of an isentropic expansion. Depending on the nozzle efficiency it can increase slightly since higher pressure ratios are needed to reach sonic conditions.

Here are some examples of calculation for an unchoked nozzle with complete expansion:

$$\frac{P_{0,7}}{P_\infty} \leq \Pi_{cr} \quad T_{0,7} - T_{jet} = T_{0,7} \cdot \eta_{is,jet} \left[1 - \left(\frac{P_\infty}{P_{0,7}} \right)^{\frac{k_g - 1}{k_g}} \right]$$

$$P_{jet} = P_\infty$$

$$V_{jet} = \sqrt{2c_{p,g}(T_{0,g} - T_{jet})}$$

$$FN = \dot{m} \cdot (V_{jet} - V_0)$$

$$\begin{aligned} \dot{m} &= \rho A v \\ p &= \rho R T \\ v &= M \sqrt{k_g R T} \end{aligned} \quad \begin{aligned} T &= T_0 \left/ \left(1 + \frac{k_g - 1}{2} M^2 \right) \right. \\ p &= p_0 \left/ \left(1 + \frac{k_g - 1}{2} M^2 \right)^{\frac{k_g}{k_g - 1}} \right. \end{aligned}$$

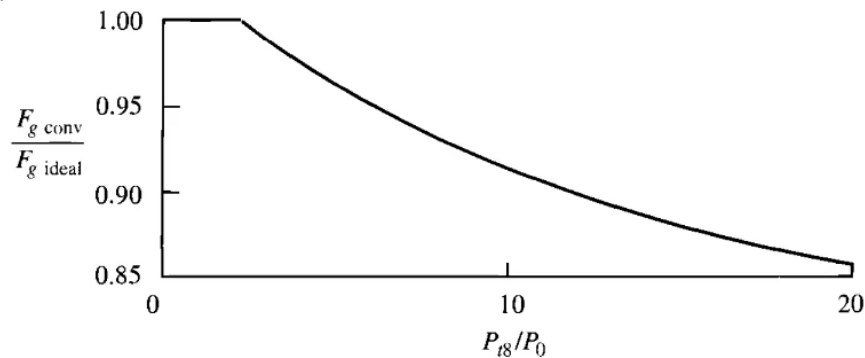
$$A = \frac{\dot{m} \sqrt{T_{0,g}}}{P_{0,g}} \sqrt{\frac{R}{k_g}} \frac{\left(1 + \frac{k_g - 1}{2} M^2 \right)^{\frac{k_g + 1}{2(k_g - 1)}}}{M}$$

For a choked nozzle with incomplete expansion:

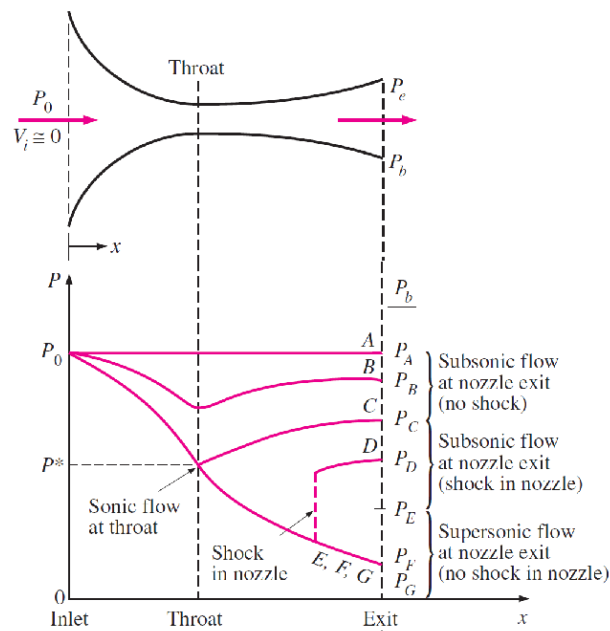
$$\begin{aligned}
 p_8 &= p_{0,7} / \Pi_{cr} \\
 T_8 &= T_{0,7} / TR_{cr} \\
 v_{jet} &= \sqrt{k_g R \cdot T_8} \\
 A_{Nz} &= \frac{\dot{m} \cdot R \cdot T_8}{p_8 \cdot V_{jet}}
 \end{aligned}
 \quad
 \begin{aligned}
 \frac{p_{0,7}}{p_0} > \Pi_{cr} &= \frac{1}{\left[1 - \frac{k_g - 1}{\eta_{is,jet} \cdot (k_g + 1)} \right]^{\frac{k_g}{k_g - 1}}} \\
 TR_{cr} &= \frac{k_g + 1}{2}
 \end{aligned}$$

$$\begin{aligned}
 F_N &= \dot{m} \cdot (V_{jet} - V_0) + A_{Nz} \cdot (p_8 - p_0) \\
 F_N &= \dot{m} \cdot (V_{eff} - V_0) \quad \text{solve for } v_{eff}
 \end{aligned}$$

For choked nozzles if the pressure ratio continues to increase we see that we start getting losses of thrust that are accounted by the thrust coefficient. This is due to the fact that the expansion that occurs outside of the nozzle does not contribute to thrust generation. Some of this energy is still recuperated from the pressure thrust term, however it is not as efficient as directly converting that energy into kinetic energy.



Let's now look at how the flow behaves in a convergent divergent nozzle:



When the pressure ratio across the nozzle is 1 there is no pressure gradient to drive the flow and the mass flow rate is zero. Once we start rising the pressure ratio the flow starts flowing through the nozzle, until it reaches a maximum velocity at the throat and slows down again in the divergent part. This situation persists for all pressure ratios under the critical pressure ratio, with increasing velocities at the throat and exit as the pressure ratio increases. At the critical pressure ratio (or slightly below it), the flow reaches sonic conditions at the throat and then slows down again in the divergent part.

For higher pressure ratios there is one design pressure ratio for which the expansion is complete. In this situation the flow reaches the throat at sonic conditions and keeps expanding in the divergent part. In this case we have that the pressure at the exit of the nozzle is equal to the ambient pressure.

For every pressure ratio between the critical one and PE however we have that a normal shock occurs in the nozzle. The flow keeps expanding in the divergent part of the nozzle until it cannot manage the pressure difference between the flow itself and the ambient pressure. Therefore a rapid compression occurs in the nozzle that brings the flow to subsonic conditions. After that the flow slows down further in the divergent part (since it's subsonic) and the pressure rises to the ambient one). This condition has to be avoided at all cost since the normal shock introduces a lot of noise, structural damages and losses.

In this region, as the pressure ratios keep increasing the normal shock moves downstream, until we reach PE, at which point the normal shock is on the exit section of the nozzle. For pressure ratios in between the ideal pressure ratio PF and PE we have that the flow will follow the ideal expansion curve and reach the nozzle exit with a pressure that is lower than the ambient pressure. This is why, for this and the previous regime the nozzle is said to be **overexpanded**. The compression of the flow occurs outside of the nozzle through a system of oblique shocks.

Finally if the pressure ratio is higher than the ideal pressure ratio the nozzle is **underexpanded**, since the flow will still follow the ideal expansion curve and have a higher pressure than the ambient one. Therefore, the flow will continue to expand outside the nozzle through expansion fans which flare the plume. This is then followed by the formation of shock diamonds (**Mach disks**) further downstream to realign the flow.

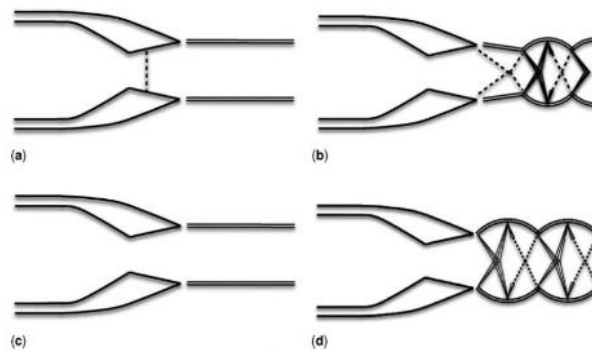


Figure 4. Nozzle expansion. (a) Overexpanded - normal shock inside, $NPR \ll NPR_{design}$; (b) overexpanded - oblique shocks at exit, $NPR < NPR_{design}$; (c) perfectly expanded, $NPR = NPR_{design}$; (d) underexpanded, $NPR > NPR_{design}$.

If we look in detail the sequence of compression and expansion waves for an over and under expanded nozzle we see:

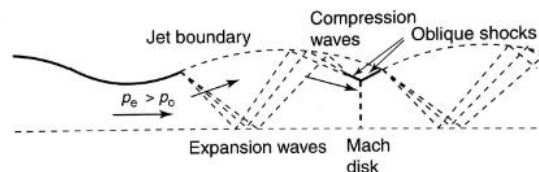


Figure 4.14 Flow pattern in under-expanded nozzle jets.

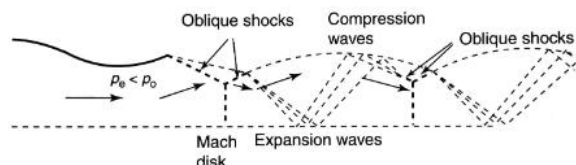
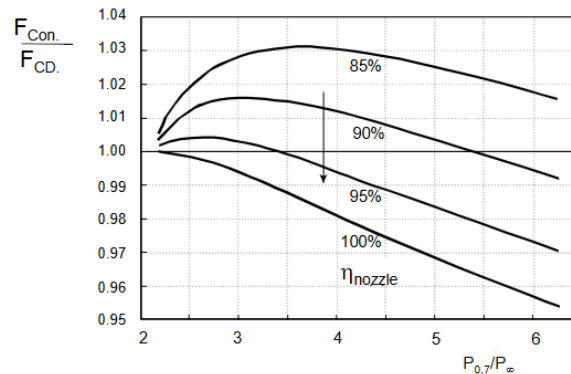


Figure 4.15 Flow pattern in over-expanded nozzle jets.

The main difference is that for an over expanded nozzle the first thing that occurs outside the nozzle are oblique shocks that attach to the nozzle lip and converge into a Mach disk. This is then followed by a succession of expansion and compression fans. When compression fans converge a Mach disk is formed.

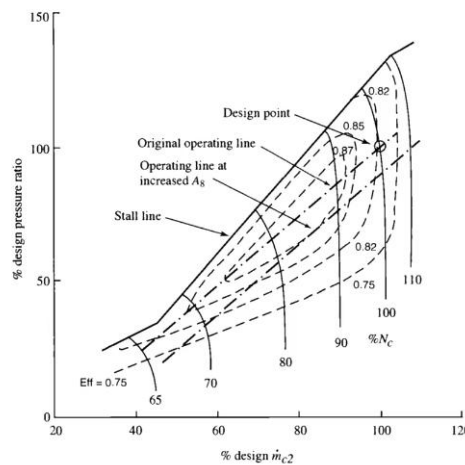
For an under expanded nozzle the first thing the flow encounters are expansion fans, which get reflected in an unlike sense as compression waves on the exhaust plume boundary and coalesce into oblique shocks and Mach disks.

From the graph below we can see that convergent nozzles have better thrust coefficients at low pressure ratios (or lower flight Mach number. Typically until $M \sim 1.5$). However, their efficiency drops once the pressure ratios surpass pressure ratios that are a bit higher than the critical pressure ratio (pressure ratios of around 5). That is supersonic aircrafts and fighters use convergent-divergent nozzles.



We can see that as the isentropic efficiency of the nozzle increases the pressure ratio after which the C-D nozzle becomes better gets lower and lower. This is because as the isentropic efficiency increases the pressure ratio required for the flow to become supersonic decreases.

Note: a variable area nozzle, as we have already said, can be used to control the mass flow rate through the engine. We can see this from the compressor map:



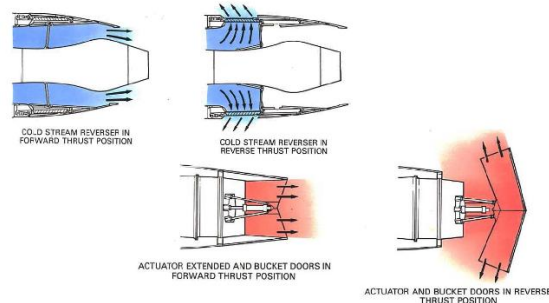
By opening the area of the nozzle the mass flow rate through the engine is increased and therefore the operating line moves away from the surge line, increasing the operational margin.

Thrust reversers

Thrust reversers are used to reduce break wear during landing and add a layer of safety especially for wet and icy runways. They cannot be used for the certification of the landing distance in dry conditions due to their low reliability.

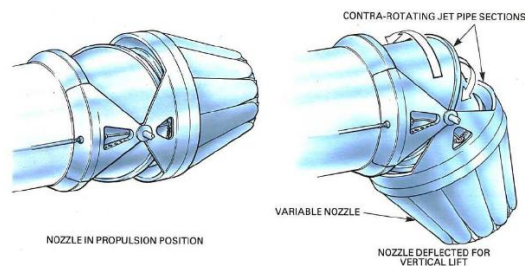
The basic idea is do redirect the exhaust gasses forward in order to produce negative thrust. To do this there are three different types of mechanisms:

- **Bucket type reversers:** two panels close over the exhaust and reverse both the core and bypass flows. Not used anymore if not only for business jets.
- **Cascade type reversers:** the nozzle cowl shifts backward and the engine nacelle opens up, revealing cascade grids which redirect the gasses forward. Blocker doors block the flow from going out the nozzle.
- **Petal type:** doors open all over the nacelle.



Thrust vectoring

Thrust vectoring means that the axis along which the thrust is applied is changed. This is only used for military aircrafts and can be used for short take-off or even vertical take-off.



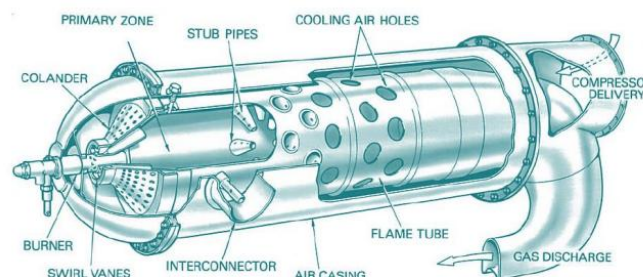
Lecture 8

Combustion chamber

The combustion chamber is located between the compressor and turbine in a gas turbine and is required to convert the chemical energy of the fuel into thermal energy with the smallest possible pressure loss and with the least emission of undesirable chemicals. In other words, the combustor provides the heat input into the gas turbine cycle. It receives air from the compressor, introduces a stream of fuel into it, creates the conditions for the fuel and air to mix and react and, eventually, delivers a mixture of hot post-combustion gases to the turbine. Such a process is commonly referred to as **internal combustion**.

For the sake of analysis, combustion systems are generally classified as either those of constant pressure or constant volume.

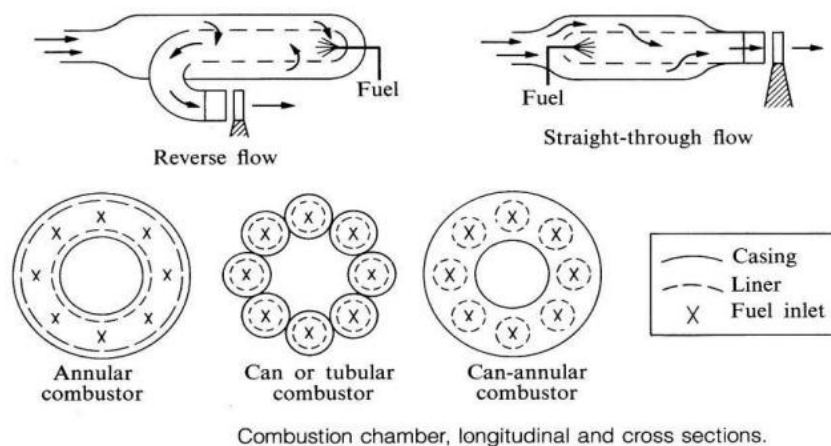
One of the first combustors for a gas turbine engine was the whittle combustor:



For the first combustors the only objective was to introduce heat into the cycle, without having to worry about emissions. Nowadays the combustion chamber design is not only focused on burning the fuel efficiently but also introducing the minimum amount of pollutants as possible.

Combustion chambers can be classified depending on:

- Direction of the flow:
 - **Reverse:** the Whittle combustor is a reverse flow combustion chamber, where the flow comes in from one end, heat is added and the flow then reverses its direction and goes out the same way. This is done to achieve a satisfactory residence time. Typical of old combustion chambers.
 - **Straight:** flow comes in one end, heat is added in the middle and then exhaust gasses come out the other end.
- Combustor type: the geometry of the combustor
 - **Annular:** the combustion chamber shape is cylindrical. This type is the most used in modern engines.
 - **Can of tubular** combustors: used in the first generations of engines. They are made up of smaller cans placed in a circle.
 - **Can-annular** combustor: mix of the two previous solutions.



The early gas turbine engines used can type combustors, which had holes in the sides to connect all of the cans that were positioned in a circle. This was to equalize the pressure in all of the cans in case of an issue or to reignite one of the flames.

In the 70's the design shifted towards an annular architecture. This has the advantage that flame propagation is much easier and less cooling air is required due to the reduced surface area of the combustor. The **liner** (material that "lines" the combustion chamber walls) has holes that allow air unused in the combustion to form a thin film which takes away heat from the liner allowing temperatures that are higher than the melting point of the material.

Enthalpy

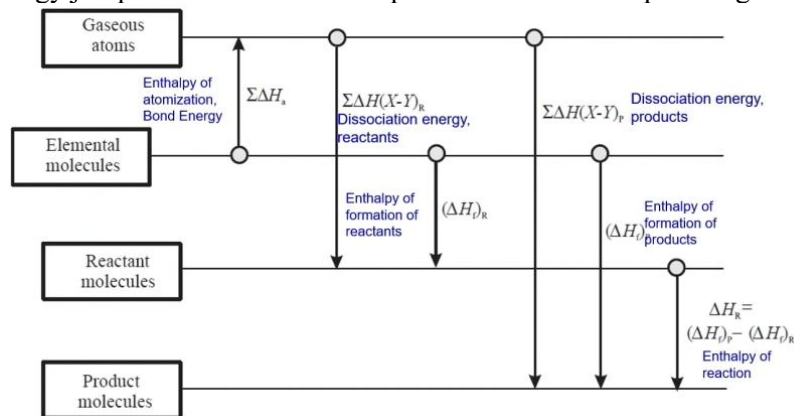
Enthalpy is one of the most used properties to study the process of combustion. There are three types of enthalpy we can use:

- **Sensible:** energy related to temperature increase w.r.t. a reference temperature
- **Latent:** energy required for phase transformation
- **Reaction enthalpy:** energy that is obtained from an exothermic chemical process or energy required for an endothermic process. For the study of combustion processes, reaction enthalpy is the property of interest.

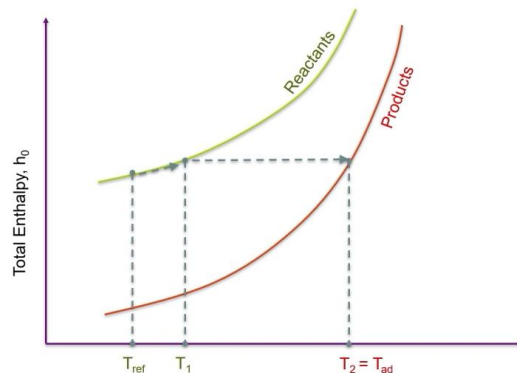
Note: typically sensible enthalpy is lower than latent enthalpy, which is itself lower than the reaction enthalpy.

A type of reaction enthalpy is the **enthalpy of formation**, which is the amount of energy released at a given temperature when 1 mole of an element is produced (for reference elements like H₂, O₂ and N₂ the formation enthalpy is zero). This can be used to evaluate the amount of heat released from a chemical reaction.

On a molecular level, as we go from elemental molecules to gaseous atoms we need to give an amount of heat that is equal to the **energy of atomization** (which is equal to the bond energy). The jump from the enthalpy of reactant molecules and products molecules is defined as the enthalpy of reaction. This can be positive or negative depending on if the reaction is exothermic or endothermic. The jump from the elemental molecules and the reactant molecules is quantified by the **enthalpy of formation** of the reactants. The same is true for the products. The **dissociation energy** is instead defined as the energy jump from the reactants or products and their respective gaseous atoms.



If we chart the total enthalpy of both the products and the reactants we see that they both increase with temperature (sensible enthalpy increases). If we were to have an adiabatic combustion, no energy would leave the system, therefore the total enthalpy would remain constant (formation enthalpy is transformed into sensible enthalpy). The new thermodynamic state would be on the reactants curve at a higher temperature.



The **total** (or **absolute**) **enthalpy** at a given temperature can be defined as the sum of the formation enthalpy and the sensible enthalpy.

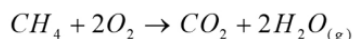
$$h_i(T) = h_{f,i}^0(T_{ref}) + \Delta h_{s,i}(T),$$

or

$$\Delta h_{s,i}(T) = h_i(T) - h_{f,i}^0(T_{ref}) = \int_{T_{ref}}^T c_{p,i} dT;$$

h_i - mass or molar specific enthalpy of species i.

If we consider a process of combustion involving methane and oxygen we would get CO₂ and water vapor. The **heat of reaction** (which is just the reaction enthalpy of a combustion process) of this process is just the difference in formation enthalpy of the products and the reactants.



$$(Q_P)_{NTP} = (\Delta H_f)_{CO_2} + 2(\Delta H_f)_{H_2O} - (\Delta H_f)_{CH_4} - 2(\Delta H_f)_{O_2}$$

$$(Q_P)_{NTP} = (-393522) + (-241827) - (-74897) - 2(0)$$

$$(Q_P)_{NTP} = -802279 \text{ kJ/k mol}$$

From this the heating value of a fuel can be evaluated as the ratio between the heat of reaction and the molar mass of the fuel. We can define two types of fuel calorific values (amount of heat released by combustion of one unit mass of fuel) the **Lower Heating Value** LHV and the **Higher Heating Value** HHV. The first is defined as the amount of heat released by combustion of 1kg of fuel, where all the water in the products is in gaseous form (has absorbed part of the heat released by the reaction to change phase), while the HHV is defined as the amount of heat released by combustion of 1kg of fuel, where all the water in the products is in liquid form.

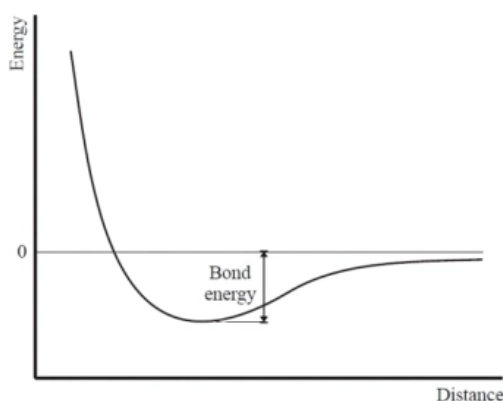
Bond energy

We're considering the transformation of chemical energy into heat. Chemical energy is the energy that is conserved in chemical bonds: when a bond is broken energy is released and vice versa.

Table 11.1 Some Atomisation, Dissociation and Resonance Energies (Based on 25 °C)

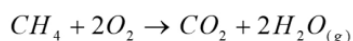
Bond Atomisation (ΔH_a)	Energy MJ/kmol	Bond Dissociation ($\Delta H(X-Y)$)	Energy MJ/kmol	Resonance	Energy MJ/kmol
H—H	435.4	C—H	414.5	Benzene: C ₆ H ₆	150.4
C (graphite)	717.2	N—H	359.5	Naphthalene: C ₁₀ H ₈	255.4
(O = O) _{O₂}	498.2	O—H	428.7	Carbon dioxide: CO ₂	137.9
N≡N	946.2	H—OH	497.5	—COOH group	117.0
		C—O	351.7		
		C=O	698.1		
		C—C	347.5		
		C=C	615.5		
		C≡C	812.2		

The bond energy is defined as the energy required to separate two atoms, in other words the minimum potential energy relative to when the atoms are infinitely separated. The minima coincides with the equilibrium state. To move away from this equilibrium, energy needs to be provided; this is the bond energy.



If a isobaric or isochoric reaction is carried out in stages, the algebraic sum of the heat evolved in separate stages is equal to the total amount of heat released when the reaction occurs directly, which means that the heat liberated in a reaction is independent of the path of the reaction between the initial and the final states. This is known as the **Hess Law**.

Let's consider the oxidation of methane (combustion). The simple reaction is:



However, in reality there are approximately 50 species involved and more than 300 reactions occur. Thanks to the Hess law we can still evaluate the enthalpy of the reaction from the initial and final states, independently of the stages in between (since combustion can be assumed to be occurring at constant pressure or at constant volume).

The fuel used in aviation are all different types of hydrocarbons, specifically alkanes, aromatics (characterized by an aromatic group similar to benzene) and alcohols (characterized by an hydroxyl group). From the following table the heating values and the stoichiometric air to fuel ratio of these fuels can be computed. The **stoichiometric air to fuel ratio** is the ideal ratio between air and fuel moles to have a complete combustion, where all the reactants are transformed into products, with no leftover reactants.

Fuel	Chemical Formula	Molecular Weight (m_w) (integrals)	Enthalpy of Formation, Δh_f (kJ/kmol)	Higher Enthalpy of Reaction, Δh_R (MJ/kmol)	Lower Enthalpy of Reaction, Δh_R		Stoichiometric Air-Fuel Ratio, e_{st}
					(kJ/kg)	(MJ/kmol)	
Alkanes							
Methane	CH ₄	16	-74,898	-891	-50,047	-802.9	17.16
Ethane	C ₂ H ₆	30	-84,724	-1561	-47,519	-1428.8	16.02
Propane	C ₃ H ₈	44	-103,916	-2221	-46,387	-2045.4	15.60
Butane	C ₄ H ₁₀	58	-124,817	-2880	-45,771	-2660.2	15.38
Pentane	C ₅ H ₁₂	72	-146,538	-3538	-45,384	-3274.3	15.25
Hexane	C ₆ H ₁₄	86	-167,305	-4197	-45,134	-3889.3	15.16
Heptane	C ₇ H ₁₆	100	-187,946	-4856	-44,955	-4504.4	15.10
Octane	C ₈ H ₁₈	114	-208,586	-5515	-44,820	-5119.5	15.05
Cetane	C ₁₆ H ₃₄	226	-454.3	-10,711	-44,000	-9963.4	14.88
Aromatics							
Benzene	C ₆ H ₆	78	+82,982	-3304	-40,605	-3172	13.20
Toluene	C ₇ H ₈	92	+50,032	-3950	-40,967	-3774	13.43
Xylene	C ₈ H ₁₀	106	+18,059	-4598	-41,276	-4378	13.60
Naphthalene	C ₁₀ H ₈	128	+150,934	-5233	-39,455	-5057	12.87
Alcohols							
	C _n H _{2n+1} OH	C _n H _{2n+2} O					
Methanol	CH ₃ OH	32	-201,301	-764	-21,114	-676	6.44
Ethanol	C ₂ H ₅ OH	46	-235,466	-1410	-27,742	-1278	8.85
Propanol	C ₃ H ₇ OH	60	-235,107	-2069	-31,504	-1893	10.30

Some observations:

- The enthalpies of reaction for many hydrocarbon fuels on mass basis is similar, between 40-50 MJ/kg.
- The stoichiometric air-fuel ratio for many hydrocarbons lie between 13:1 to 17:1
- Some fuels have positive formation enthalpies
- All fuels have negative enthalpies of reaction (they release heat when they participate in a reaction)
- The enthalpies of reaction for alcohols is less than non-oxygenated fuels (as oxygen does not provide any energy of reaction)
- Commonly used fossil fuels are a mixture of several hydrocarbon compounds and do not have a precise chemical formulation

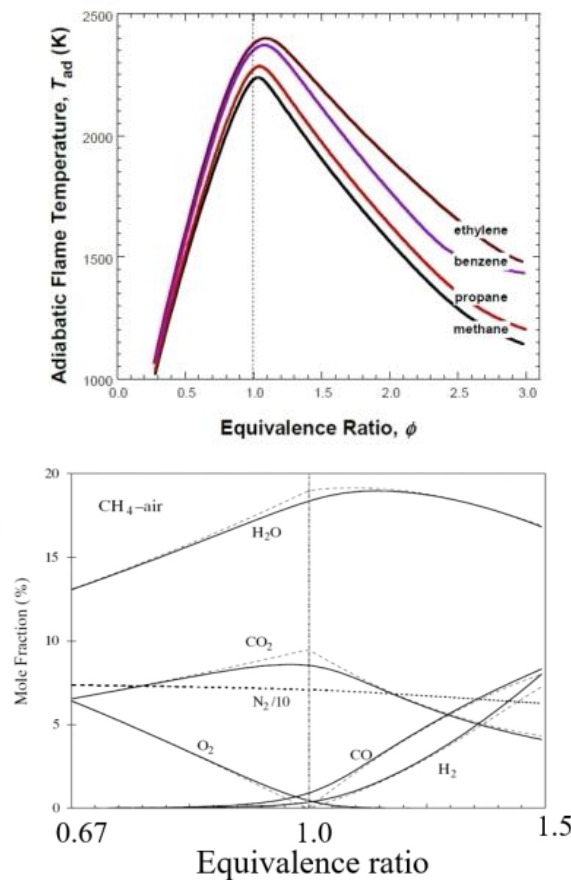
When a mixture of air and fuel burns with a fuel to air ratio lower than the stoichiometric ratio, the combustion is called **lean combustion** since there is a lower than ideal quantity of fuel in the mixture and some air will not participate in the reaction and will instead absorb some of the heat released during the reaction. Thus the final temperature of the exhaust gasses will be lower.

If the mixture of air and fuel burns with a fuel to air ratio higher than the stoichiometric ratio, the combustion is called **rich combustion** since there is a higher than ideal quantity of fuel in the mixture and some of this fuel will not be burned during the reaction.

If a fuel-air mixture burns adiabatically at constant pressure, the absolute enthalpy of the reactants at the initial state equals the absolute enthalpy of the products at the final state. The final-state temperature is defined as the constant-pressure **adiabatic flame temperature**.

$$H_{prod}(T_{ad}) - H_{reac}(T_0) = 0$$

The adiabatic flame temperature will be highest when the air to fuel ratio is close to the stoichiometric one, since there won't be any unused reactants absorbing heat.

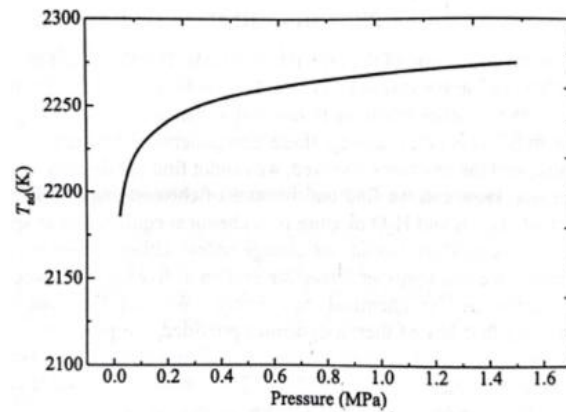


In these graphs the **equivalence ratio** appears. This is just the ratio between the real fuel to air ratio FAR and the stoichiometric fuel to air ratio FAR,stoich. When it's lower than one, the mixture is lean, while when it's higher than one the mixture is rich.

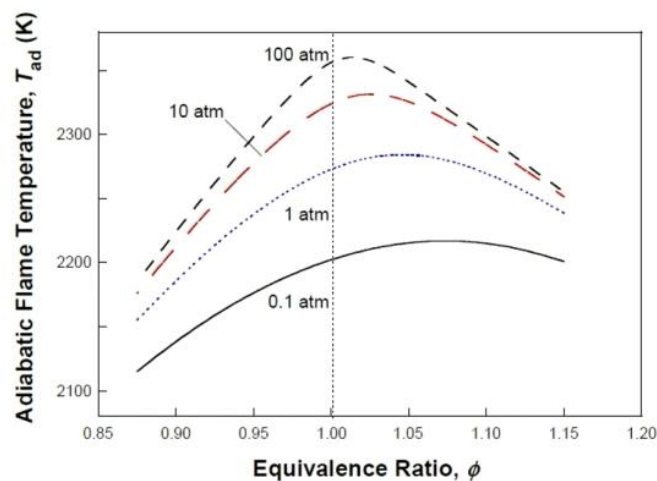
$$\phi = \frac{\varphi_{real}}{\varphi_{stoichiometric}}$$

We see that the peak in adiabatic flame temperature is slightly shifted to the right of the equivalence ratio of 1. This shift is due mainly to **dissociation** which occurs at high temperatures. From the second graph we see that the CO₂ and H₂O are the main products of combustion at an equivalence ratio of 1. Since the temperature is so high, many of these products will dissociate absorbing energy and releasing oxygen. At a slightly richer equivalence ratios the equilibrium shifts towards less dissociation, leading to higher temperatures. Also the excess oxygen released from dissociation can be consumed by the extra fuel, leading to a slightly higher temperature.

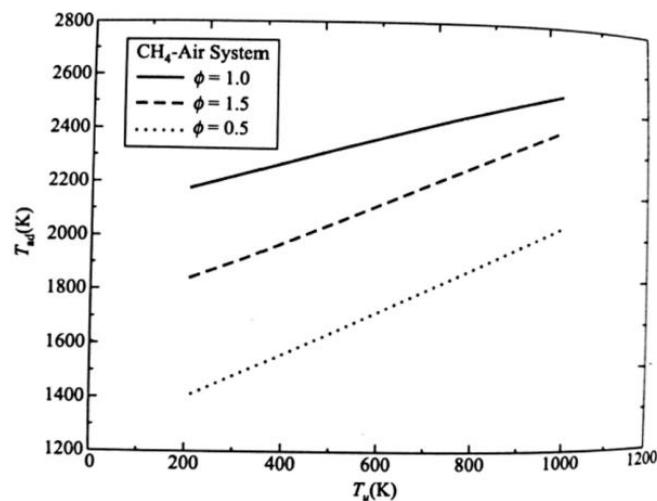
Let's now look at how the pressure in the combustor effect the adiabatic flame temperature.



We see that the adiabatic flame temperature increases with pressure until a plateau is reached. Higher pressure increases the “speed” of the chemical reactions, so it shifts chemical equilibrium toward more complete combustion (for the same residence time) and more dissociation suppression, so less energy is spent breaking molecules apart. This raises the flame temperature. As pressure continues to increase, dissociation becomes minimal and the reactants/products are already near their equilibrium limits, so further pressure increases add little change, producing a plateau.



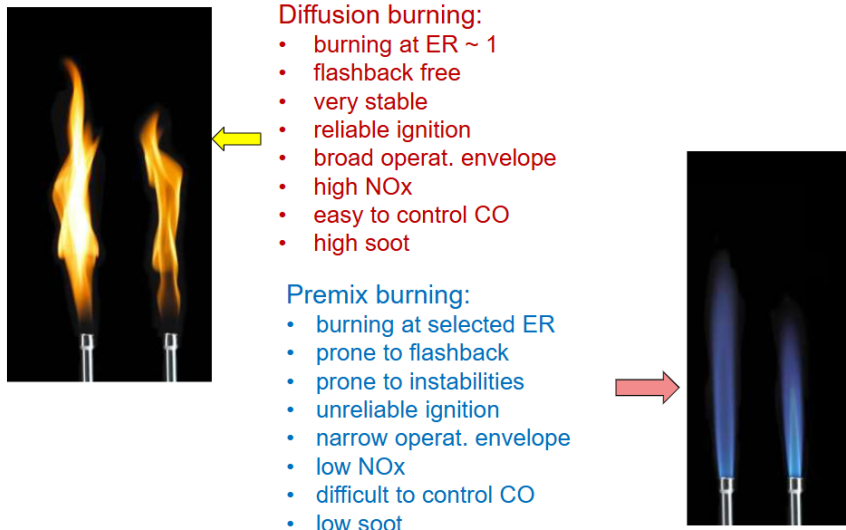
If we instead consider the effects of the inlet temperature we see that as the inlet temperature increases, the adiabatic flame temperature increases but not 1 to 1.



This is because part of the added inlet energy is consumed by greater product dissociation at the higher flame temperature. Dissociation absorbs energy, reducing how much of the inlet-temperature rise shows up as temperature gain in the adiabatic flame.

Combustion can be classified in different ways:

- Based on the mixing of fuel and the oxidizer (main distinction):
 - **Pre-mixed:** fuel and oxidizer are mixed before the reaction. This solution burns at a selected equivalence ratio and are therefore prone to **flashback**, other instabilities and is hard to ignite. Since the temperatures reached for this kind of combustion are not too high, the NO_x emissions are also contained as well as soot production. CO production becomes hard to control.
 - **Non pre-mixed:** also called diffusion burning (occurs naturally at an equivalence ratio of 1 on the boundary of the flame). It is flashback-free and also very stable. They allow for a broad operating envelope and have reliable ignition. However, these flames produce high amounts of NO_x and soot due to the high temperature gradients that form on the boundary between the fuel and the air. Here dissociation of nitrogen and oxygen occur, creating different types of nitrogen oxides. Since the NO_x emissions are high, the CO emissions are easy to control
 - **Partially pre-mixed:** most combustors are of this type. Only a portion of the oxidizer is mixed with the fuel.

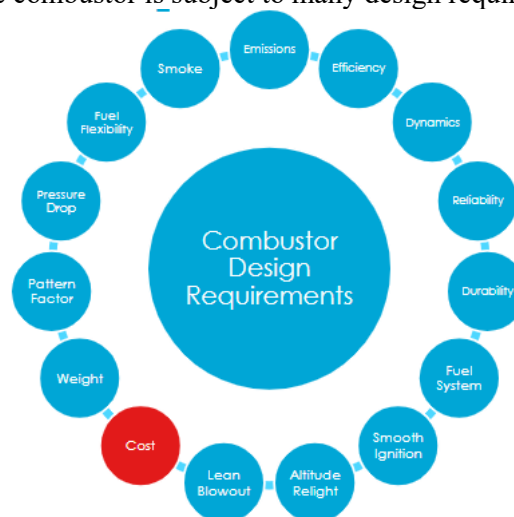


- Nature of flow:
 - **Laminar**
 - **Turbulent**
- Phase of reactants:
 - **Homogeneous:** the reactants are all in the same phase
 - **Non homogeneous:** the reactants are in different phases such as kerosene (liquid) and air (gas). This requires apposite fuel injectors.
- Temporal variation:
 - **Steady:** typical of laminar flow flames
 - **Non-steady:** typical of turbulent flows
- Speed of the combustion:
 - **Deflagration:** subsonic flame front propagation. It is also typically characterized by a decrease in pressure.
 - **Detonation:** supersonic flame front propagation. Characterized by a large pressure increase
- Speed of the flow:
 - **Compressible**
 - **Non compressible**

- Heat transfer:
 - **Adiabatic**: gas turbine combustion is adiabatic since heat is not being extracted from the system, but only used to increase the temperature of the products
 - **Non adiabatic**: typical for industrial combustion chambers (e.g. furnaces)
- Convection:
 - **Natural**
 - **Forced**
- Gravity:
 - **Micro-gravity**: this concerns only rocket propulsion
 - **Normal gravity**
- Spatial dependence (strongly dependent on the geometry of the combustor):
 - **1-D flame**
 - **2-D flame**
 - **3-D flame**
- Reaction rate:
 - **Equilibrium**
 - **Finite rate**

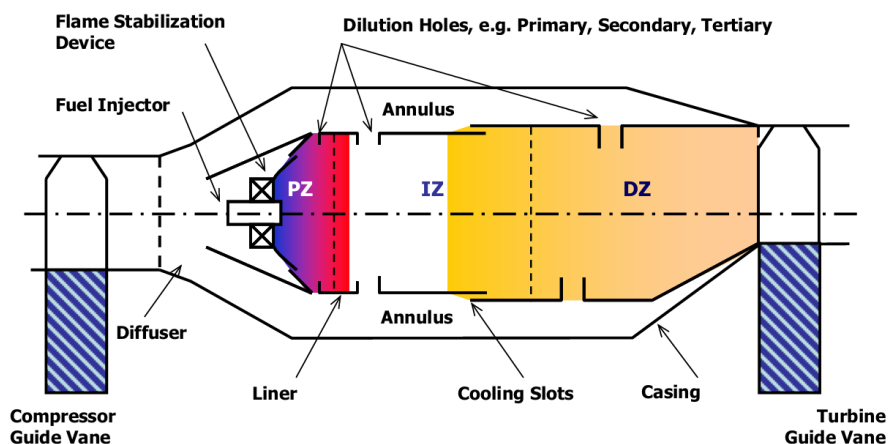
Combustor components

The design of an aero engine combustor is subject to many design requirements.



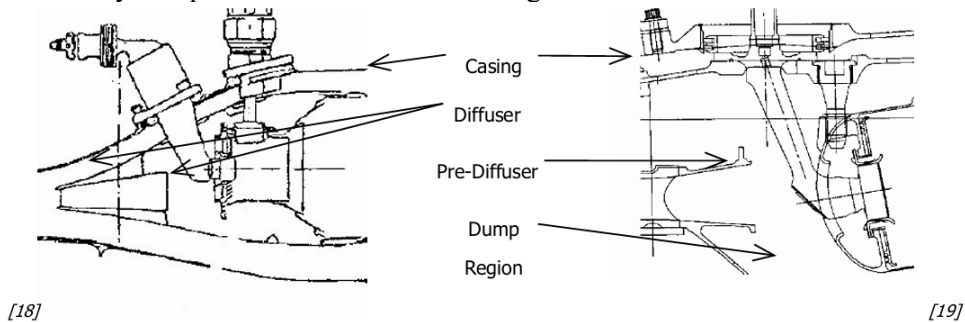
Note: **pattern factor** is determined by the temperature profile at the combustor exit. A good pattern factor is required for longevity of the first stages of the turbine.

The main components of a combustor are: diffuser, casing, liner, fuel injector, some type of a flame stabilization device and a cooling arrangement.



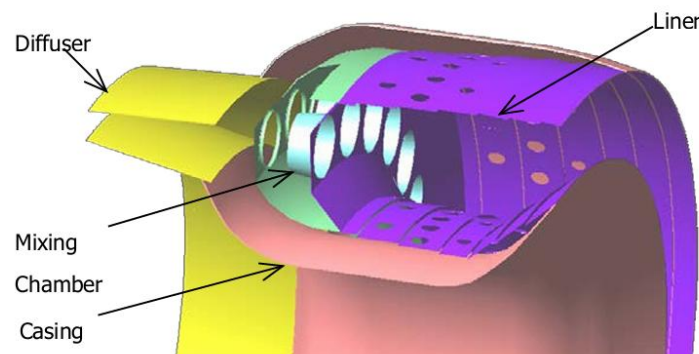
Quantitatively, the drop in total pressure associated with combustion is proportional to the dynamic pressure of the flow. Therefore, the compressor discharge velocity should be reduced to minimize the loss. It is customary to do so by incorporating a diffuser at the combustor inlet. Additional functions of the diffuser are to recover the flow dynamic pressure by raising its static pressure and smoothen flow instabilities.

There are two main design philosophies for the diffuser. One is to employ a relatively **long duct diffuser** to achieve a gradual flow deceleration. The other is the so-called “**dump**” diffuser, which consists of a short smooth pre-diffuser where the air velocity is reduced to about half its inlet value. At exit, the air is literally dumped into the combustor casing.



The aerodynamic diffuser has smaller pressure losses but results in a considerable length. The dump diffuser is exactly opposite. Its shorter length made this type the automatic design choice for modern aircraft engines.

Next are the **casing** (supports the pressure load) and **liner** (supports the thermal load). The simplest possible form of combustor would be a straight-walled duct connecting the compressor to the turbine. Unfortunately, this arrangement is impractical due to the fact the pressure loss incurred would be excessive. Simply fitting a diffuser, however, is not sufficient to obtain a viable combustion system. First of all, the air velocity remains too high for a flame to sit. Secondly, pouring fuel directly into the available airflow would result in no flame, as the air to-fuel ratios are too high for the mixture to ignite. The airflow should therefore be partitioned and the flame stabilized. Arrangements for that are provided inside the liner located inside the casing and downstream of the diffuser. The space between the liner and casing is called **annulus**.



The liner is divided into divided into a **primary zone (PZ)**, **secondary zone (SZ)** and **dilution zone (DZ)**.

A **fuel injector** and a **flame stabilization device** mark the front boundary of the primary zone. In some cases the PZ may also be preceded by an additional mixture preparation device: a **mixing chamber** (for pre-mixed combustion). The main function of the PZ is to anchor the flame and provide sufficient time, temperature and mixing to achieve complete combustion of the incoming fuel air mixture. If the PZ temperature is around or above 2000 K, dissociation reactions will result in significant amounts of CO in the gases. If the residence time of the fuel-air mixture in the PZ is too short, appreciable amounts of UHC and CO will be present due to incomplete combustion. These products are major polluting species and also absorb a lot of the heat released.

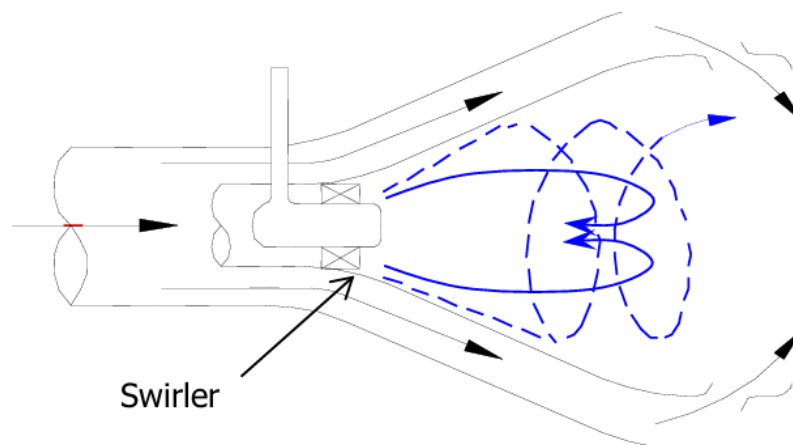
To provide additional time and space for the burnout of CO and UHC, the secondary zone succeeds the PZ. The right conditions are created in the SZ by adding extra air to reduce the temperature, while not quenching the combustion.

At low altitude operation (higher pressure), the SZ needs to recover dissociation losses and complete the burning of any leftovers from the PZ. Some of the soot that is produced in the PZ is also oxidized in the SZ.

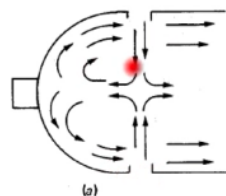
At high altitude operation (lower pressure) the reaction rate in the PZ is lower due to the lower pressure and so the SZ needs to act as an extension to the PZ allowing for higher residence time. The remaining air is admitted into the dilution zone, mixing it with the core air. This is done to reduce the temperature of the outlet stream and make its temperature pattern acceptable to the turbine.

Another important element is the fuel injector. Fuel can be in gaseous form or in liquid form. A common method of supplying a gaseous fuel into combustion chamber is forcing it through a specially designed orifice. For liquid fuels it is more complicated since the fuel needs to be vaporized through atomization, otherwise the combustion will occur heterogeneously around the big droplets of fuel. The fuel injectors used for liquid fuels like kerosene are therefore called **atomizers**.

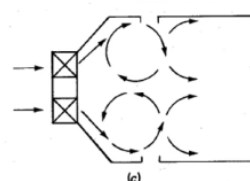
Air leaves the compressor at a speed on the order of 150 to 200 [m/s] for the case of an industrial gas turbine or a large aircraft engine. The velocity is reduced about twice in the combustor diffuser. It is further decelerated down to 25 to 35 [m/s] in the primary zone of the liner. However the flow velocity still remains far greater than the speed of the flame. To prevent the flame from being blown away, it is therefore necessary to set up local regions with much smaller velocity. The most common solution is to generate a flow reversal in the PZ, which can be obtained by imparting swirl to the flow. Recirculation is created in the core region of the swirl where the amount of rotation is high.



Somewhere near that boundary, the velocity assumes the value that matches the value of the flame speed. This is the region where the flame “anchors”. **Swirlers** also contribute to fuel-air mixing and provide effective ways to control the stability and intensity of combustion. Other ways to supply flame stabilization are:



Stabilization by opposing jets



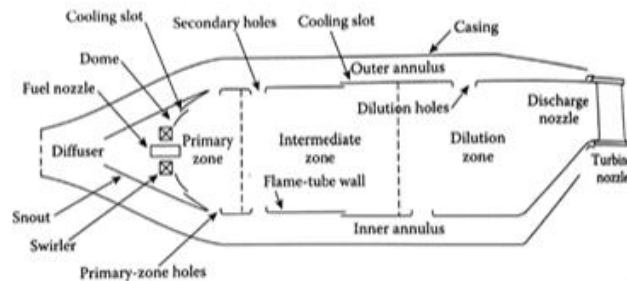
Stabilization by swirling flow assisted by secondary injection

The stabilization by swirling flow assisted by secondary flow injection is the most common solution and it is a mix of the two previous solutions.

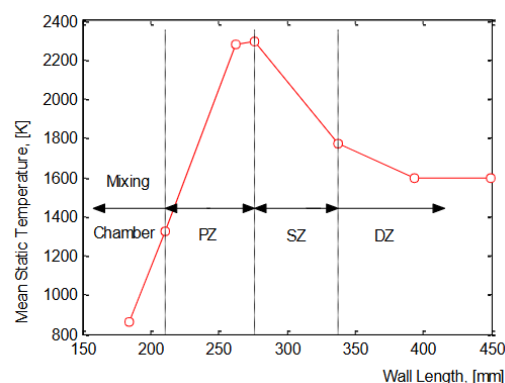
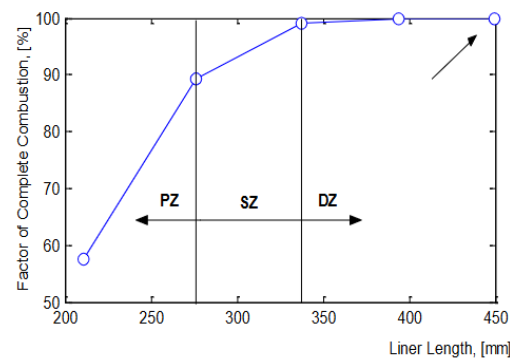
The combustor liner is exposed to very high temperatures during gas turbine operation. To ensure the life of the liner, it is necessary to remove the heat transferred to it and prevents contact with the hot combustion gases. An efficient way to do this is **film cooling**. This technique employs holes through which air is injected in the axial direction along the inner surface of the liner wall to provide a protective cooling film. This film is being gradually destroyed downstream. Therefore, normal practice is to provide a succession of cooling-hole bands along the length of the liner.

In summary, the incoming airflow is split into two flows since there is too much air for a complete combustion; 20-25% of the air goes into the core while the rest flow in the annulus. The core air then enters the diffuser where it is slowed down. Fuel is then injected into the core flow and the swirler aid with mixing. This all occurs in the primary zone. The flame stay in between the primary and intermediate zone. Meanwhile the liner walls are cooled by the secondary flow that enters through small holes and forms a protective film.

Then in the dilution zone the rest of the air is mixed to ensure the correct temperature profile to the turbine inlet.



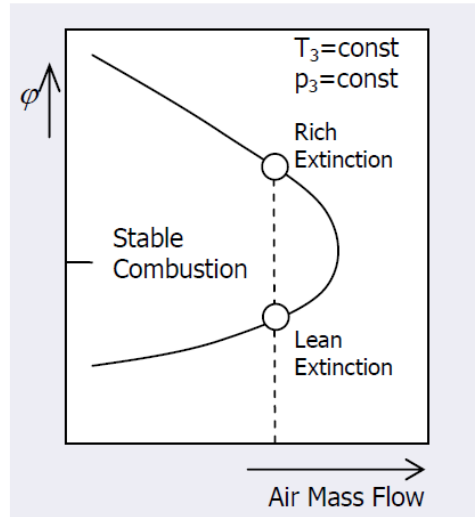
The evolution of combustion and the temperature variation in a typical combustion chamber are plotted in the graphs below.



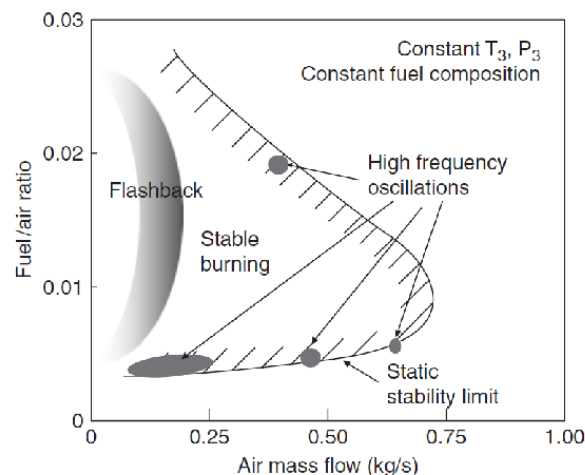
Most of the temperature rise occurs in the PZ, since that's where the flame sits. Then in the SZ and the DZ temperature decreases again since secondary flow is being mixed with the core flow.

Flammability limits

Gas turbine combustion is supposed to be a self-sustaining continuous process under broad operating conditions. We can define a region of equivalence ratio and air mass flow (or air velocity) outside of which combustion cannot take place.



We see that as the flow velocity increases the equivalence ratio range over which combustion can occur decreases. The curve is defined by a **rich extinction limit** and a **lean extinction limit**. However, a combustor must operate well within these limits, since there are many effects we want to avoid close to the boundaries.

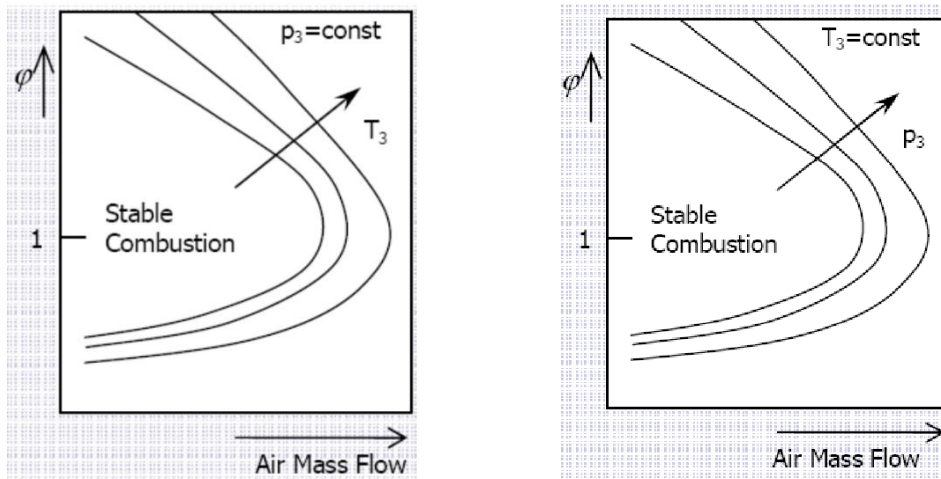


Flashback is a phenomena in which the flame propagates upstream and can only occur at low velocity flow. This can greatly damage the combustor. Near the limits high frequency oscillations in the flame occur, which can cause a flame-out. These acoustic oscillations then lead to unsteady heat release, which in turn causes flow perturbation that cause acoustic oscillations. This cycle continues and it creates pressure fluctuations which decrease the combustors efficiency and structural vibrations. This is what we refer to with **combustor instability**.

Just like for the adiabatic flame temperature, inlet temperature and pressure have an effect on these limits. As we increase these two properties the limits move to the right, increasing the possible operating range of the combustor.

Higher pressure and higher inlet temperature both make ignition and reaction kinetics faster and easier, so mixtures that were previously too lean or too rich can now sustain a flame.

This is particularly advantageous during take-off, when the pressure is much higher. The critical condition is relight at cruise, since pressure and temperature are both very low.



If we look at the below chart for example we see that engines tend to have overall equivalence ratios that are outside of these limits, especially at take-off. Therefore the design of the combustion chamber needs to endure higher local equivalence ratios so that we stay in the flammability limits. Also bigger engines have higher overall equivalence ratios since they operate at higher altitudes.

Engine Family	Aircraft	Take-Off Thrust, [kN]	Overall AFR at TO	Overall ϕ at TO
CFM56-7	B737 NG	91.6	54.0	0.27
RB211-535	B757	163.3	52.3	0.28
CF6-80E1	A330	297.4	49.3	0.30
PW4000-112"	B777	396.6	43.1	0.34

The stability of the flame is also a function of other combustion parameters such as:

- **Ignition delay:** the time between when a fuel–air mixture is exposed to ignition conditions and when rapid heat release (combustion) actually begins. The delay increases with inlet temperature.
- **Activation energy:** the minimum energy the reactants must acquire for a chemical reaction to proceed at an appreciable rate. As the equivalence ratio increases and approaches 1, the activation energy decreases. Varies depending on the type of fuels.

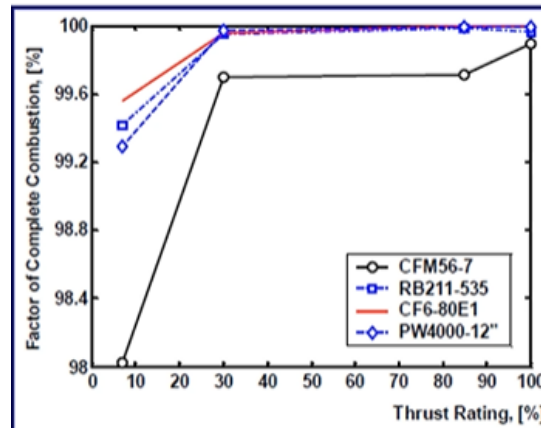
Combustor efficiency

Since the purpose of the combustor is to provide heat to the flow, the combustor efficiency can be defined as the ratio between the actual enthalpy rise and the ideal enthalpy rise, where the enthalpy rise is equal to the heat added to the system since the combustion is isobaric.

$$\eta_b = \frac{\bar{c}_p \left[(\dot{m}_a + \dot{m}_f) T_{t4} - \dot{m}_a T_{t3} \right]}{\dot{m}_f \cdot h}$$

Typically combustor efficiencies are very high and tend to increase with pressure and temperature due to the increased reaction rate (for the same residency time, combustion is more complete). The opposite is true for the Mach number, since this decreases residency time.

Since the design point of the combustor is the cruise condition we see that the efficiency tends to increase as we increase the thrust generated.



The main losses in the combustor come from the pressure losses that occur due to friction, mixing, and heat addition as well as dissociation that occurs at high temperatures. Other factors are incomplete combustion and heat losses.

Pressure loss due to friction is a major contributor especially in the DZ since the flow is accelerating back up again.

Let's now summarize all of the characteristics of an actual aero engine combustor:

- Primarily turbulent combustion since the velocity of the flow is still very high even though the flow is slowed down significantly
- External ignition is required to light the combustion, then combustion is self-sustained unlike in automotive engines
- Constant pressure distribution (very small pressure losses)
- Pressure losses penalize performance and have to be limited
- Low relative volume and weight
- Fast volume and heat dynamics
- Low residence time (in the order of a couple of milliseconds)
- Adiabatic combustion (only very small heat losses that penalize the performance)
- High pressure and inlet temperatures (40 (large engines)-60 (smaller engines) bar and $T > 900\text{K}$)

Note: gas turbines for power generation operate at lower pressures ($< 30\text{bar}$) and lower temperatures ($< 750\text{K}$), while micro turbines operate at very low pressures (3-5 bar) and quite high temperatures ($T \sim 1000\text{K}$).

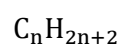
As fuels the aviation industry mainly uses liquid kerosene although different fuels are being studied both in liquid form and gaseous form.

We also observe that there are different types of combustor types:

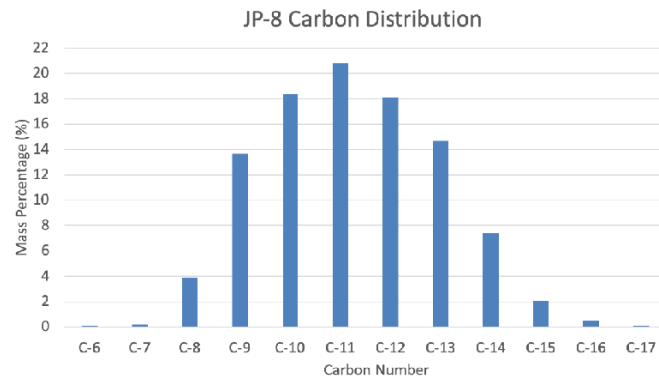
- Conventional
- RQL, lean burn, staged combustor, etc...
- Alternative technologies
- Use of catalysts and inhibitors (not currently used, active area of research)

Fuel

The fuel used by aviation is kerosene, which is a product of the distillation of crude oil together with gasoline, diesel and many other petrol derivatives. Kerosene is a mix of tens of different compounds, whose relative concentration vary depending on the use cases of the fuel. The compounds with the highest concentration are **alkanes** (paraffines) which have the following chemical formula:



Aromatics are also present with very small concentrations to reduce the soot formation in the combustion chamber. Many other chemical species are present and their concentrations are specifically determined to obtain the specified requirements. The most common carbon chains present in kerosene are the C-11 and C-10/C-12 chains as we can see from this distribution.

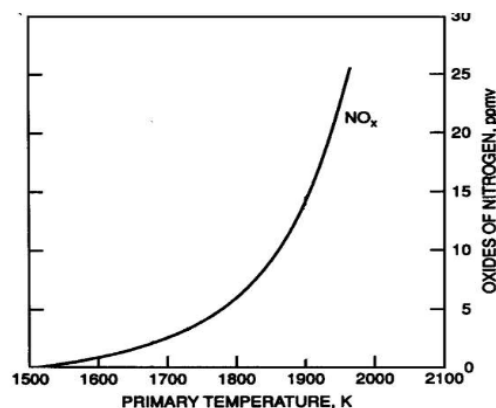


Engine emissions

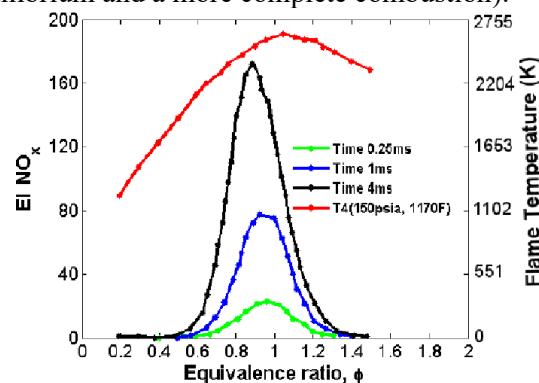
Engine emissions are strictly regulated for the Landing and Take-off cycle LTO. This cycle is made up of 4 phases: the approach, the taxi, the take-off and the climb. The cruise condition is not regulated. This is mainly to reduce local air pollution near airports. All of the products of the combustion, such as carbon monoxide, nitrogen oxides, smoke and unburned reactants are highly harmful for human beings.

The component of the emissions that is most regulated are the NO_x emissions since they can have the most dangerous effects.

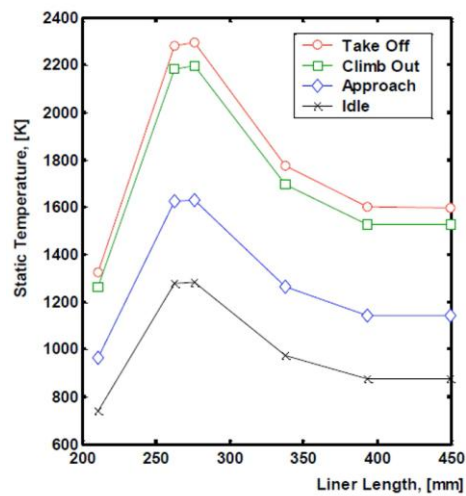
There are many different ways that nitrogen oxides can form, however we focus on **thermal NO_x**, which is created inside the combustion chamber especially at high temperatures.



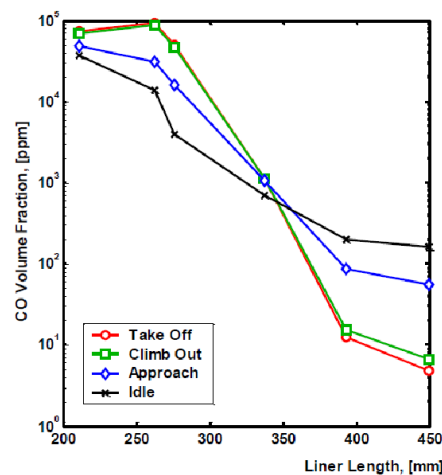
Factors that influence the NO_x emissions are all the factors that influence the adiabatic flame temperature, since the emissions increase exponentially after a certain temperature is surpassed. Also residence time plays a role (higher residence time corresponds to higher NO_x emissions, since the reaction shifts towards equilibrium and a more complete combustion).



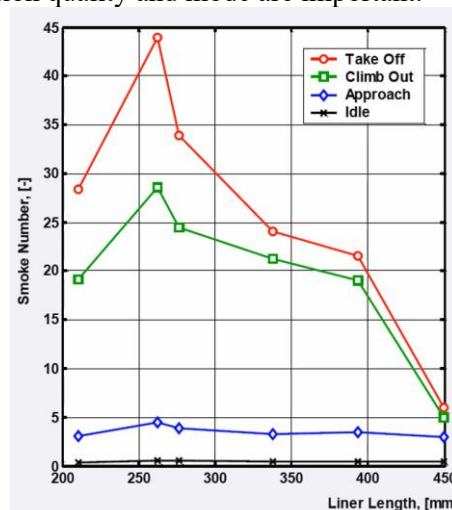
Even though the peak of the adiabatic flame temperature is slightly on the rich side, we see that peak emissions are slightly on the lean side. This is because on the rich side the availability of oxygen is less. We can see that different temperatures in the combustor are reached for different phases of the mission:

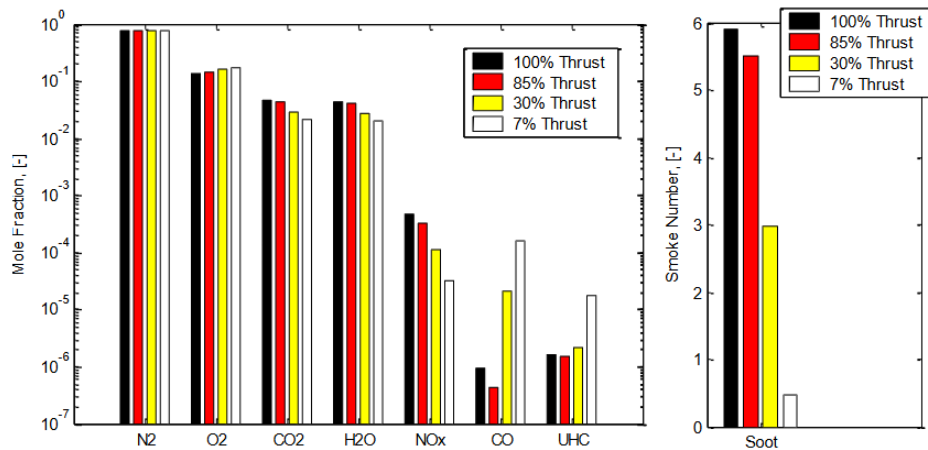


The CO production is opposite of the NOx production: it is highest for idle conditions and lowest for take-off and climb.



Soot emissions seem to follow the same trend as NOx emissions. The controlling factors are the fuel to air ratio, pressure, temperature, quality of fuel injection and mode of the fuel injection. The equivalence ratio is particularly important for the soot emissions (soot is formed mainly in rich conditions), therefore the injection quality and mode are important.





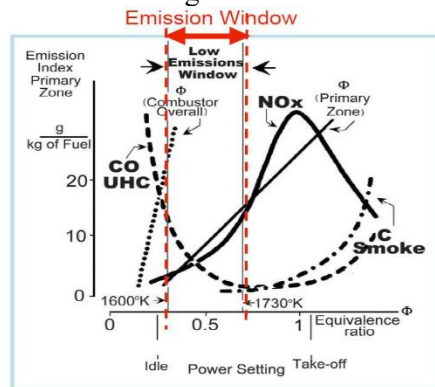
The main emissions from aero engines are molecular nitrogen and molecular oxygen since the combustion typically occurs in lean conditions, some of the air does not participate in the combustion process.

For higher thrust settings the amount of CO₂, H₂O, NO_x and soot produced are higher since more fuel is being burned, while the oxygen concentration decreases since it's being used up. For the CO₂ and H₂O it's because the amount produced per kg of fuel burned stays pretty much constant. CO emissions follow the opposite trend since it is formed at lower temperatures (it cannot be oxidized into CO₂) and is also a product of incomplete combustion (at lower temperatures the reaction is further from being a complete one).

Unburned hydrocarbons are highest when the thrust is minimum.

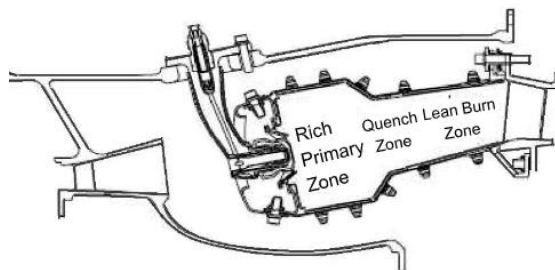
Note: UHC and CO behave similarly, as do NO_x and soot emissions.

By superimposing all of these trends we get a pretty small low emissions window around low equivalence ratios. That is why all of modern engines have an overall lean equivalence ratio.

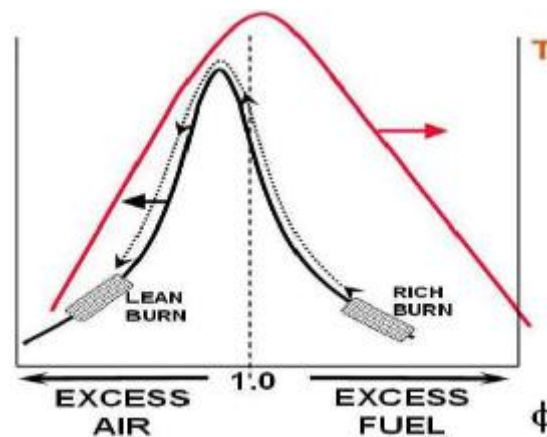


RQL

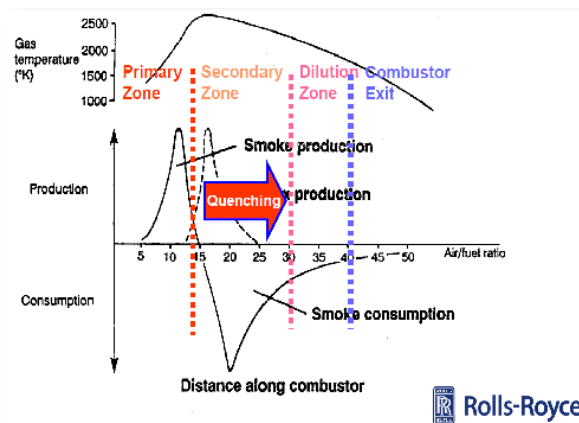
Rich burn-Quick quench-Lean burn engines are a combustor architecture used to reduce the NO_x emissions.



The equivalence ratio is kept quite rich in the PZ (close to 2), which leads to relatively low temperatures in the PZ. This is followed by a quench zone where a lot of air is added from outside the liner, leading to a last zone of lean burn. This way the region where NO_x emissions are very high is avoided by avoiding the very high temperatures that occur at stoichiometric FARs, without sacrificing too much the efficiency. In the figure below we can see that NO_x emissions (in black) are very low at high and low equivalence ratios. The quench process occurs so rapidly that the existence of near-stoichiometric solutions is kept to a minimum, therefore the temperature does not surpass the threshold for NO_x generation.



Most of the modern engines use the RQL technology. A big amount of soot is produced in the rich burn zone since the equivalence ratio is high, however it is usually oxidized in the SZ and DZ.



TAPS

A **Twin Annular Premixing Swirler** is an alternative technology used by GE for their combustors. In this case the primary zone is already lean since almost 70% of the incoming airflow goes in the core of the combustor. This is a huge difference compared to the RQL combustor where only 25% of the air actually goes in the core. Since the amount of air for TAPS combustors is very high there is also a need for very complex swirlers to slow down the flow and stabilize the flame.

Note: we can compare some different characteristics between industrial and aero combustion chambers.

Criteria	Industrial	Aero
Weight & Size	Not critical	Critical
Operating conditions	Sea level	Wider than the flight envelope
No. Of cycles	30,000-50,000	6000-8000
Equivalent Operating Hours (EOH)	100,000-150,000	30,000-40,000
Emission regulations	Very stringent	Stringent
Critical point in design	Lean blowout	Relight
Fuel flexibility	Important	Not critical
Fuel type	Mostly Natural gas	Kerosene
Active air distribution	Possible	Not possible
Residence time	$\cong 10\text{-}30$ ms	$\cong 5$ ms
Combustor Type	Mostly Can Type	Mostly Annular type
Combustion philosophy	Mostly Lean, reheat	RQL/TAPS/ Staged

Lecture 10

Turbomachinery

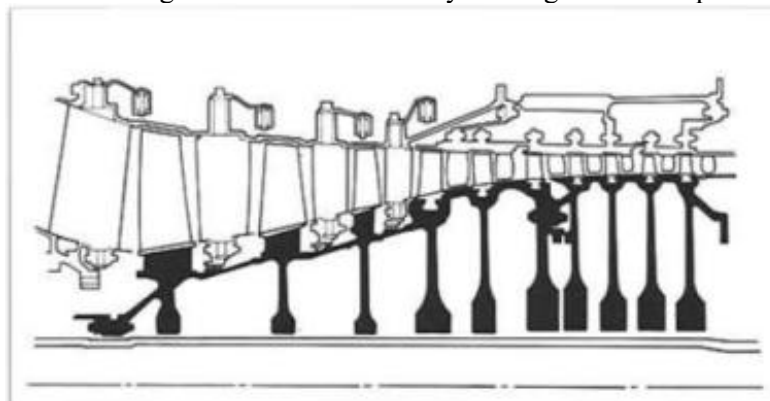
An aero engine is mainly made up of turbomachinery: both turbines, both compressors and the fan are all examples of turbomachinery.

Let's first look at the compressor. There are many different types of compressors and they can be classified into:

- **Mechanical compressors:** they use moving mechanical parts to compress the air. Can be further classified into:
 - **Positive displacement:** compression of the fluid is achieved by compressing the fluid in a closed volume that reduces.
 - **Dynamic:** compression is achieved by doing work on a moving fluid without changing its volume. Can be then classified based on the direction of movement of the fluid:
 - **Axial**
 - **Centrifugal**
 - **Diagonal:** mix of the previous two

Axial compressor

The most used architecture for compressors in aero engines are **axial compressors**. In an axial compressor stages are positioned in succession in order to achieve a high-pressure ratio in a gradual way. Each stage is made up of a **rotor** followed by a **stator**. Mechanical power is used to increase the kinetic energy of the fluid, which is then converted in pressure by a diffusion process. Work is done on the fluid by the rotors, which are connected to the shafts that drive them and provide them the power harvested by the turbine stages. The stator (or diffuser) transforms this kinetic energy into an increase in pressure through diffusion. This part is also used to modify the direction of the fluid and make it as it was before entering the next rotor. This way the stages can be repeated.

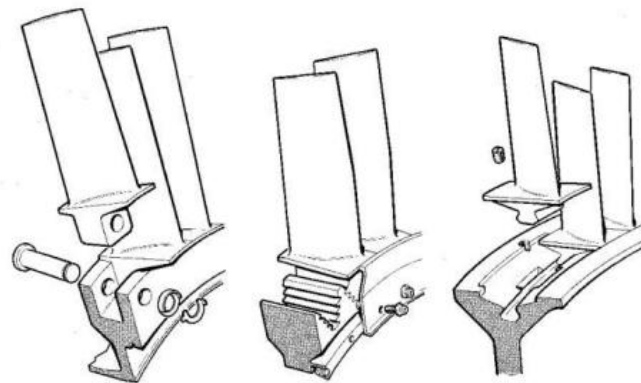


From the figure above we can also see that the stator blades are also connected to a mechanism, which allows them to rotate around their axis. These are called **variable stator blades** and allow the compressor to function even in the harshest operating conditions without stall.

Another thing to note is that the size of the blades decreases progressively as we move through the compressor. This is done to maintain the axial velocity constant (to have constant velocity triangles and therefore a constant loading and work distribution of the compressor stages); since pressure is increasing, density is also increasing, thus to maintain the axial velocity, the cross-sectional area needs to decrease given that the mass flow rate is constant.

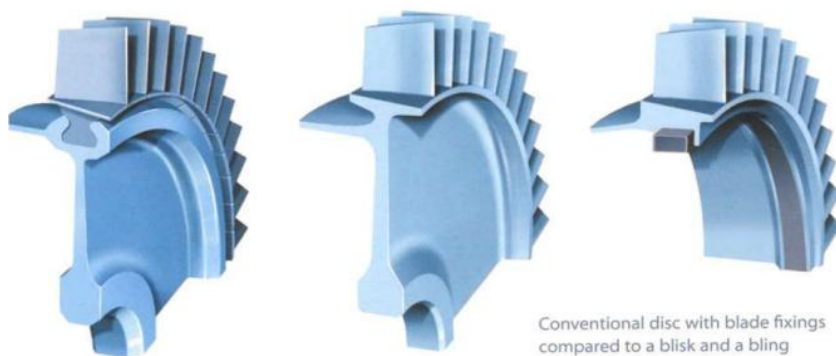
In most cases, before the first stage of the compressor, a IGV **Inlet Guide Vane** is present, which impart a swirl component to the axial flow, aligning it so that the relative velocity the first rotor sees is equal to the relative velocity all the other rotors see.

The root of the blades are made to slide into place around a disk that is connected to the shaft. They are then locked in place through different mechanisms, such as:



a. Pin lock b. Pine tree locks c. Bird tail lock

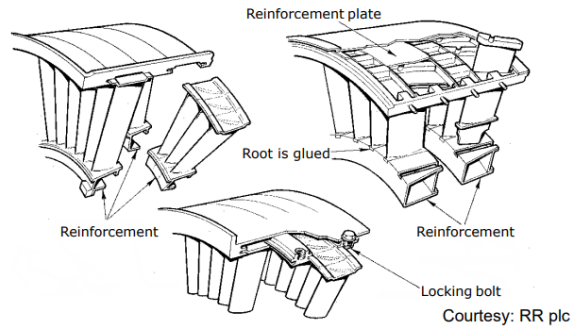
For high performance engines, the blades are machined with the disk so to reduce mechanical complexity and weight. These kinds of disks are called **bladed disks** (also called **blisks**) and are much harder to manufacture, much more expensive and are impossible to repair (they can only be replaced).



Bladed disks are typically never used in civil aircrafts and are only used in some military engines.

For non-variable stator blades, they are anchored both at the root and at the tip. In some cases to save on weight the stator blades are only connected to the outer shell of the compressor and are only glued to the core.

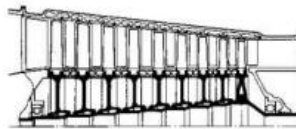
For smaller engines sometimes the saving in cost of only anchoring one side of the stator blades are enough to outweigh the increase in losses due to the presence of a gap.



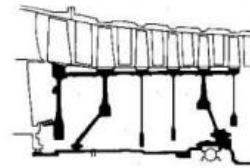
For the connection between the rotor blades and the shaft there are some different solutions:

- **Disk rotor:** generally used only for older engines. A series of individual disks connected together connect the blades of the rotor stages to the shaft. They are easy simple and easy to manufacture but they have poor stress distribution.
- **Drum rotor:** a hollow cylindrical drum is mounted on the shaft and on its surface all of the rotor blades are mounted. The stator blades are mounted on the outer casing and have a certain tip clearance with the drum's surface. They have a better centrifugal stress distribution, but can be heavier than disks.
- **Drum rotor with variable vanes:** in this case the cylindrical surface has indents which allow the presence of the mechanism which holds the VSVs.

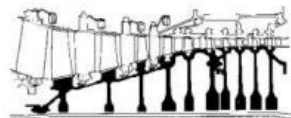
a. Disk rotor



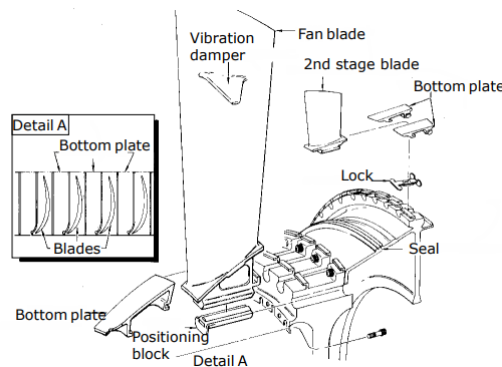
b. Drum rotor



c. Drum rotor with variable vanes



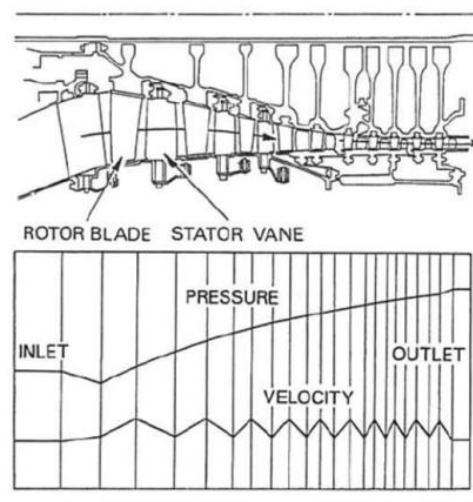
The design of these disks are very important since they are subject to very high centrifugal stresses and can harmfully contribute to the propagation of vibrations that can lead to structural damage. Unlike the compressor blades, the fan blades are hollow inside and have an internal honeycomb structure to reduce weight. Moreover, fan blades typically have airfoil sections and supports towards the tip that allow them to rest on one another, greatly reducing vibrations. These supports are typical of older turbofans and are not used anymore in modern turbofans.



Modern fan blades are much more complex than older fan blades. They have more exotic 3D geometries and are usually made of carbon fibre composites with titanium tips. Additionally, they have very low hub-to-tip ratios and so the flow is dominated by 3D effects. For this reason, together with the fact that each section sees completely different angles of attack, is why their 3D geometry is very important.

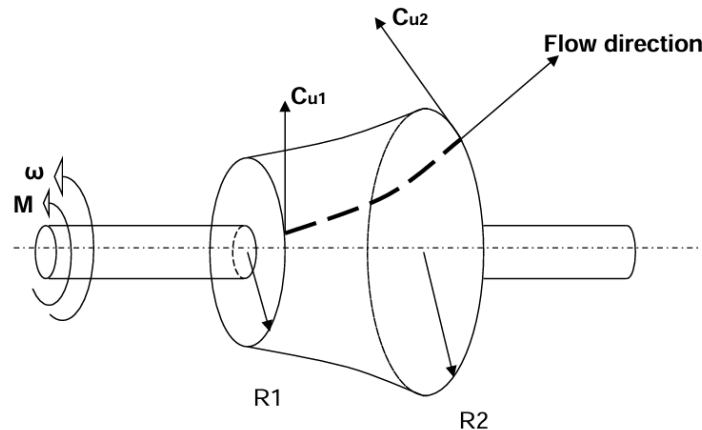
Let's now consider more in depth how an axial compressor functions. For our considerations we assume that there are no 3D effects and that the flow is completely axial.

The variation of pressure and velocity along a generic axial compressor is as follows:



Pressure increases gradually, while the absolute velocity increases in the rotor and decreases in the stator.

We now consider a compressor and a fluid element along its path.



For the whole compressor holds:

$$\text{Power} = P = M \cdot \omega \quad (7.1)$$

$$F dt = d(m c) \quad (7.2)$$

Where c stands for the absolute velocity of a fluid particle. From this follows

$$F = \frac{d}{dt}(m c) = \dot{m} c + m \dot{c} \quad (7.3)$$

Because of the blading arrangements, the flow inside a turbo-machine rotor is unsteady and asymmetric. However, the unsteadiness is periodic so that, on average, we may omit the contribution of the time derivative of the velocity vector to the torque equation.

Assuming $\dot{c} = 0$ it follows

$$F = \dot{m} c \quad (7.4)$$

For the fluid element holds:

$$dF = \dot{m} dc \quad (7.5)$$

$$dF_r = \dot{m} dc_r \quad (7.6)$$

$$dF_u = \dot{m} dc_u \quad (7.7)$$

As the radial component gives no contribution to the torque M , we obtain:

$$dM = r dF_u \quad (7.8)$$

$$dM = r \dot{m} dc_u \quad (7.9)$$

The torque required for the compression is:

$$M = \dot{m} \int_1^2 r dc_u \quad (7.10)$$

From which follows that

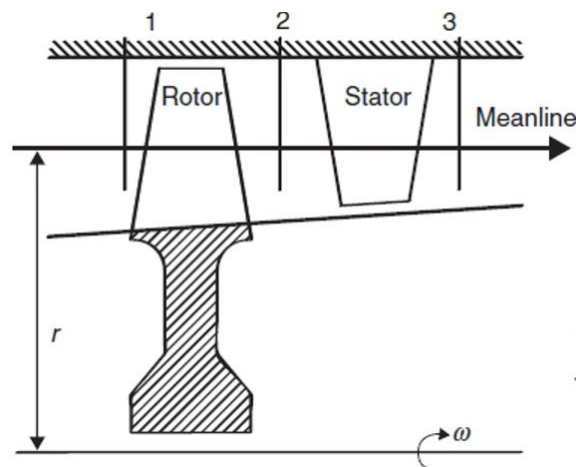
$$M = \dot{m} (c_{u2} r_2 - c_{u1} r_1) \quad (7.11)$$

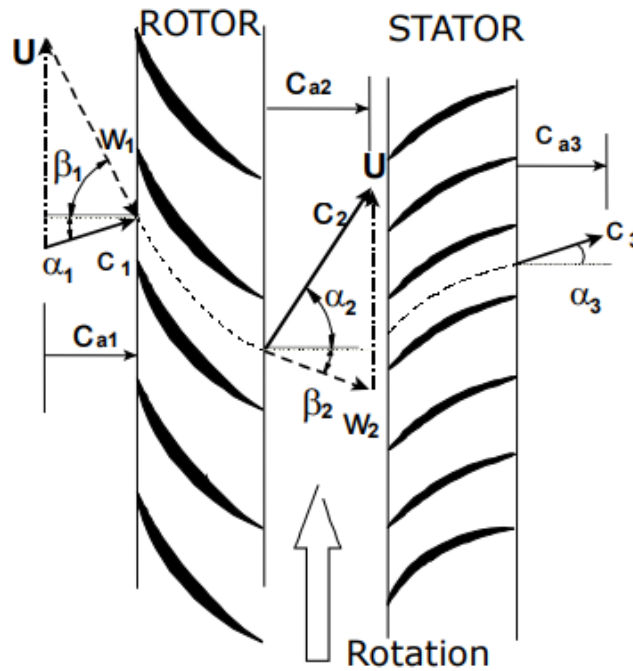
$$P = M\omega = \dot{m} (c_{u2} r_2 - c_{u1} r_1) \omega \quad (7.12)$$

Using $\omega = u_1 / r_1 = u_2 / r_2$, we obtain the **Euler's Equation for Turbomachinery**:

$$P = \dot{m} (c_{u2} u_2 - c_{u1} u_1) \quad (7.13)$$

Euler's formula is used to evaluate the power needed by a compressor stage. This can be applied to all types of turbomachinery, including centrifugal compressors and turbines. The power can also be expressed as the total enthalpy jump across the compressor stage. We can study a single stage by looking at a cross-section along its mean line:

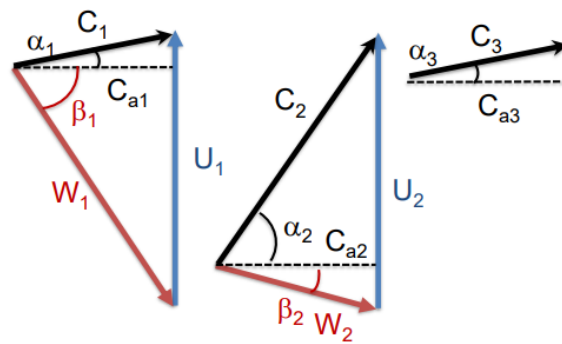




Note the direction of rotation w.r.t. the camber of the airfoils, since work is being done on the flow. To study turbomachinery blades **velocity triangles** are used, which represent the following velocity relation:

$$W + U = C$$

Where U is the rotational speed of the blade section we're considering, W is the flow relative velocity to the blade and C is the flow's absolute velocity. For an axial compressor:



We assume that the angle of the relative velocity of the flow leaving a blade is the same as the angle of the trailing edge of the blade. Also to reduce losses, we assume the stages are designed so that the incoming velocity (relative for the rotor and absolute for the stator) is parallel to the leading edge of the blade.

To repeat stages the absolute velocity coming out of the stator needs to be equal to the incoming absolute velocity to the rotor.

Looking at velocity triangles of the rotor and assuming that the axial velocity has the same value throughout, we obtain:

$$\begin{aligned} u = c_{u1} + c_{w1} = c_a (\tan \alpha_1 + \tan \beta_1) &= c_{u2} + c_{w2} = c_a (\tan \alpha_2 + \tan \beta_2) \Rightarrow \\ \Rightarrow (\tan \alpha_1 + \tan \beta_1) &= (\tan \alpha_2 + \tan \beta_2) \end{aligned} \quad (7.14)$$

$$P_s = \dot{W}_s = u(c_{u2} - c_{u1}) \quad (7.15)$$

$$P_s = \dot{W}_s = u(c_{u2} - c_{u1}) \quad (7.15)$$

$$c_{u1,2} = c_a \tan \alpha_{1,2} \Rightarrow \dot{W}_s = u \cdot c_a (\tan \alpha_2 - \tan \alpha_1) \quad (7.16)$$

$$c_{u1,2} = c_a \tan \alpha_{1,2} \Rightarrow \dot{W}_s = u \cdot c_a (\tan \alpha_2 - \tan \alpha_1) \quad (7.16)$$

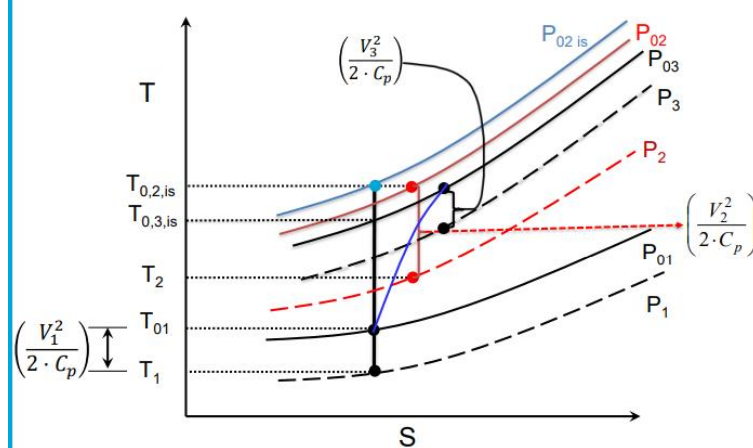
$$(\tan \alpha_1 + \tan \beta_1) = (\tan \alpha_2 + \tan \beta_2) \quad (7.17)$$

$$(\tan \alpha_1 + \tan \beta_1) = (\tan \alpha_2 + \tan \beta_2) \quad (7.17)$$

$$P_s = \dot{W}_s = u \cdot c_a (\tan \beta_1 - \tan \beta_2)$$

$$P_s = \dot{W}_s = u \cdot c_a (\tan \beta_1 - \tan \beta_2)$$

From the T-S diagram of a compressor stage we have:



The total temperature increases from 1 to 2 since work is being done on the fluid, while it stays constant between 2 and 3 since no heat or work is exchanged.

■ Increase in total temperature per stage: $T_{0,3} - T_{0,1} = T_{0,2} - T_{0,1} = \frac{W_s}{C_p}$

■ Increase in total temperature per stage: $T_{0,3} - T_{0,1} = T_{0,2} - T_{0,1} = \frac{W_s}{C_p}$

■ Pressure ratio per stage:

$$\frac{p_{0,3}}{p_{0,2}} = \left(1 + \eta_{is} \frac{\Delta T_{0,1-3}}{T_{0,1}} \right)^{\frac{k}{k-1}} =$$

$$= \left(1 + \frac{\eta_{is} \cdot \lambda \cdot u \cdot c_a (\tan \alpha_2 - \tan \alpha_1)}{c_p T_{0,1}} \right)^{\frac{k}{k-1}}$$

86

do not allow extremely high rotational velocities; we also would like to have high axial flow velocity, but flow separation and losses limit this term; we would like to have a high $\tan \alpha_2 - \tan \alpha_1$ (which is proportional to how much the rotor blades turn the flow), but there is a limit to the blade curvature caused by flow separation problems.

Typically diffusion takes place in both rotor and stator, so we can define a **degree of reaction**, which provides a measure of the extent to which the rotor contributes to the overall static pressure rise in the stage. It is normally defined as:

■ Definition of degree of reaction:

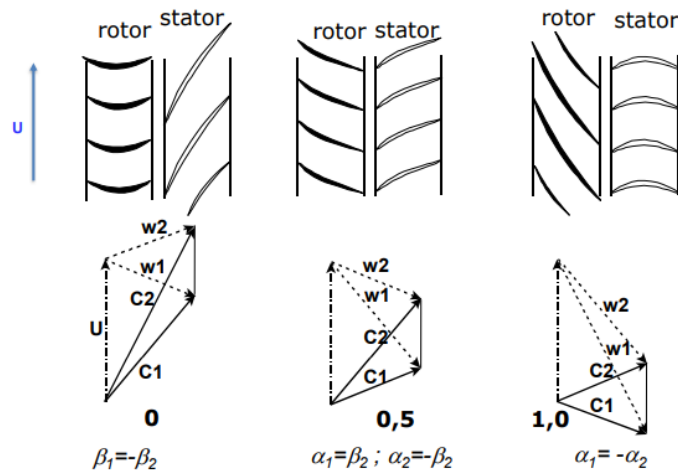
$$\Lambda = \frac{\text{static enthalpy rise in the rotor}}{\text{static enthalpy rise in the stage}}$$

$$\Lambda = \frac{\Delta T_{\text{static rotor}}}{\Delta T_{\text{static rotor}} + \Delta T_{\text{static stator}}}$$

■ So for an axial compressor:

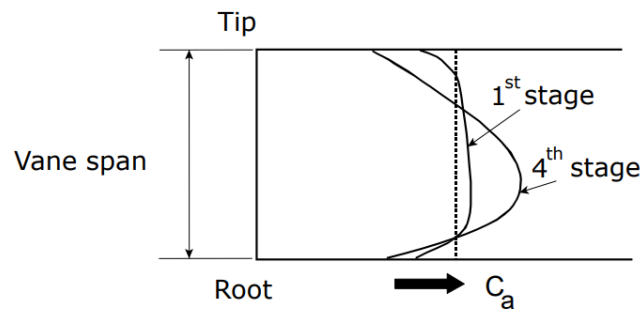
$$\Lambda = \frac{c_a}{2u} (\tan \beta_2 + \tan \beta_1) = 1 - \frac{c_a}{2u} (\tan \alpha_1 + \tan \alpha_2)$$

For degree of reaction close to zero, the rotor (called a “**bucket**” rotor or **impulse rotor**) is only increasing the velocity of the flow, while all of the diffusion occurs on the stator, vice versa, for degree of reaction of 1 all of the diffusion occurs in the rotor. Typically we want a degree of reaction of around 0.5 for a more gradual pressure increase across the compressor. The higher is the degree of reaction, the more is the amount of energy increased in the rotor and so the higher is the temperature rise in that part of the compressor. Different values of the degree of reaction are responsible for a different shape of the vanes.



All the considerations that have been made refer to an imaginary plane section of the blade. The velocity triangles have been drawn in a tangential plane at the mean radius. At the front end of the compressor the blades are much longer, so there are marked variations in blade speed from root to tip, and it becomes essential to consider 3D effects in analysing the flow. Moreover, in the real condition, the axial velocity changes between root and tip of the blades due of the presence of the boundary layer along the annulus walls. This effect becomes more evident in the later stages, because the boundary layer thickens as the flow progresses, and the area available for the flow is reduced below the geometric area.

Also the clearance between the blade tips and the blades themselves is constant in absolute value. However, the chord of the later stages is smaller, so the space in between is relatively bigger and tip effects losses are greater.

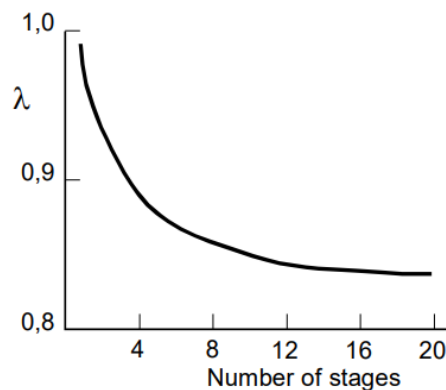


A so called “**Work-Done**” factor is introduced to account for the reduction in work capacity caused by the change in axial velocity. Integrating the specific power along the blade height and using the “Work-Done” factor λ , we obtain:

- Integrate specific power along the blade height → “work done”-factor, λ

- Power:

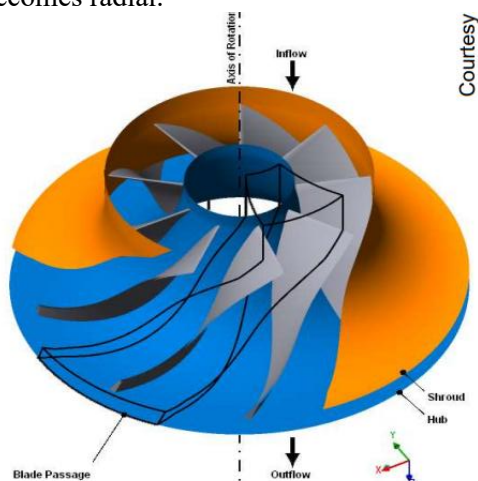
$$\dot{W} = \lambda \cdot \dot{m} \cdot W_s$$



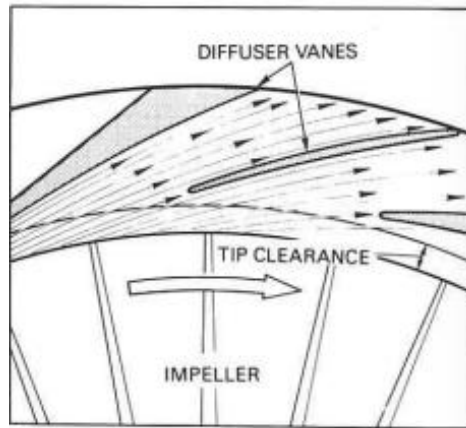
As the number of stages increases the work being done on the fluid is being wasted more and more on losses and increasing the total temperature of the fluid instead of the total pressure.

Centrifugal compressors

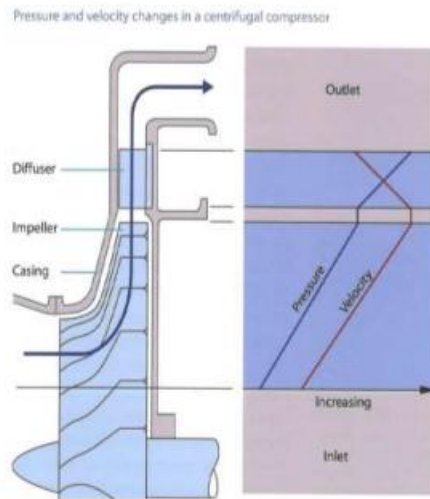
Centrifugal compressors, unlike axial compressors, change the direction of the incoming axial flow. The flow, which is initially parallel to the axis of rotation, leaves the compressor in a radial manner. The first part of the compression occurs due to work being done on the flow by the compressor blades, however most of the compression actually occurs due to the centrifugal forces that act on the fluid particles when the flow becomes radial.



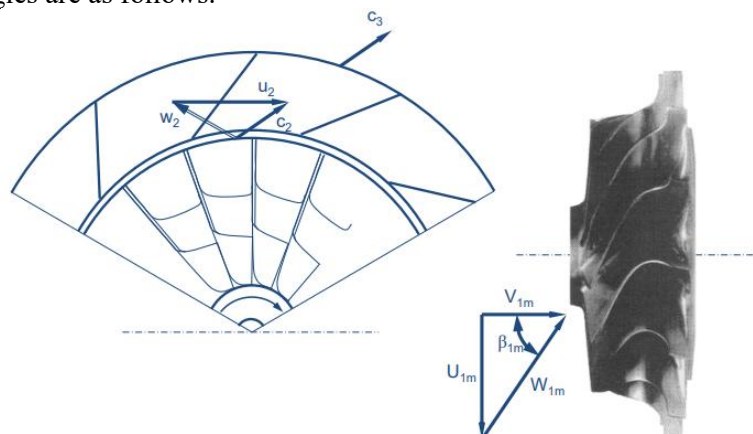
When the flow comes out of the **impeller** (rotating component) it goes into a **radial diffuser** which slows down the fluid and further increases pressure.



Similarly to an axial compressor, in the rotor the flow's velocity and pressure are increased thanks to the work being done on the fluid, while in the stator the velocity decreases as the pressure increases. However, the work in this case is mainly being done by centrifugal forces.



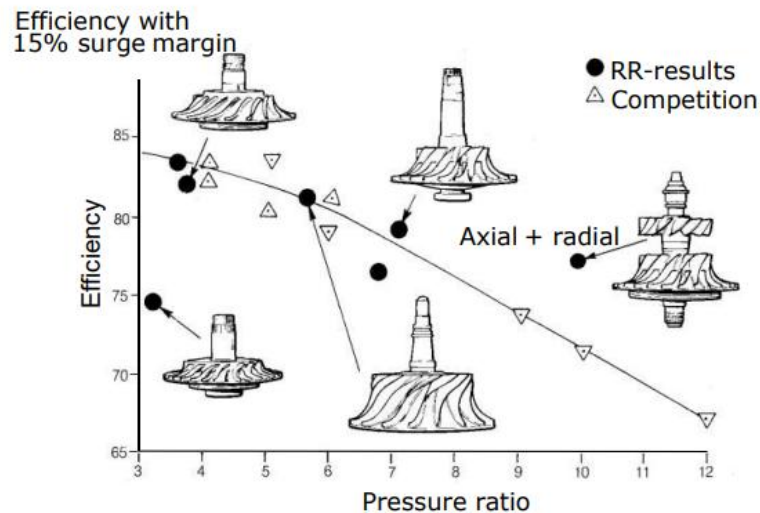
The velocity triangles are as follows:



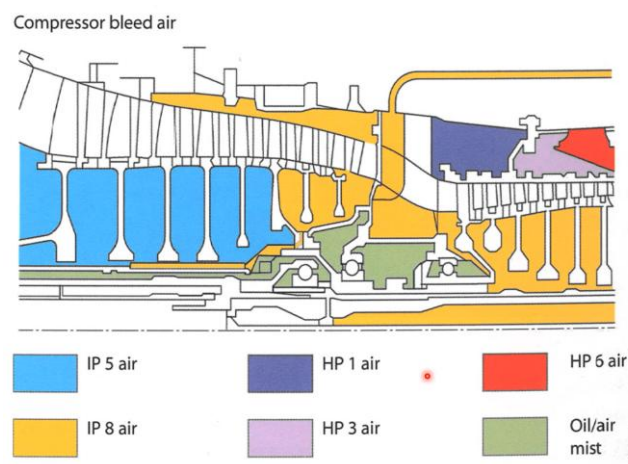
Centrifugal compressors are almost never used in civil aircraft engines but are sometimes used for turboshafts for helicopters or APUs.

The advantages of the centrifugal compressors is that have very high compression ratios for a single stage. Typical pressure ratios for modern centrifugal compressors are around 7-8 for one stage.

Therefore, if the required compression ratio is low, radial compressors are lighter and smaller than axial compressors, however after a certain compression ratio axial compressors become much more efficient from a space and weight point of view. Radial compressors also have the issue that they have a much larger frontal area, while axial compressors can achieve very high pressure ratios without ever changing their frontal area.



Note: **compressor bleed air** can be up to 20% of the total mass flow rate through the engine. This bleed air can be used for many functions like: cabin environmental control, tip clearance control, thrust balancing, etc. One of the main functions of the compressor bleed air is to pressurize the cavities in that are present in the engine. This is done so that the oil present in the shaft, which is at extremely high temperatures, does not get inside the engine and stays inside the shaft. Since it's always at a high temperature it can evaporate and the mist produced is highly inflammable. This mist is typically expelled at the back of the engine.



The bleed air taken from later stages is much more expensive from an energy POV, therefore typically we try to use lower pressure bleed.

Compressor characteristics

For any gas compressor we can express the dependence of the total pressure at the compressor outlet p_{02} and adiabatic efficiency on the other important variables:

$$\Delta h_{is}, P = f(\dot{m}, \Omega, k, R, \nu, \rho, design, D, a)$$

The relevant gas properties are specified by the kinematic viscosity ν , the density ρ , the specific heat ratio k , and the gas constant R . Design is the complete specification of the geometric shape of the machine and D is the characteristic size, while a is the speed of sound. Δh is the enthalpy variation across the compressor and P is the output of the compressor.

Dimensional analysis can be used to simplify the representation of these relations. This is a method which combines the variables involved to form a smaller and more manageable group of dimensionless variables.

The first step is to check that all the variables on the right-hand side of the relation are actually independent. The second step is to reduce the relation to one in which each dimensionless dependent variable is expressed as a function of a lower number of independent dimensionless variables.

By following these steps we get:

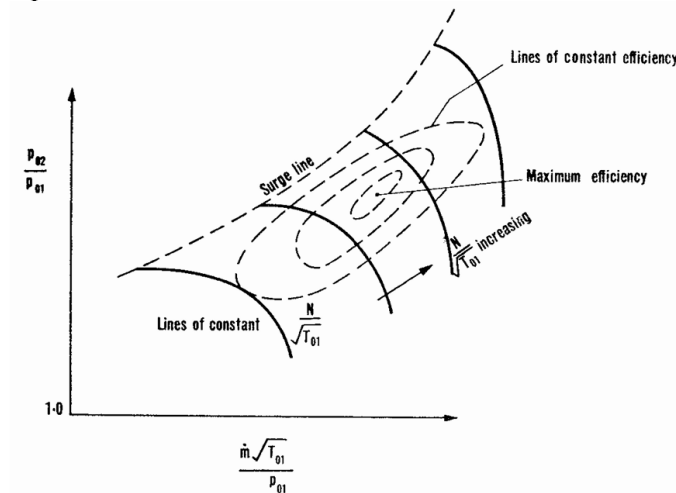
$$\frac{p_2}{p_1}, \frac{\Delta T}{T} = f \left[\frac{\dot{m} \sqrt{RT_0}}{D^2 p_0}, \frac{\omega D}{\sqrt{RT_0}}, Re, \kappa \right]$$

Pressure ratio Π
Efficiency η
Mass flow $\dot{m}$
Shaft speed ω

Radius and diameter can all be taken out this relations since they are constant. From this relation we can see that the pressure ratio and temperature ratio (directly related to the efficiency of the compressor) in the compressor can be expressed as a function of the non-dimensional mass flow rate, the non-dimensional rotation and the flow properties (Re and heat capacity ratios). Assuming constant flow properties we get:

$$\frac{p_2}{p_1}, \frac{\Delta T}{T} = f \left(\frac{\dot{m} \sqrt{RT_{0,1}}}{D^2 p_{0,1}}, \frac{\Omega D}{\sqrt{RT_{0,1}}} \right)$$

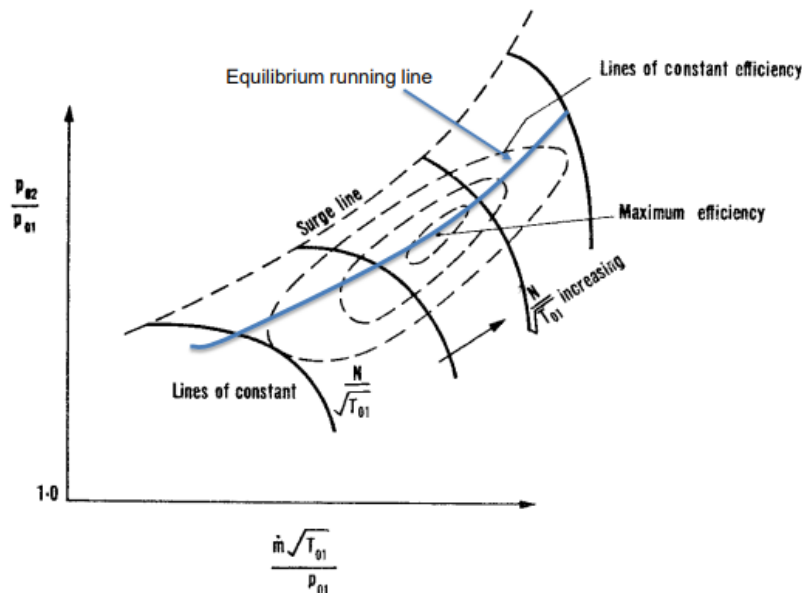
We can represent this relation in a diagram called the **compressor map**. This “map” represents the performance of the compressor.



The compressor characteristic shows the most important quantities that influence the performance of the compressor. These quantities are grouped in dimensionless parameter groups:

- a) $\frac{\dot{m} \sqrt{RT_{0,2}}}{p_{0,2} D^2}$; mass flow parameter group
- b) $\frac{p_{0,3}}{p_{0,2}}$; the pressure ratio
- c) $\frac{ND}{\sqrt{RT_{0,2}}}$; shaft speed parameter group
- d) η ; efficiency (isentropic or polytropic)

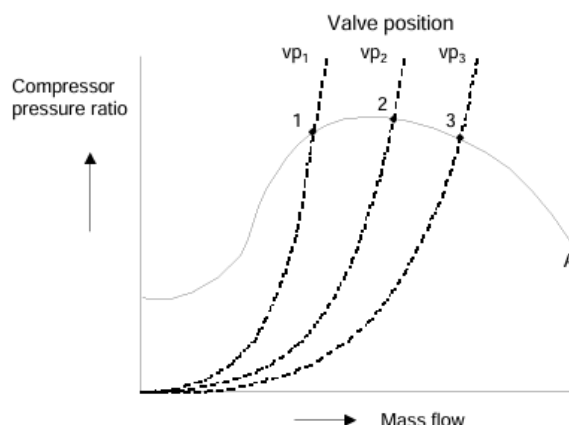
The design point is typically close to the maximum efficiency region. By changing the rpms of the compressor, the operating point shifts along the blue line beneath called **equilibrium line**. This design operating line needs to be distant enough from the surge line. This distance is called the **surge margin**.



Operational limits

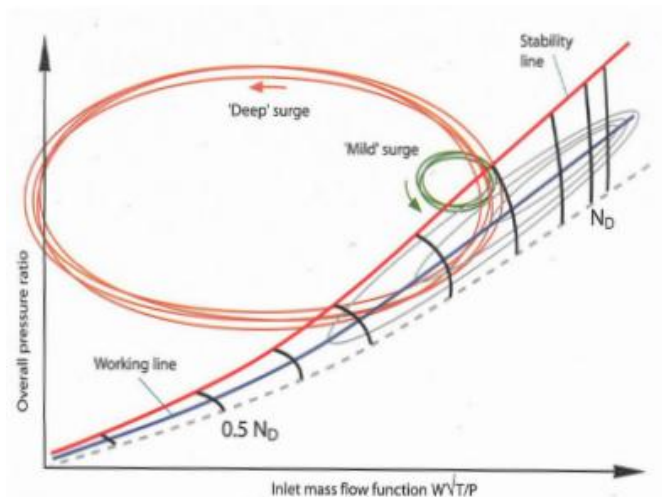
Surge

To explain “surge”, we assume a compressor running at a fixed speed having an adjustable valve at the exit channel. By adjusting the position of the valve, the mass flow and the pressure ratio over the compressor can be controlled. The pressure ratio whats can be plotted as function of the mass flow rate by slowly adjusting the valve positions. By opening the valve we get a curve similar to the “curve A”. When the valve is opened slowly, the pressure ratio will increase at first since the stator vanes’ contribution to the pressure ratio will increase and the inflow conditions to the blades will be better. This will continue until an optimum is reached (at operating point 2), after which the pressure ratio will decrease because the efficiency decreases as a result of an increasingly more negative angle of incidence on the blades.



The curve enclosed by points 2 and 3 is considered to represent stable operation of the compressor in contrast to the “unstable” curve up to point 1. The curve before point 1 is unstable because a disturbance can cause the mass flow to decrease, which would result in a pressure drop at the compressor outlet. This decrease in pressure will result in an adverse pressure gradient at the compressor outlet since the pressure in the next component (the combustion chamber) does not decrease immediately. This adverse pressure gradient slows down the flow and results in a further

decrease in mass flow rate. This continues until global flow reversal occurs in the compressor. In the mean time, since the compressor is not working anymore, the pressure downstream of it also drops, facilitating the compressor to pick up again, repeating the whole cycle. This whole cycle creates high frequency oscillations and it's called **compressor surge**. From the figure below the “operating points” of a compressor in surge follow the red cycle, going from **mild surge** (no flow reversal) to **deep surge** (global flow reversal) again and again. The difference with the stall phenomenon is that the stall is a local disturbance of airflow with local flow reversal, while surge is characterised by a flow reversal of the entire compressor. However, the two are related as typically surge is preceded by stall. The position of the surge line is highly dependent on the dynamic character of the surge phenomenon and is not only compressor dependent but also depends on the whole engine system. The surge line defines a hard lower limit for compressor operations that cannot be crossed.



We can also see that at lower rotational speeds the operating margin of the compressor reduces quite drastically and surge becomes a bigger risk.

Surge in compressors has to be avoided because it causes periodically oscillating forces of high magnitude, which can set individual blades into vibration. The vibration of the blades will ultimately result in fracture (fatigue). Since the flow is reverted continuously during surge, hot compressor air at the end comes back to the first stages to get compressed again, causing the compressor to warm up. The warming up also contributes to the fracture in the form of thermal fatigue.

Some ways to avoid surge are:

- **Variable bleed valves:** bleed air from the compressors reduces the flow resistance downstream and increases the airflow in front of the blow-off.
- **Variable stator vanes:** The front stages are fitted with variable stator vanes to provide optimal angles of airflow incidence for low spool speeds.
- **Multi-spool compressor:** splitting up the compressor in different parts, with different rotating speeds allows to control the rotational speed of the front and back stages.

Stall

Airflow separation occurs when the angle of attack becomes too large for the airflow to follow the blade profile. This referred to as **stall**.

When a disturbance in the flow is observed stall can occur. In the figure below, for example, flow breakdown in channel B results in deflection of the inflow incidence at channel A and C. Channel A receives the fluid at an increased angle of incidence (positive direction), while channel C receives the fluid at a reduced angle of incidence (negative direction). This will result in flow breakdown channel A, therefore the flow to channel B will now be more favourable. It appears that the blockage has moved opposite to the direction of rotation. This phenomenon is called “**rotating stall**”.

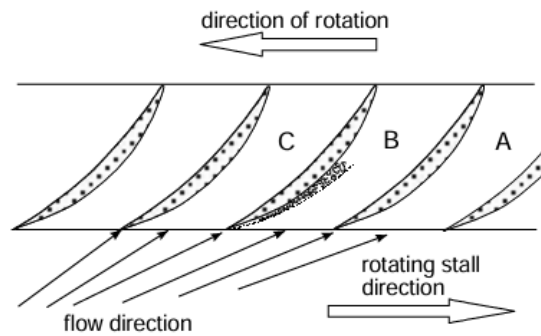
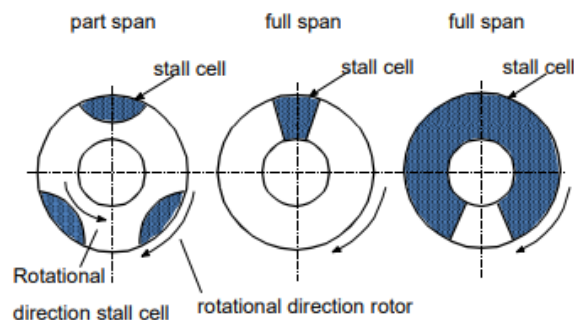
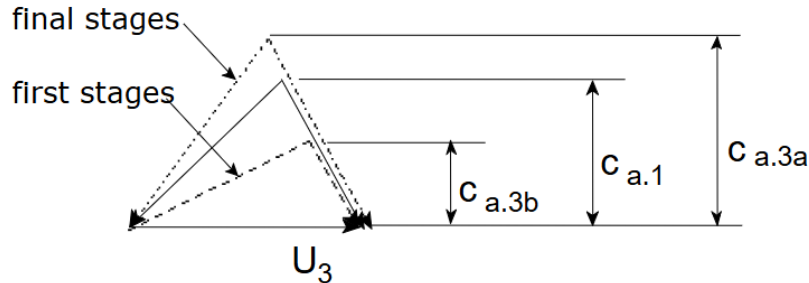


Figure 8-4 - Rotating stall

Rotating stall usually appears at the blade tips since the speeds are at maximum. In case the flow worsens, the stall areas will expand to grow to the length of the blade, which is called “full span stall”. This can then devolve into full on compressor surge.



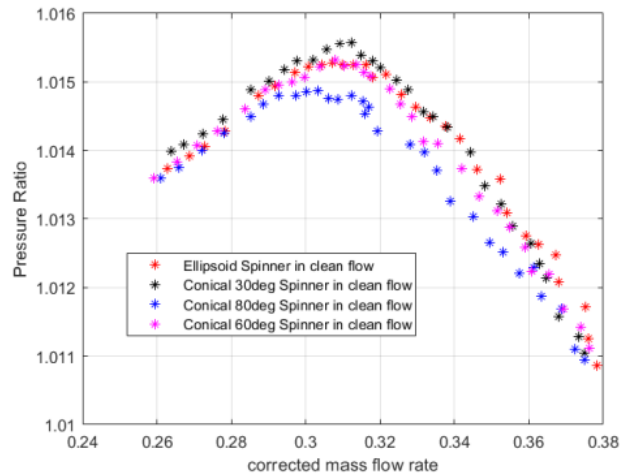
Stall typically occurs at low rotor speeds because in these conditions we have the following velocity triangles:



When the rotor velocity is low, the pressure ratio is lower than what it typically is, which means that the density of the fluid at the final stages is much lower than usual. For the same mass flow rate, this means that the last stages see an axial velocity which is much higher than normal. This reduces the angle of attack that the blades of the last stages see, which can cause negative stall. The opposite occurs in the first few stages (same density, therefore similar axial velocity with a much lower rotational speed), leading to positive stall.

The solution for these issues are the same as the ones for the surge.

Note: one important element that determines the inflow conditions for the compressors is the shape of the **spinner** of the fan and the amount of distortion present at the fan face. We see that typically the spinner has a conical or elliptical shape, since these shapes help reduce the amount of distortion. Also we see that they have very similar performances in clean flow.

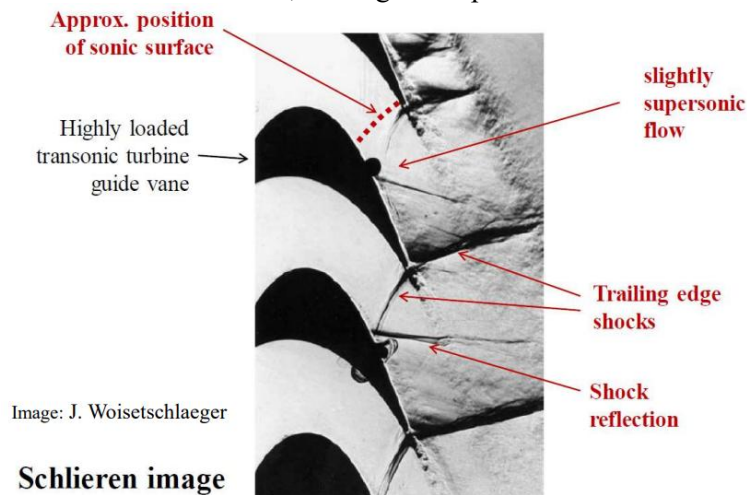


Lecture 11

Axial turbine

The axial turbine has a very similar architecture to the axial compressor. In this case however, it is the fluid that does work to move the turbine blades. This work is then used to operate the compressor. A typical turbine stage is made up of:

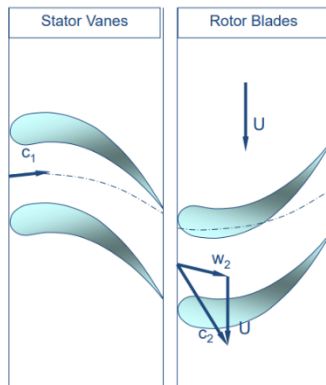
- **Stator:** fluid expands and accelerates, so pressure decreases while velocity increases. Instead of the compressor stators, which act as diffusers, turbine stators act as nozzles. This is why they are sometimes called **Nozzle Guide Vanes** NGVs. The converging channels within the stator are accelerating the flow, usually to sonic and in some cases, supersonic velocities. This occurs because the flow in the turbine is already in the transonic regime and, after the expansion, it can reach sonic conditions, leading to the possible formation of shocks.



- **Rotors:** the fluid moves the turbine blades, work is extracted so velocity, total temperature and total pressure decrease.

The flow inside the turbine is generally less complex than the flow in a compressor, since in this case the flow is expanding, so a favourable pressure gradient is present. This pushes the air through the turbine, unlike the adverse pressure gradient that is present in the compressor stages. For this reason turbines do not suffer from surge as compressors do.

The velocity triangles of a turbine stage are as follows:



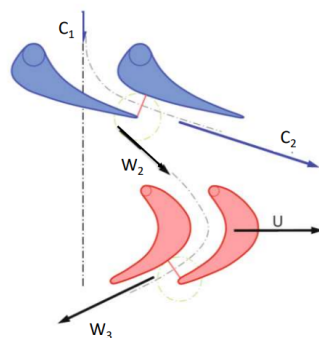
[Note: the second velocity triangle in this figure is wrong since they should be the velocities relative to the stator, so either C1 and W1, or C3 and W3]

We see that another difference with the compressor is that in this case the tangential velocity is much lower than the relative velocity, while in the compressor it's the opposite.

Also since in this case the rotor is being “pushed” by the air and not the other way around, the relative velocity it sees is much lower. This is the main reason why the turbine blades are much blunter than the compressor blades, since stall is less of an issue. Also the airfoils can be thicker, allowing for the cooling channels that bring the compressor bleed air to form a thermal protective film around the turbine blades.

Just like for the compressor, since we typically want repeating stages, the exiting velocity of one stage has to be the same as the entering velocity.

Finally we have to remember that the axial component of the absolute velocity is constant throughout and this is ensured by having an increasing cross section.



- Airfoil shapes are determined initially from flow inlet and exit angles, which essentially fix the camber line.
- For a vane, the inlet metal angle is obtained from the absolute inlet flow angle; the exit metal angle follows from the turning needed.
- For a rotor blade, the inlet metal angle matches the relative inlet flow angle closely; the exit metal angle aligns with the relative exit flow angle, which is determined by the turning angle needed to provide the power

The variation of total pressure, static pressure and absolute velocity we can see that:

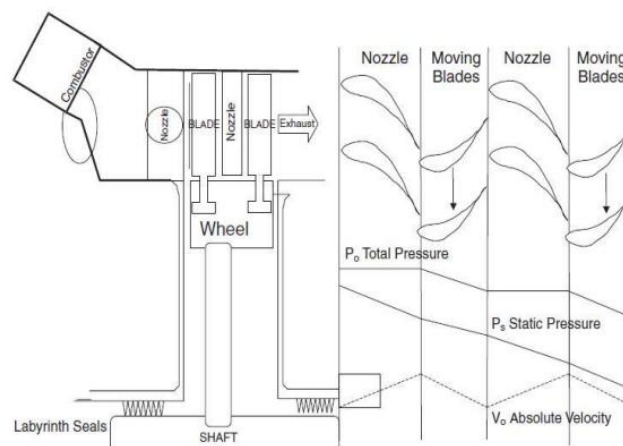
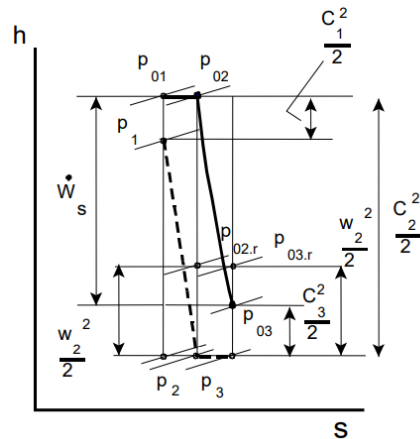


Figure 9-10. Schematic of a reaction-type turbine showing the distribution of the thermodynamic and fluid mechanic properties.

The total pressure decreases slightly in the stator due to losses and then decrease in the rotor due to the extraction of work, the static pressure decreases overall (the amount by which it decreases in each

section is determined by the degree of reaction) and the velocity increases in the stator and decreases in the rotor.

Let's now observe the T-S diagram of a turbine stage:



[Note: this figure represents a turbine with a degree of reaction equal to 1 since the static enthalpy does not change in the rotor stage]

Due to the losses the total pressure from P01 to P02 decreases a bit. Due to the expansion, the static pressure from 1 to 2 decreases. This decrease is equal to the increase in kinetic energy of the gasses between stage 1 and 2. Then in the rotor the static pressure changes slightly depending on the degree of reaction, while the total pressure decreases drastically due to the extraction of work. The kinetic energy of the gasses at station 3 is the same as the kinetic energy of the gasses at station 1 since absolute velocity has to be the same.

The power extracted can be evaluated from the Euler equation:

$$\dot{W} = H_{02} - H_{01} = \dot{m} \cdot (C_{U2} \cdot U_2 - C_{U3} \cdot U_3)$$

To extract/impart work to a fluid a change in momentum has to occur. For axial compressor and axial turbines, the change in total enthalpy is entirely due to change in tangential flow speed, while for radial compressors or radial turbines the change in total enthalpy is due to a change in radius (directly related to the centrifugal effect).

Let's analyse the turbine stage more closely:

Here also: assumption: $c_a = \text{const}$

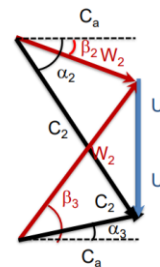
Looking at velocity triangles:

$$\Rightarrow \tan \alpha_2 - \tan \beta_2 = \tan \beta_3 - \tan \alpha_3$$

Specific power:

$$W_s = u \cdot c_a (\tan \alpha_2 + \tan \alpha_3)$$

$$W_s = u \cdot c_a (\tan \beta_3 + \tan \beta_2)$$



Other important turbine parameters:

■ Total temperature drop per stage: $T_{0,1} - T_{0,3} = T_{0,2} - T_{0,3} = \frac{W_s}{c_p}$

■ Pressure ratio per turbine stage:
$$\Pi_{turb} = \frac{p_{0,1}}{p_{0,3}} = \left(1 - \frac{1}{\eta_{is}} \frac{\Delta T_{0,1-3}}{T_{0,1}} \right)^{\frac{k}{k-1}} = \left(1 - \frac{UC_a (\tan \alpha_2 + \tan \alpha_3)}{\eta_{is} c_p T_{0,1}} \right)^{\frac{k}{k-1}}$$

The first relation can be simplified only because the total temperature does not change in the stator. The total temperature drop can then be used, together with the isentropic efficiency term to determine the pressure ratio per turbine stage.

To study turbine performance three important parameters need to be defined:

- **Flow coefficient:** ratio between the axial velocity of the flow and its tangential velocity. In a way it's a measure of the amount of flow that can pass through the turbine stage.

$$\phi = \frac{C_a}{U}$$

- **Loading factor:** ratio between the enthalpy drop for a stage and the square of the tangential velocity. It defines the “skew” of the velocity triangles, since the more skewed they are (the higher the deflection of the flow is), then the higher the specific work that is being extracted per stage is.

$$\psi = \frac{\Delta h}{U^2}$$

$$\psi = \frac{(C_{U2} \cdot U_2 - C_{U3} \cdot U_3)}{U^2}$$

$$\psi = \frac{(C_{U2} - C_{U3})}{U}$$

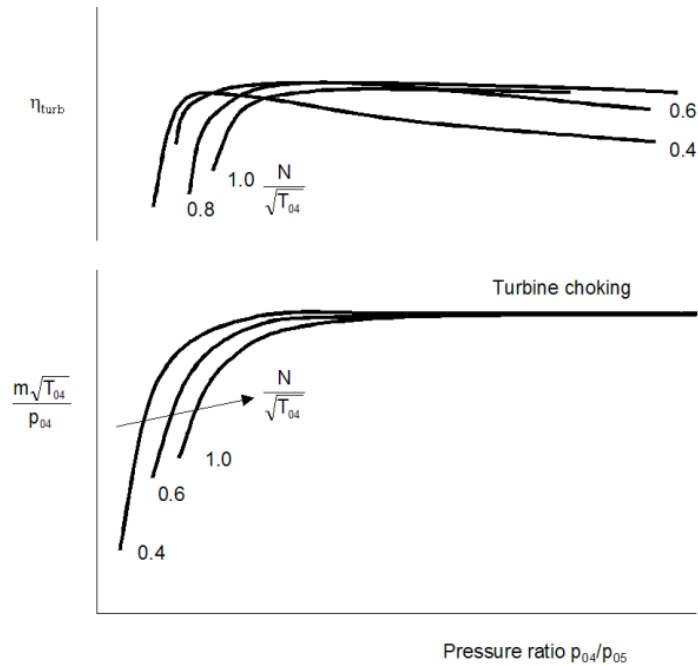
$$\psi = \frac{C_a (\tan \alpha_2 - \tan \alpha_3)}{U}$$

- **Degree of reaction:** same definition as for the compressor. For an axial turbine a relation can be derived to evaluate it from the velocity triangles:

$$\Lambda = \frac{C_a}{2u} (\tan \beta_3 - \tan \beta_2) = 1 - \frac{C_a}{2u} (\tan \alpha_2 - \tan \alpha_3)$$

Typical aero-engine turbines have a degree of reaction of around 0.5.

Just like for the compressor, the turbine performance can be evaluated from dimensionless parameters. The same dimensionless parameters used for the compressor map are used for the turbine. The **turbine map** is:



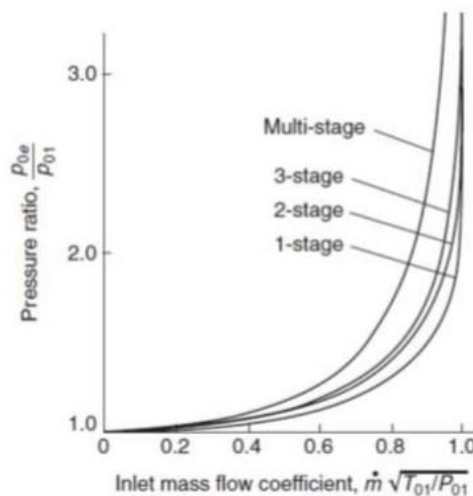
The performance characteristics of a turbine are determined by the choking of the stator vanes. As the pressure ratio increases, the corrected mass flow rate increases until the turbine becomes choked and the mass flow rate plateaus. As the rotation of the turbine increases, more work is being extracted by the rotors (the decrease in velocity in the rotor is higher) and a higher pressure ratio is needed for the mass flow to become choked.

Since the turbine is operating in a favourable pressure gradient we also see that its efficiency is not as dependent on the operating conditions as the efficiency of the compressor. This is also because once the flow becomes choked the incidence angles basically do not change.

Since the turbine and the compressor are connected to each other and have drastically different performance characteristics, a compromise has to take place to determine the operational point, while keeping in mind that the rotational speed and the corrected mass flow rate are the same for both. Also the power and torque balance has to be taken into account.

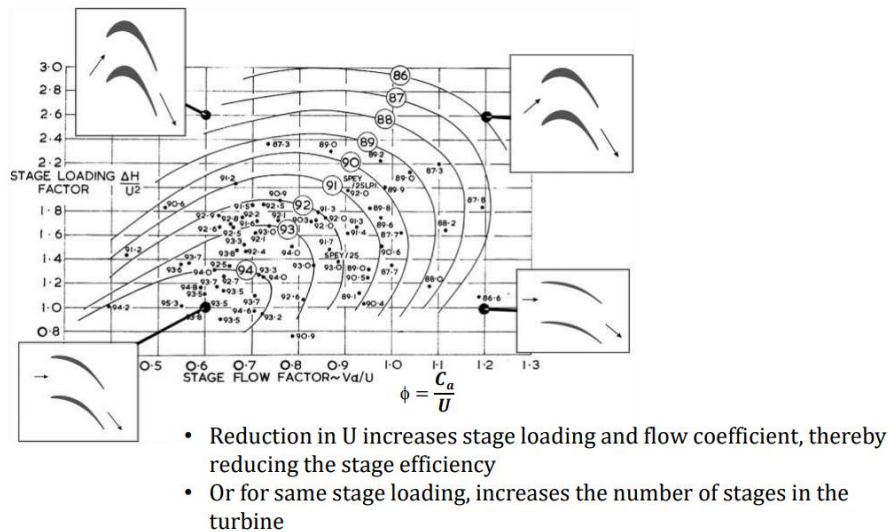
These three constraints determine the operating point of the turbomachinery.

As we increase the number of stages in a turbine increases we see that the corrected mass flow rate (also called “**swallowing capacity**”) tends to decrease, since it is determined by the stage that can handle the lowest mass flow rate before choking.



This is not really relevant for the high pressure turbine, since it typically has 1-2 stages, however it is relevant for the low pressure turbine since it usually is a multistage turbine.

One way to determine the performance of turbines is to look at the **Smith chart** of the turbine, which represents the stage efficiency as a function of loading factor and flow coefficient:



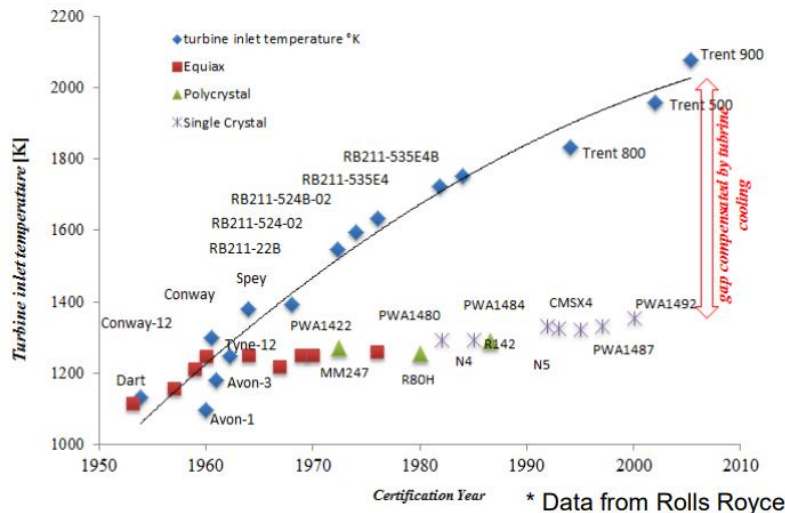
A lower RPM increases the stage loading and the flow coefficient, while a higher RPM reduces them (for a given enthalpy jump and axial velocity). For a high pressure turbine the RPM is very high so it has a high efficiency regardless. However, the low pressure turbine typically has to rotate at a lower speed since it is also connected to the fan, which cannot have high rotational speed due to tip shocks. From the chart we can observe that a high loading factor corresponds to the flow being “turned” a lot (high skew of the velocity triangles), therefore extracting a lot of work per each stage, while also introducing losses. On the other hand a high flow factor corresponds to high passage area, since it is a measure of the swallowing capacity of the stage.

High stage loading means that the number of stages needed can be reduced, thus reducing the total weight of the engine. However this does come at the disadvantage of lower turbine efficiency. In aero-engines, since the weight is one of the most critical design parameters, blades that turn the flow a lot are typically used to have higher loading factors at the cost of lower efficiency.

The reason why the GTF is such an efficient engine is that the tangential velocity of the low pressure turbine can be much higher, since it is basically decoupled from the fan rotational speed. This allows it to have much higher turbine efficiencies even with high work extraction per stage.

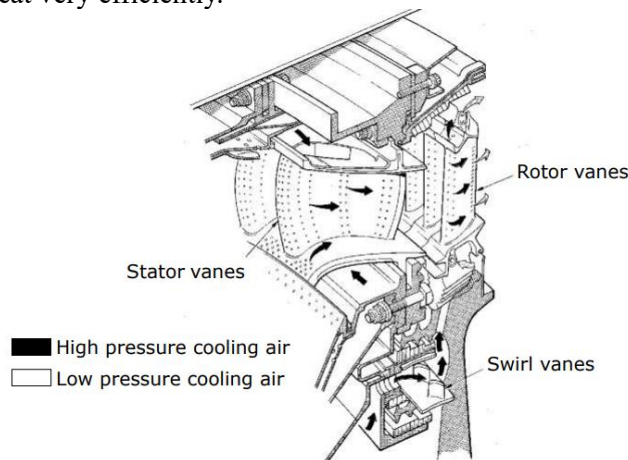
Turbine cooling

The maximum temperature that is allowed inside the engine is determined by the high pressure turbine blades, that receive the hot gasses from the combustion chamber at the TIT. Since we want to have the high temperature ratio to increase the thermodynamic efficiency, the turbine blades are built to withstand incredibly high temperatures. From the chart below we actually see that the TIT is typically much higher than the melting point of the material out of which the turbine blades are made out of.



Advancements in material technology have made possible the manufacturing of turbine blades that have a **single crystal structure** without any grain boundaries. This eliminates any imperfections in the molecular structure of the material and greatly decreases the effects of high temperature **creep** which can lead to failure at high temperatures, especially when coupled with high stresses (centrifugal forces).

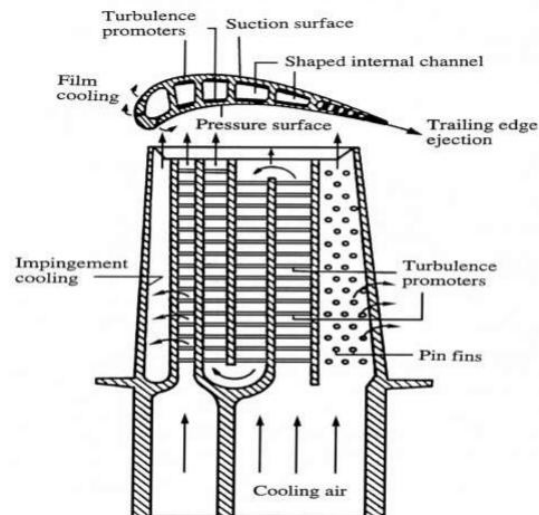
The main reason for the discrepancy between the melting point of the material and the operating temperature is cooling, which is achieved by supplying compressor bleed air to the turbine blades. This air is released from tiny holes on the surface of the turbine blades and forms a protective thermal film which dissipates heat very efficiently.



One element present in some turbine blades is the **shroud**, which interlocks with the shroud of neighbouring turbine blades, forming a platform which reduces tip losses. It is not always used since it does increase the centrifugal force acting on the blades (worse mass distribution), effectively reducing their lifespan.

The most critical region of the blade that needs to be cooled is the leading edge, since that is where the stagnation point of the incoming flow is and where the temperature reaches the maximum. To cool this part of the blade **impingement cooling** is used, where a channel dedicated to the leading edge blows air on the back of it to promote heat dissipation. These jets create a very thin boundary layer on the surface and achieve extremely high local heat transfer coefficients.

Note: the channels inside the blades that transport bleed air have all **turbulence promoters** to create turbulence and enhance heat transfer away from the walls.



Jet impingement is also used in the **active clearance control system**, which is made up of small ducts in which compressor bleed air is run through. These ducts blow air on the casing of the turbine blades and can either expand the casing, allowing for more tip clearance when the engine is running at cruise (high temperatures and high centrifugal forces) or contract the casing, allowing for less tip clearance when the engine is running idle for example.



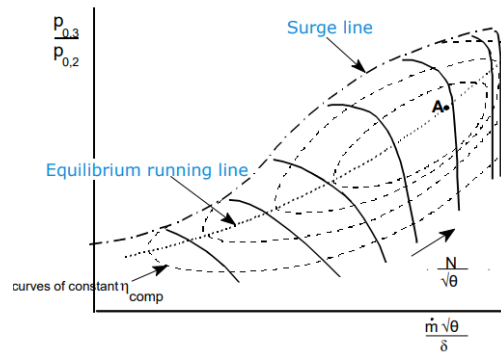
Lecture 12

The performance of a gas turbine aero engine is determined by:

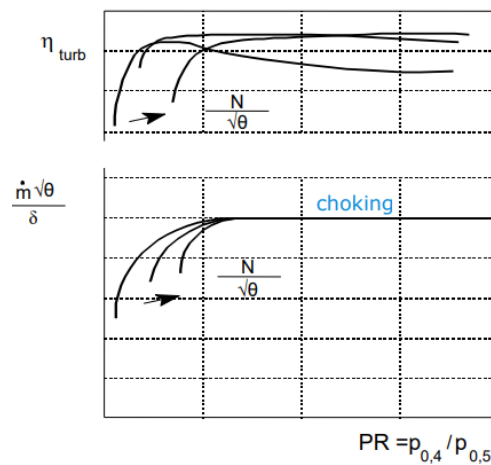
- Components characteristics: performance is determined by all the components we have seen so far as well as their limits (e.g. stall, temperature, ...)
- Engine characteristics: the operating conditions obviously determine the engine's performance
- Ambient conditions

The range of engine operating conditions is determined by superimposing the operational envelopes of all the components that make up the engine. The FADEC (**F**ull **A**uthority **D**igital **E**ngine **C**ontrol) keeps the engine within its operational limits. It can do this by controlling different variables and systems like: the fuel flow, variable stator vanes, variable bleed valves, active clearance control, nozzle geometry, after burner fuel flow, thrust vectoring, ...

Let's start with the compressor. As previously seen, the compressor's characteristics are represented by the compressor map.



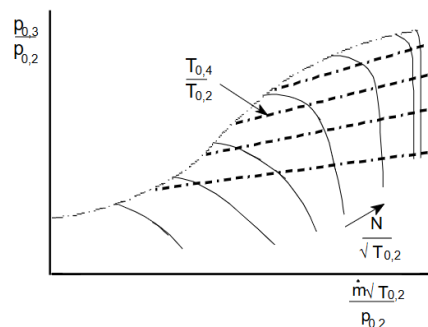
The operating conditions of the compressor are typically represented along an equilibrium line. However, this is an ideal representation, which is valid only if the compressor is changing speed very slowly without strong accelerations (in other words if we do not take into account system dynamics). The operating conditions of a turbine is determined from the turbine map.



The turbine characteristic is less complex than that of the compressor since the turbine becomes choked after a certain pressure ratio, independently of the speed. During normal operations, the turbine is almost always choked. Moreover, the efficiency of the turbine does not change much once it becomes choked.

Other components such as the ducts and inlets have much simpler characteristics. In this type of component, the only thing that matters is the pressure loss, which is generally a function of the flow velocity. For all components heat loss effects are also taken into account through sophisticated models to evaluate their characteristic performance.

By superimposing all of the component characteristics we can obtain the characteristic of the whole gas turbine.

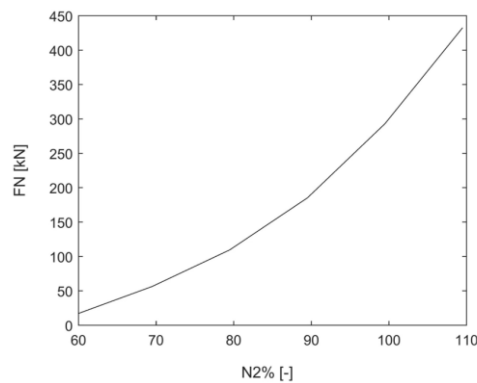


From this we can check the required temperature ratio for the gas generator to be self-sustaining, given an operational point of the compressor (I think).

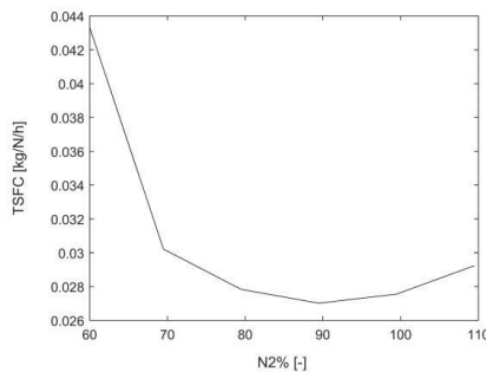
The characteristic performance of a component can be adapted to the operating condition by changing the geometry of the component. For example, in a compressor it is possible to avoid the surge line when changing the operational point by implementing variable stator vanes (adapt the angle of attack of the stators, avoiding stall and surge) and variable bleed valves, which are typically positioned between the booster and the HPC (opening these valves reduces the back pressure and allows for an increase in mass flow rate, moving away from the surge line). The IGV can also be variable to improve the inlet condition of the flow entering the booster.

Although stall is not very common in turbines (since there is a favourable pressure gradient), the same thing can be applied to turbine through variable stator vanes. This has been tried in the past, leading to improved SFC but has not been implemented widely.

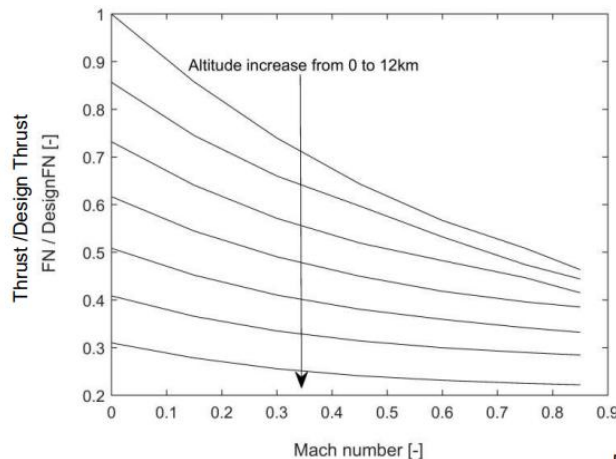
Let's now see how the performance of the whole engine (thrust and fuel consumption) changes with different component operating conditions. Regarding the compressor, an increase in speed directly correlates to an increase in thrust.



This is due to the fact that an increase in spool speed is directly correlated to an increase in pressure ratio and therefore energy available (i.e. thrust). For the TSFC we also see improvements as the spool speed increases at first, however after a certain threshold the trend changes.

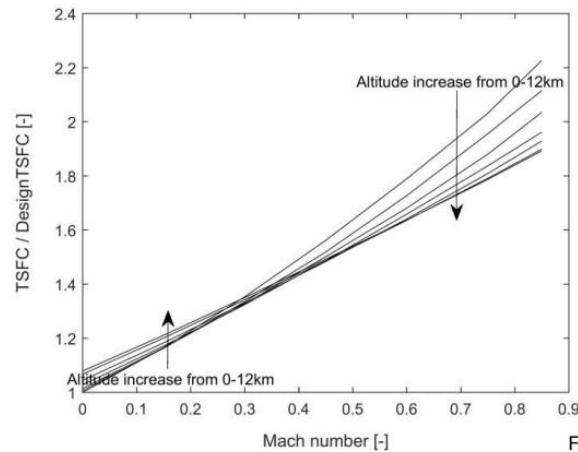


The cruise spool speed is determined by this graph and is set at around 80-90%. With respect to Mach number and altitude:



As we can see, when the altitude increases the thrust produced also reduces w.r.t. the design thrust. This is mainly due to the reduction in air density and therefore mass flow rate. The trend with increasing Mach number is the same. This is instead due to the increase in ram drag; from the thrust equation we know that the thrust is directly proportional to the difference in jet velocity and flight velocity. As the flight velocity increases it becomes harder and harder to provide the same velocity increase to the exhaust gasses.

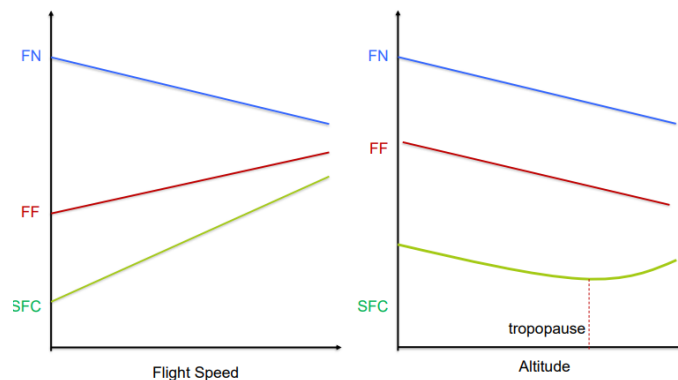
For the TSFC:



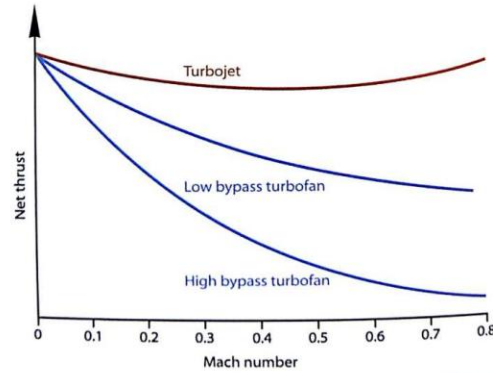
The behaviour of the thrust specific fuel consumption is very different from the thrust. The TSFC increases with Mach number always since it becomes harder to provide the same exhaust velocity increase w.r.t. the flight speed at higher Mach.

However, it decreases with altitude until the tropopause is reached (after which the external temperature stays roughly constant). External temperature decreases, which corresponds to an increase in the temperature ratio between the turbine inlet temperature and the ambient temperature. From a thermodynamic point of view this is very beneficial for the efficiency, which leads to a decrease of TSFC. This is not the case for lower Mach numbers, where the decrease in outside temperature (higher thermodynamic efficiency) is outweighed by the decrease in pressure (lower thrust). Lower pressure and dynamic compression means that more work has to be done by the compressor to achieve the same thrust, therefore the TSFC actually increases with higher altitude (up until $M \sim 0.3$).

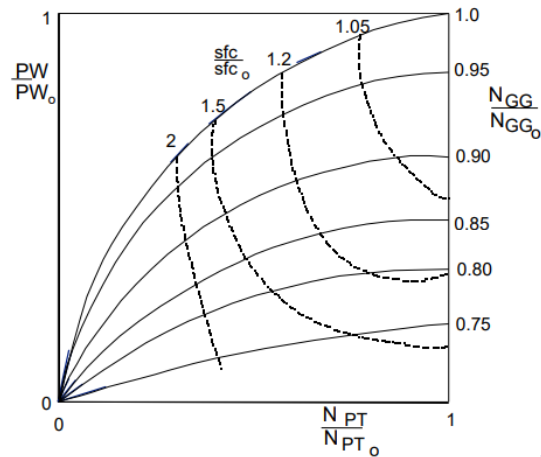
In summary:



To evaluate engine performance w.r.t. operating conditions we can define the **thrust lapse rate**. This parameter quantifies the reduction in thrust of an engine when compared to its design value at sea level and at $M = 0$ (uninstalled thrust). It can be defined w.r.t. Mach number or altitude:

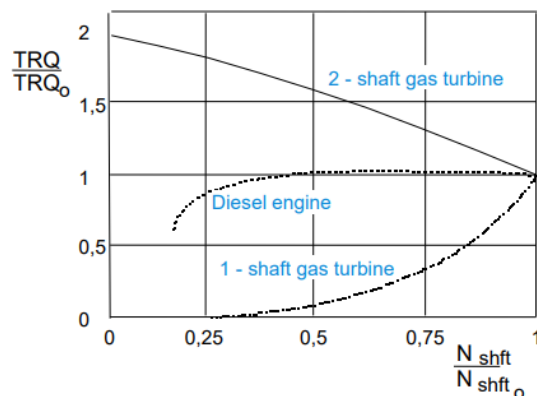


For a turbofan the thrust reduces with Mach number, while for a turbojet it first reduces and then increases again (after a certain point all the compression needed comes from ram compression). Let's now consider the characteristics of the whole engine for a 2 shaft power turbine:

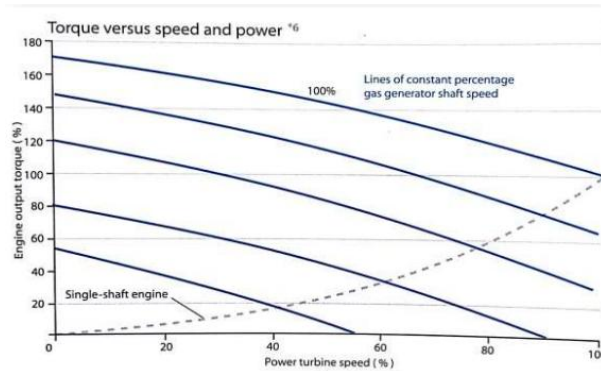


Where NPT is the speed of the power turbine that drives the load (the propeller) and NGG is the speed of the gas generator (other turbine and compressor). The operating point is determined by the rotating speed of the power turbine and the required power to drive the load PW. We can see that at low speeds for the power turbine the specific fuel consumption is quite high and it gets much better at higher NPT values. Same goes for the NGG.

Another advantage of having a two shaft turboshaft is that it's **torque characteristic** is much better than a one shaft turboshaft, especially at lower shaft speeds.



The gas turbine shaft can generate power (or torque) when it's running at high speeds to drive the loaded shaft even if the power turbine is not running. This is the great advantage of two spool turboshafts. For one spool turboshafts starting up the power turbine takes a great amount of torque, which the turboshaft struggles to generate at low speeds.



Transient operations

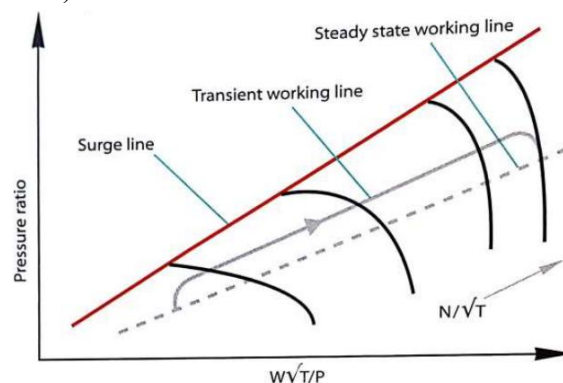
Although all the ideal operational conditions lie on the engine equilibrium line (steady-state), in reality the transition between one operating state and another does not occur slowly enough so that all states in between are steady states. The real operating line moves closer/away from the surge line, depending on the change of speed.

Each spool is governed by a rotational balance, where the change in rotational speed over time is proportional to the difference in power extracted by the turbine and the power consumed by the compressor.

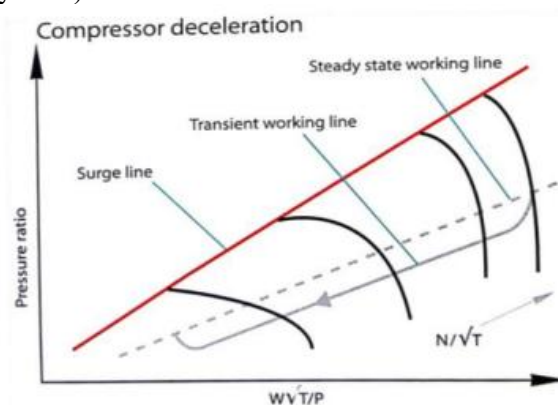
To increase the thrust generated by the engine, the fuel flow rate is increased, the TIT increases and the turbine speed increases. This means that the compressor starts working at a higher speed, while the corrected mass flow rate does not immediately increase and lags behind.

This means that during the transient, the operating point of the compressor moves up (higher pressure ratio due to higher rotational speed), while having the same mass flow rate.

This is the **acceleration line**, which occurs when there is **overfueling** (more fuel flow than needed to maintain the current steady state):

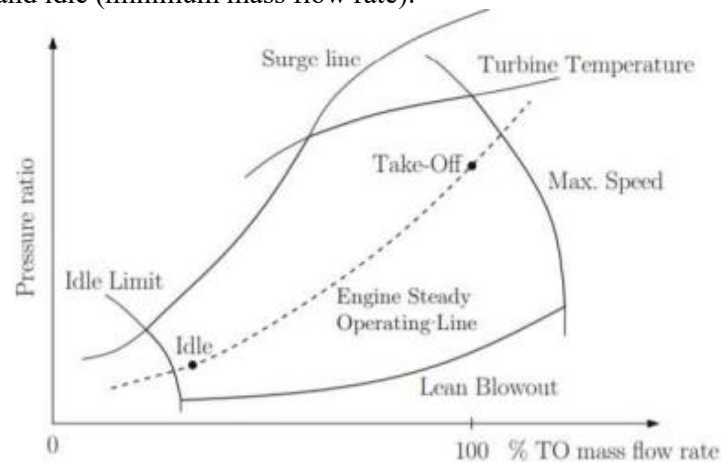


The exact opposite occurs when the thrust needs to be reduced. In this case the mass flow rate in the compressor lags behind the reduction in pressure ratio due to the reduction of spool speed. This is the **deceleration line**, which occurs when there is **underfueling** (less fuel flow is provided than needed to maintain the current steady state).



Aero engine limitations

In the chart below the main aero engine operational limitations are represented. The ideal operational conditions all lie on the engine equilibrium line between the take off condition (maximum mass flow rate for max thrust) and idle (minimum mass flow rate).



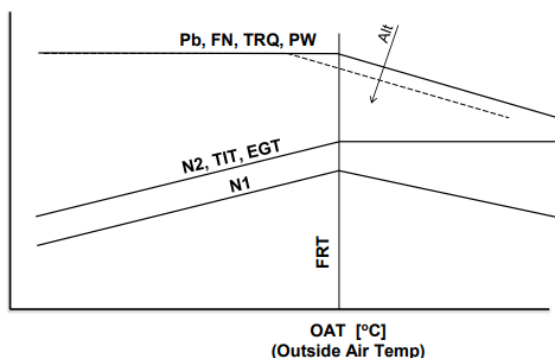
These limits are taken into account by the FADEC, especially during transient operations. The increase of fuel flow needs to be limited to avoid surge or surpassing the maximum turbine inlet temperature. For lower mass flow rates surge is the limiting factor, since stall is more likely to occur due to higher incidence angles. For higher mass flow rates the turbine inlet temperature becomes the limiting factor since an increase in fuel flow corresponds to an increase in the final combustion temperature.

The decrease of fuel flow needs to be limited to avoid lean blowout in the combustor.

The idle limit is defined as the minimum speed required for the core gas generator to sustain itself (the minimum pressure ratio required for the thermodynamic cycle to be viable), while the max speed limited is imposed by the centrifugal forces acting on the disks of the turbines.

Another way to determine the performance of an engine is to evaluate the **flat rating**.

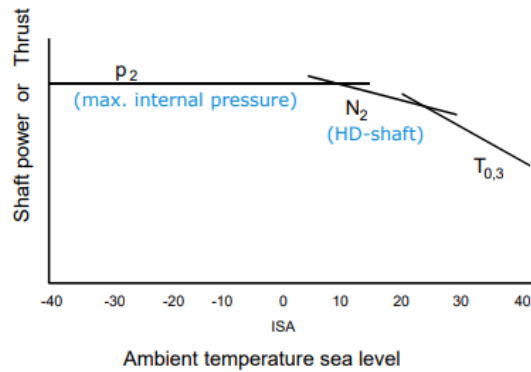
We say that an engine is flat rated if it delivers constant **rated thrust** (or shaft power) over a range of ambient temperatures and pressures, up to a specified “**flat-rating temperature**”, and becomes thermally limited only above that point.



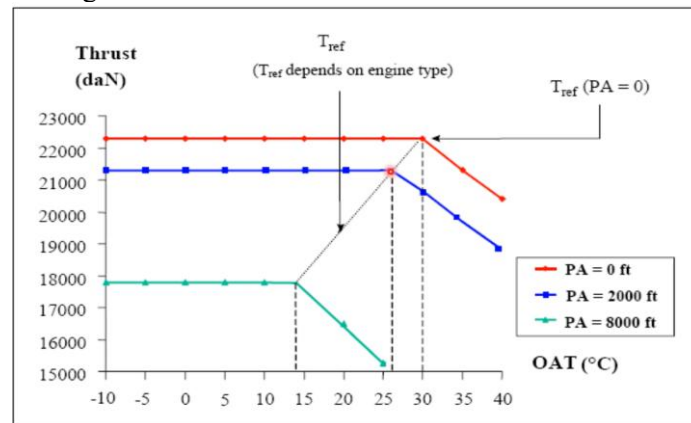
[Note: the rated thrust is not the maximum thrust the engine can produce, it is the advertised value of thrust, which is lower than the max thrust for all temperatures lower than the flat rating. Over the flat rating this is not true anymore, so this serves as a warning for the operators]

For lower temperatures the engine performance is deliberately limited by the FADEC. In this region it is the maximum pressure ratio that determines the maximum thrust that can be achieved. Still, this thrust could be obtained by increasing the TIT and the spool speeds.

For higher temperatures the engine performance decays. In this region the engine becomes temperature limited, meaning that the maximum thrust is now limited by the maximum TIT (or the maximum spool speed), so the rated thrust has to decrease accordingly.

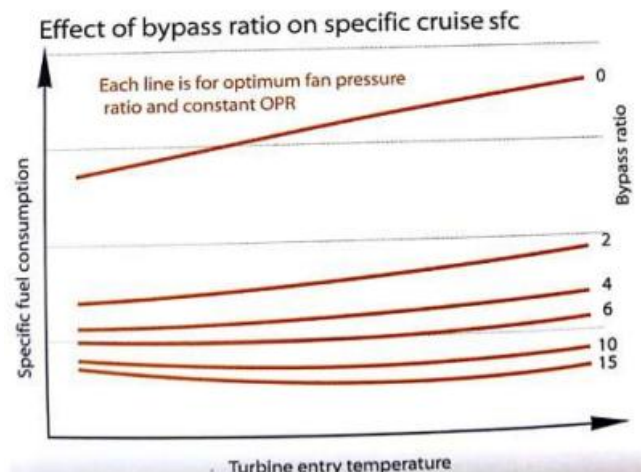


The flat rating temperature and the rated thrust both decrease with altitude since the density of the incoming air is also reducing.

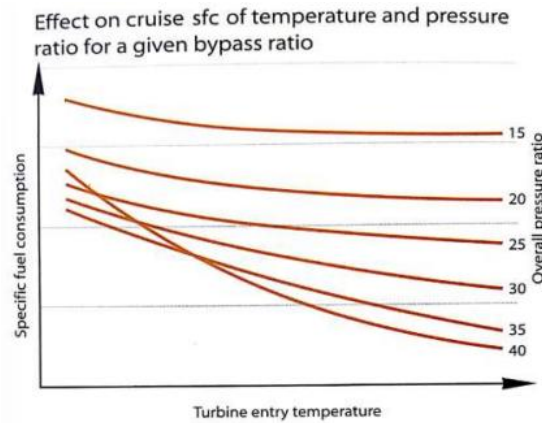


Effect of design parameters on performance

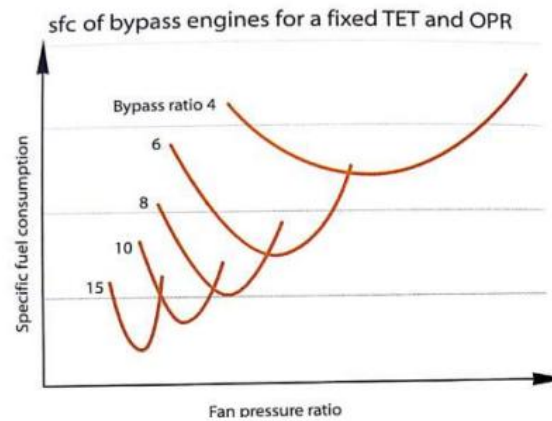
For a turbofan the 4 main design parameters are the Bypass Ratio BPR, the Fan Pressure Ratio FPR, the Overall Pressure Ratio OPR and the Turbine Inlet Temperature TIT. Let's now look at their impact on aero engine performance.



As we can see from the figure above, the fuel consumption decreases with increasing BPR, since the propulsive efficiency is increasing (to produce the same thrust the exhaust velocity of the gasses can be lower). Additionally, the dependence of the fuel consumption on the TIT decreases with increasing BPR. By keeping the OPR constant, the TSFC worsens with increasing TIT because the thermodynamic efficiency does not change much, while more fuel is being burned.



If instead we keep the BPR fixed, we can observe the opposite trend: the TSFC decreases with increasing TIT. This discrepancy is caused by the fact that as TIT increases the OPR also increases if it is allowed to change, leading to a higher thermodynamic efficiency, that coupled with more net power available results in a decrease of the TSFC. This is the same reason why the TSFC decreases with increasing OPR.

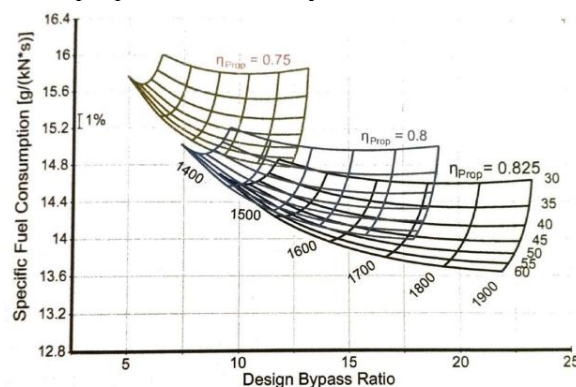


For any given BPR there is an optimum FPR which minimizes the fuel consumption, and this optimum decreases with increasing BPR. This is due to the fact that with increasing BPR more and more thrust is generated by the bypass flow, whose velocity increase is mainly dictated by the FPR. If the BPR increases there will be more mass flow rate and therefore to produce the same amount of thrust, a lower pressure increase is required.

There is a caveat however; when the FPR reduces a lot (typically to a value lower than 1.4) the bypass nozzle might not be choked in the cruise condition (since the pressure ratio in the nozzle is also decreasing), reducing the nozzle isentropic efficiency.

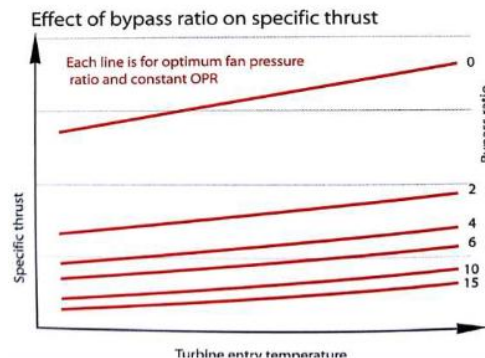
To avoid stall, especially during take off, variable blades for the fan or variable geometry for the nozzle are required.

If we also take into account the propulsive efficiency:



The propulsive efficiency is determined by the BPR and the FPR (high BPR coupled with a lower FPR). From the graph above we can see that the TSFC can also be reduced by increasing the OPR or the TIT (at a given propulsive efficiency).

Finally, we can look at the specific thrust:

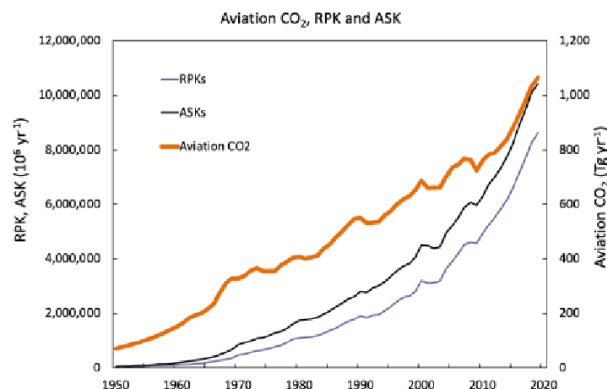


The specific thrust increases with TIT since more net power can be extracted from the cycle. Also it decreases with Bypass Ratio at a given OPR. This is caused by the fact that as BPR increases, more and more of the thrust is generated by the fan and not by the increase in jet velocity. From the thrust equation and the definition of specific thrust we can see why this leads to a decrease of the latter.

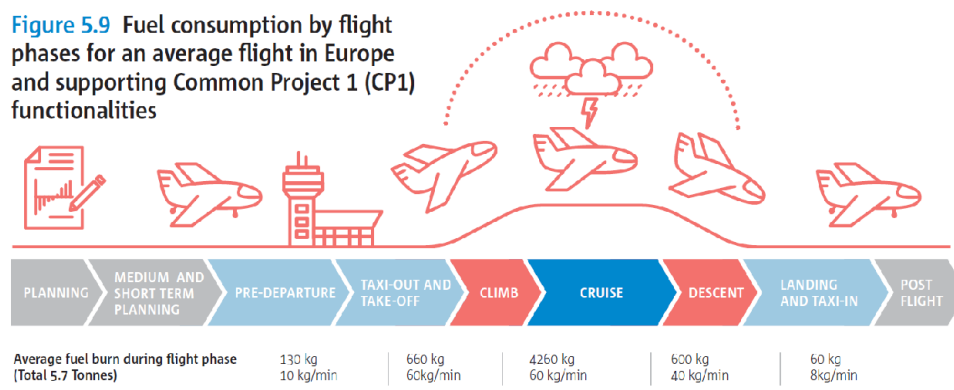
Lecture 13

Emissions

The aviation CO₂ emissions have increased by ~2.5% YoY for the past two decades. Nowadays there are on average 40 mln flights per year and this number is projected to further increase in the next decades. Since emissions are directly proportional to the total number of km flown every year, this means that CO₂ emissions will keep on growing.

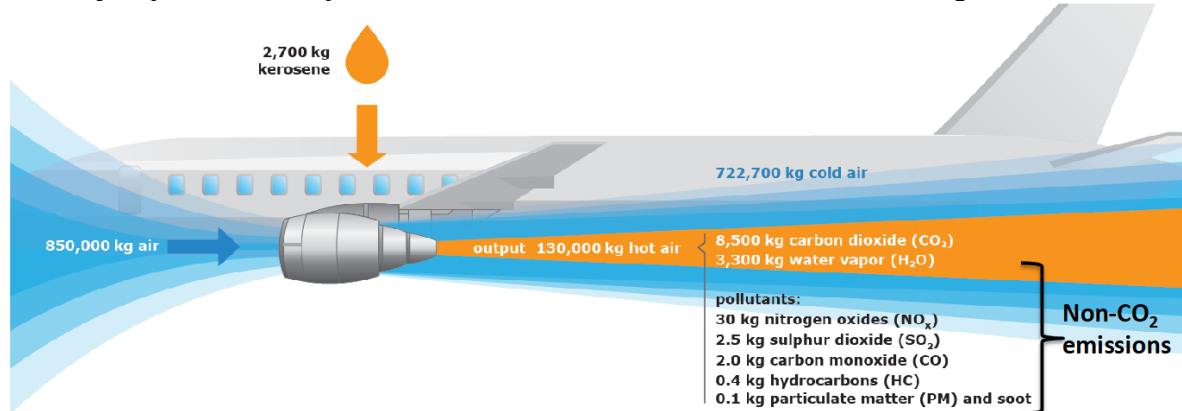


The average fuel burn over the various phases of a typical mission can be seen in the figure below:



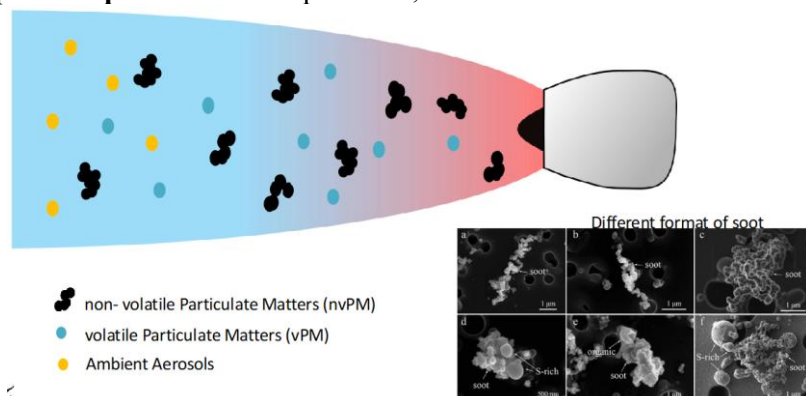
The fuel flow for the take-off and cruise conditions is the same, however this is only because the taxi phase is also included in the take off phase, so on average the fuel burn rate is the same. The highest fuel burn in absolute value occurs in cruise, while the highest fuel flow is obtained during take-off since thrust is at a maximum. The lowest fuel flow is instead reached during landing.

The major species that are produced in the combustion chamber are the following:



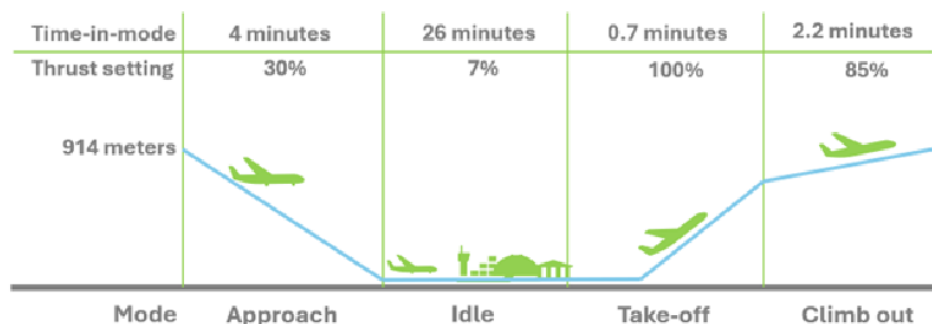
Emissions from a typical two-engine jet aircraft during 1-hour flight with 150 passengers (Source: FOCA)
The main species are CO_2 and water vapor since combustion occurs at very lean overall equivalence ratio, while everything else is a secondary emission. Although secondary species are produced at a much lower rate, this does not matter that their impact on the climate is lower than the main species produced.

On top of these products **particulates** are produced, which can be classified into:



Note: ambient aerosols are particles that are already present in the air before the combustion in the engine. Non-volatile particulate matter nvPM is primarily made up of soot.

Emissions certification for engines is done for an idealized landing and take-off at less than 3000 feet. The certification consist of four different tests at sea level, at four different thrust settings (7%, 30%, 85%, 100% of the thrust at sea level). These four thrust conditions are meant to simulate the four phases of the landing-take-off cycle:



The regulated species are NO_x, CO, unburnt hydrocarbons UHC, non-volatile particulate matter nvPM and smoke.

As we have seen before in the combustion module, NO_x and soot production is a function of flame temperature, while CO₂ and UHC production is inversely proportional to it.

To quantify the amount of emitted species the **emission index** is defined as:

$$EI(X) = \frac{m_X}{m_f}$$

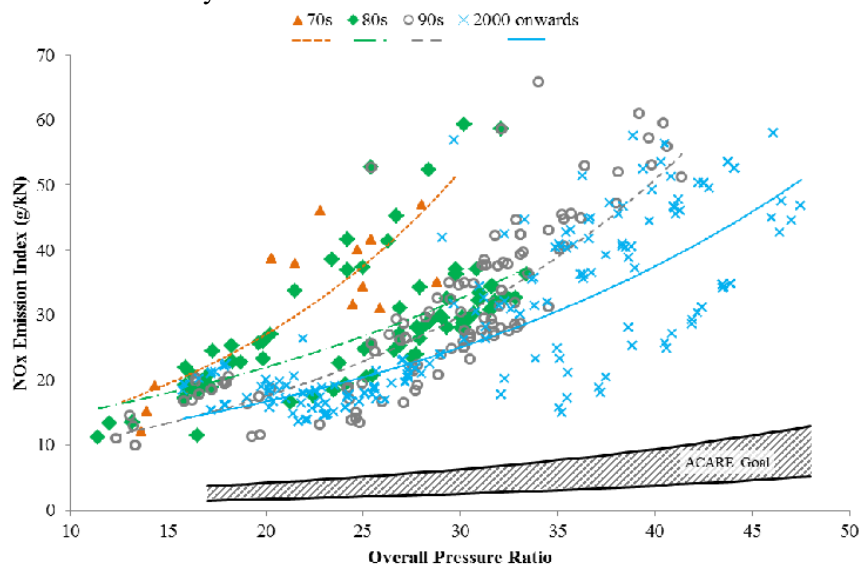
Where x is the relevant chemical specie.

The NO_x emissions are actually made up of many different nitrogen oxides that are produced during combustion. To simplify the study of emissions however it is assumed that all nitrogen is converted into NO₂.

Since NO is also one of the main products, the molar rate of NO has to be converted into molar rate of NO₂ (through their respective molar masses). This is then added to the actual molar rate of NO₂ to evaluate the EI of NO_x emissions.

For CO₂ the emission index depends strictly on the chemical composition of the fuel. Since it's almost always a hydrocarbon, the number of moles of CO₂ is equal to the number of moles of carbon present in the fuel. The issue however is that the chemical composition of hydrocarbon fuels is not constant, so the amount of CO₂ produced does change.

Specifically for NO_x emissions, ICAO regulates them w.r.t. to the engine overall pressure ratio, since NO_x emission increase with OPR. From the figure below we can see that over the years the certification limit has drastically reduced.



The EINO_x increases YoY since the OPR of the engines is increasing, together with the absolute value of NO_x emission since air traffic is increasing. However, the NO_x emission per kilometre flown has been decreasing since engines have been implementing newer combustion technologies that produce less NO_x.

In general improvements in efficiency correspond to a reduction in CO₂ emissions (since they are directly proportional to fuel burned) and an increase in NO_x emissions.

An important thing to note is that single and twin aisle aircraft have different emission rates:

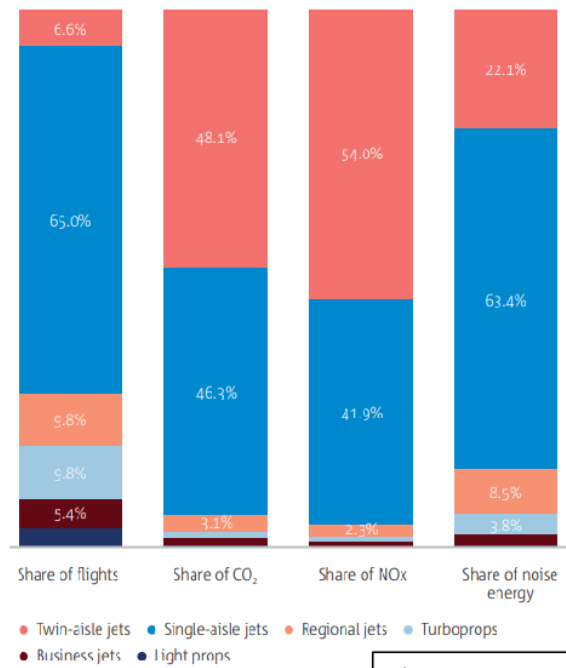
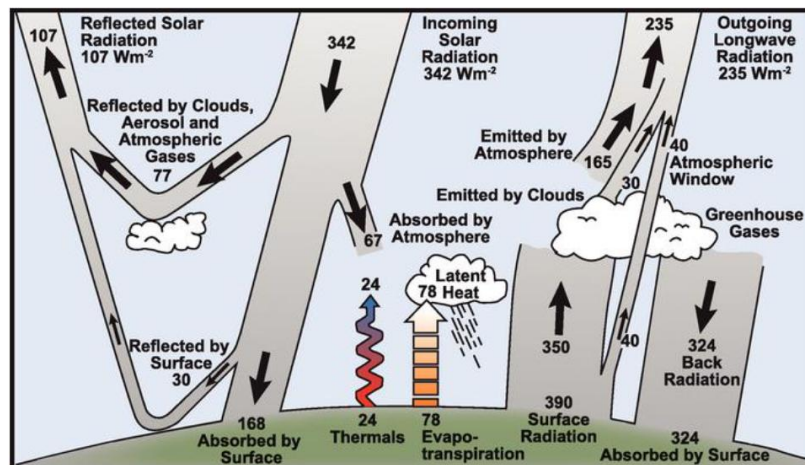


Figure source: FASA. Fi

We can see that, although single-aisle jets have a much bigger share of the total flights, their absolute emissions are very similar to the twin aisle jets for both CO₂ and NO_x. This is mainly caused by the much larger power requirement of the twin-aisle jets when compared to the single-aisle ones.

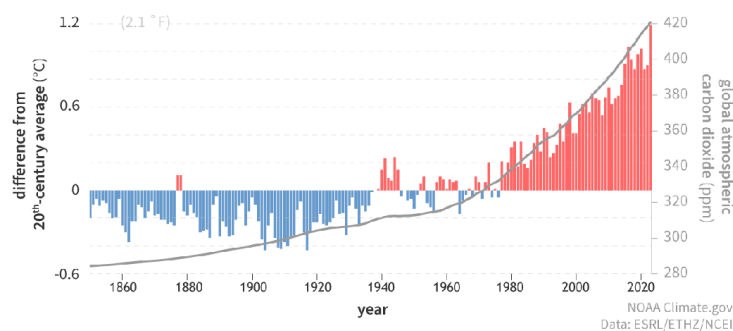
To evaluate the impact of these emissions on the climate we have to consider the global energy balance.



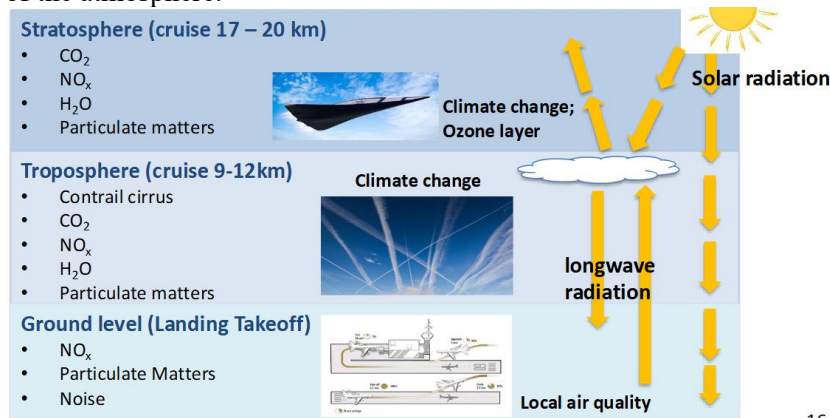
Anthropogenic activities change this balance leading to some of the energy being retained in the atmosphere. This leads to a global temperature increase and climate change.

This effect can be seen over the years in the figure below.

Earth's surface temperature and atmospheric carbon dioxide (1850-2023)



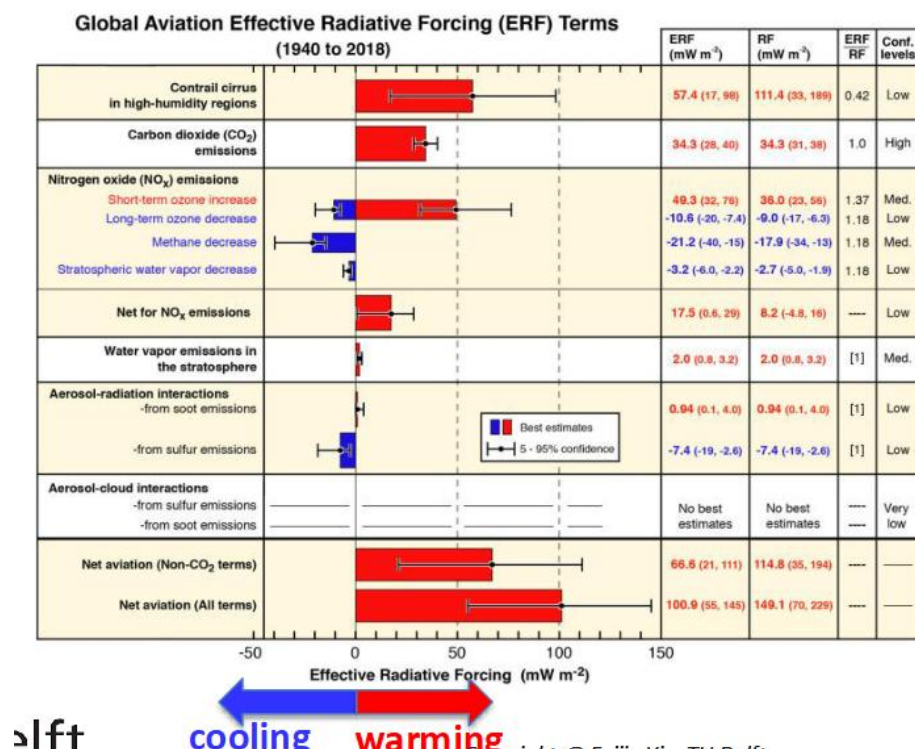
The impact of emissions can be divided based on the part of the atmosphere in which they occur. At ground level the main focus is air quality control, so the most important emissions that need to be considered are NO_x, particulate matters and noise emission. Then in the troposphere (cruise altitude) the focus shifts to greenhouse gasses, which trap long wave radiation inside of the atmosphere.



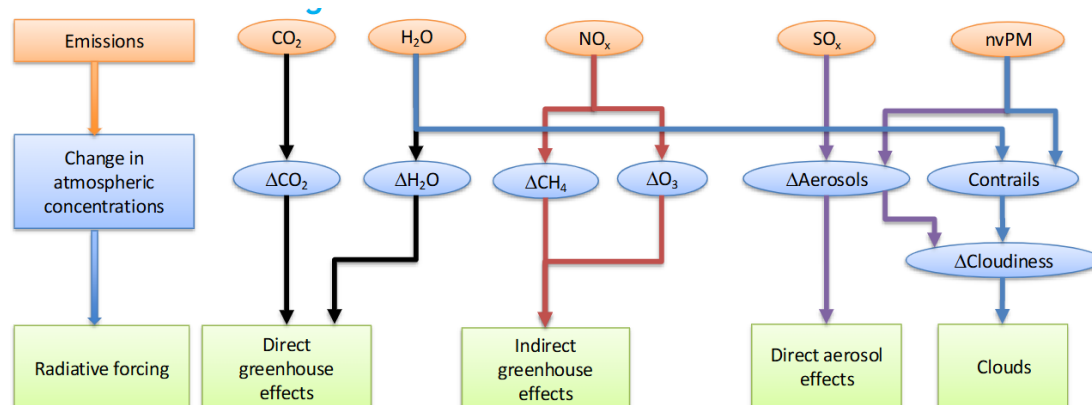
Air control is especially important at ground level since the chemical species emitted have very serious health implications.

The impact of emissions on the atmosphere can be quantified through **radiative forcing**. This parameter is a measure of how a chemical species changes Earth's energy balance by altering the difference between incoming solar radiation and outgoing infrared radiation. If this parameter is zero there is no effect on the balance, while if it's positive it means that the effect of said chemical species is to "trap" more radiation in the atmosphere than it lets out (negative effect on climate change).

The radiative forcing of the main products of aero engines can be seen below:



Although some chemical species can have a partial cooling effect (e.g. NO_x reduces the concentration of methane, which has a cooling effect), all of the emission have an overall warming effect (e.g. NO_x increases the ozone concentration, which trumps the positive effect of reducing methane concentrations). Each chemical species contributes to climate change in different ways, as can be seen below:



Note: NO_x emissions change the concentration of different greenhouse gasses but they are not greenhouse gasses themselves. It increases the concentration of ozone (heating effect) and decrease the concentration of methane (cooling effect). Nevertheless, overall NO_x emission have a heating effect.

Aerosols are suspended particles in the atmosphere and have two main effects: they act as nuclei for condensation, helping the formation of contrails and scatter/absorb incoming radiation. The radiative forcing of aerosols is unsure but is typically regarded as negative overall (warming effect).

Contrails, short for condensation trails, are vapour trail made up of ice crystals that are formed in the exhaust plume of the engine. The water concentration in the exhaust gasses is more than that in the atmosphere due to the water vapor formed as a product of combustion. This water vapor freezes at the low ambient temperature that corresponds to cruise altitudes and in the presence of nucleating particles such as soot.

Contrails change the total cloudiness of the atmosphere, breaking the global energy balance.

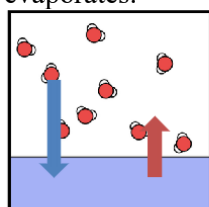
Contrail formation is governed by the **relative humidity** of the air, which is defined as the ratio between the partial pressure of water vapor and the equilibrium vapor pressure of water at the same temperature:

$$RH = \frac{p_{H_2O}}{e_l(T)} \propto \frac{\text{rate of condensation}}{\text{rate of evaporation}}$$

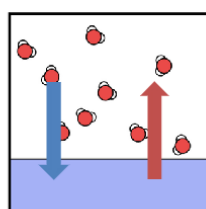
Where p_{H_2O} is the partial pressure of the water vapour and $e_l(T)$ is the saturation pressure of water at temperature T .

A mixture of water vapor and liquid water is saturated at a given temperature, if the exchange of water molecules between gas-phase and liquid phase is in equilibrium (meaning that the amount of water condensing and evaporating is equal).

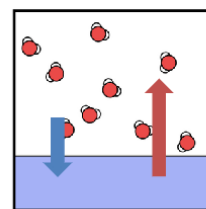
The relative humidity determines which transformation prevails: if it's higher than one it means that the partial pressure of the vapor is higher than that of an equivalent system that is at equilibrium, therefore some vapor needs to liquify. Vice versa, if it's lower than one it means that the vapor pressure is lower than that of an equivalent system that is saturated at a given temperature, so some liquid water evaporates.



Water molecule flux: "In > out" $RH > 1$
Super-saturated



Water molecule flux: "In = out" $RH = 1$
Saturated

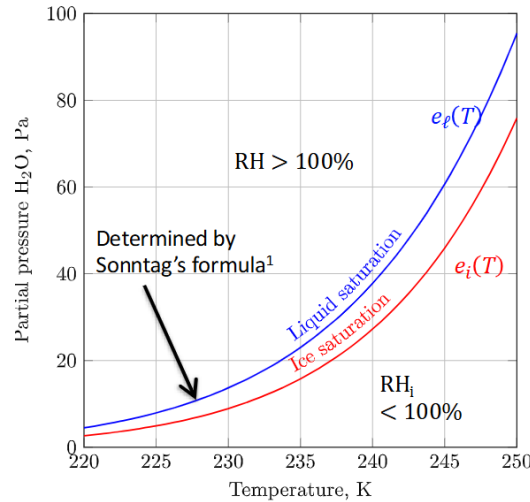


Water molecule flux: "In < out" $RH < 1$
Sub-saturated

The same process can be repeated for the ice phase. A relative humidity can be defined w.r.t. the ice phase as:

$$RH_i = \frac{p_{H_2O}}{e_i(T)} \propto \frac{\text{rate of deposition}}{\text{rate of sublimation}}$$

When the exhaust gasses and the cold outside air mix, the relative humidity changes and water vapour is transformed into ice crystals. Note that nucleating particles are important in this process; far fewer atmospheric particles can act as ice nuclei than as condensation nuclei (particles around which the transition process can occur). The relative humidity is temperature dependent: as temperature decreases the relative humidity increases, meaning that it becomes easier for ice crystals to form.

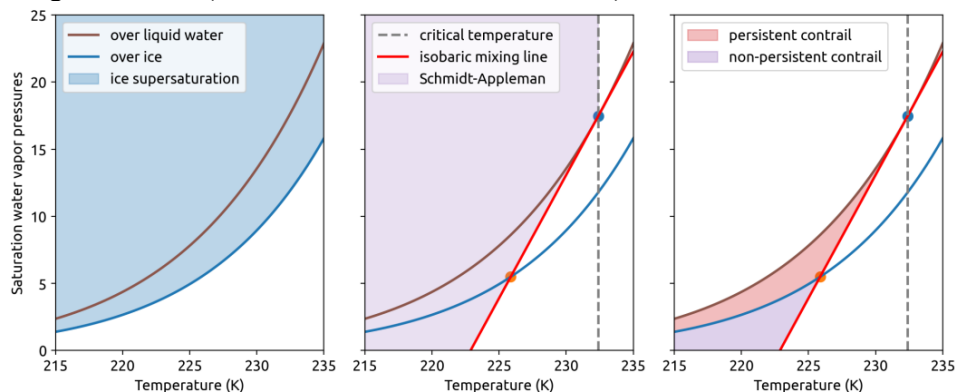


The process of contrail formation goes as following:

- Engine core exhaust is hot (~700 K) and humid.
- The ambient air during cruise altitudes is cold (~220K at 11km) and dry.
- The exhaust mixes with the environment, which constantly reduces temperature, but increases relative humidity.
- During this mixing, the air gets saturated with respect to liquid water. Droplets are formed. ($RH > 1$)
- Supercooled droplets freeze when temperature is $< -38^\circ\text{C}$.
- Contrails form

If the atmosphere is humid then the ice crystals persist and contrails remain in the atmosphere for longer periods of time. However, if the atmosphere is dry then the ice droplets quickly turn back into water vapour and the contrails dissipate.

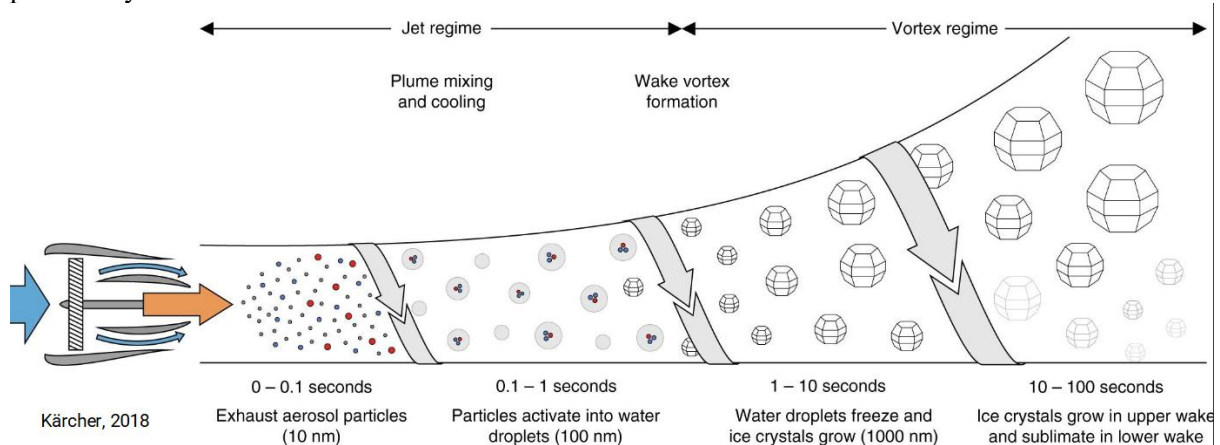
The tendency to form contrails can be evaluated through the **Schmidt-Appleman criterion**. The criterion is derived using a mixing-line analysis in a temperature–humidity (or vapor pressure–temperature) plane. The mixing line is the line along which all the possible thermodynamic states of the two mixing streams lie (i.e. the exhaust and the outside air).



In general, the higher the slope of the mixing line is, the wider the conditions for contrail formation are. The slope of the mixing line can be evaluated as:

$$G = p_a \frac{c_p E I_{H_2O}}{\varepsilon (1 - \eta) Q}$$

Where p_a is the atmospheric pressure, ε is a constant molar mass, η is engine dependent and $E I_{H_2O}$ and Q are fuel dependent parameters. As we have already said, a steeper slope corresponds to a larger contrail formation probability. If the aircraft is more effective (i.e. higher η), the temperature of the exhaust gasses is actually lower and contrail formation is promoted. On the other hand if the higher heating value of the fuel Q is larger then the slope will be lower and the contrail formation probability will be lower.



In summary:

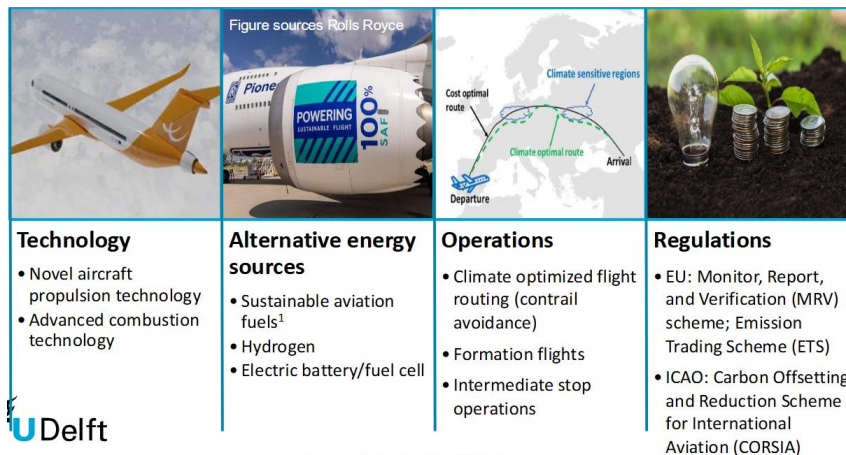
- **Contrail formation** → affected by aircraft propulsion efficiency, fuel type and ambient condition (Schmidt Appleman Criterion applies)
- **Contrail persistence** (temperature $< -38^\circ\text{C}$; relative humidity with respect to ice $> 100\%$) → affected by weather
- **Contrail properties** (optical depth, etc.) → affected by particle emissions

Non-CO₂ emission are very different from CO₂ emissions:

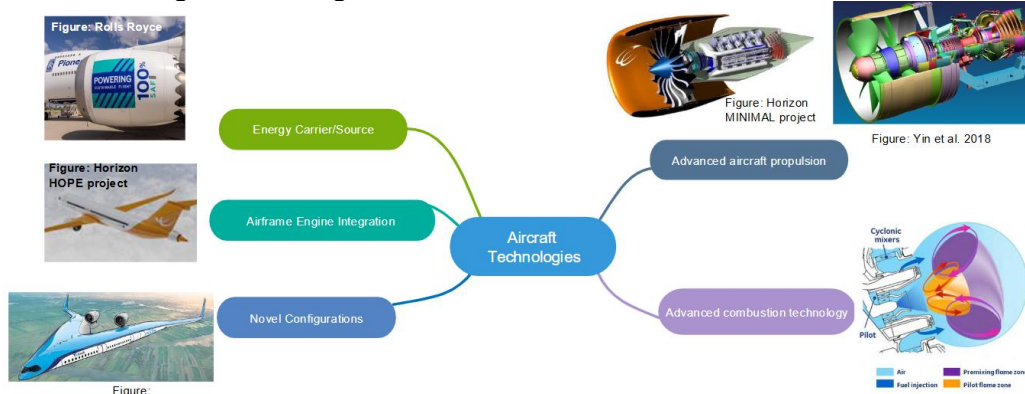
	CO ₂ effects	Non-CO ₂ effects
Time-scale	Centuries	Contrails: Hours Ozone changes: Weeks Methane changes: Years Water vapor: • Hours (troposphere) • Months (stratosphere)
Location dependence	No	Strong location dependency (the lower; the smaller)
Contribution	~1/3 of the net aviation RF.	~2/3 of the net aviation RF.
Confidence level	High	Medium/Low

Note: the high confidence level for the CO₂ emissions means that their impact is very well understood.

The main mitigation measures to combat the effect of climate change are:



When it comes to engine technologies:



In particular, for the combustion technologies lean burn technologies are able to greatly decrease the NO_x emissions and particulate emissions by reducing the maximum temperature reached in the combustion chamber with minimal impact on engine performance.

For alternative energy sources we have either drop-in solutions (e.g. SAF) or liquified hydrogen. SAF can be blended with traditional fuels at different levels with limits between 10% and 50% and can lead to a maximum reduction of 50% in climate change impact due to the non-CO₂ emissions effect.

When it comes to hydrogen, CO₂ and nvPM emission can be completely eliminated, however water vapour production is greatly increased. More research however is required since the effect are still not very well known.

To summarize, CO₂ and H₂O are greenhouse gasses since they both absorb outgoing longwave radiation. NO_x emissions enhance ozone formation and methane depletion, both of which are greenhouse gasses. However, based on the current level of understanding, the overall effect is warming. Contrail formation depends on the efficiency of the engine, fuel type and ambient conditions. There is no single solution which eliminates all of these emissions. It's always a trade-off between CO₂ and NO_x formation.

